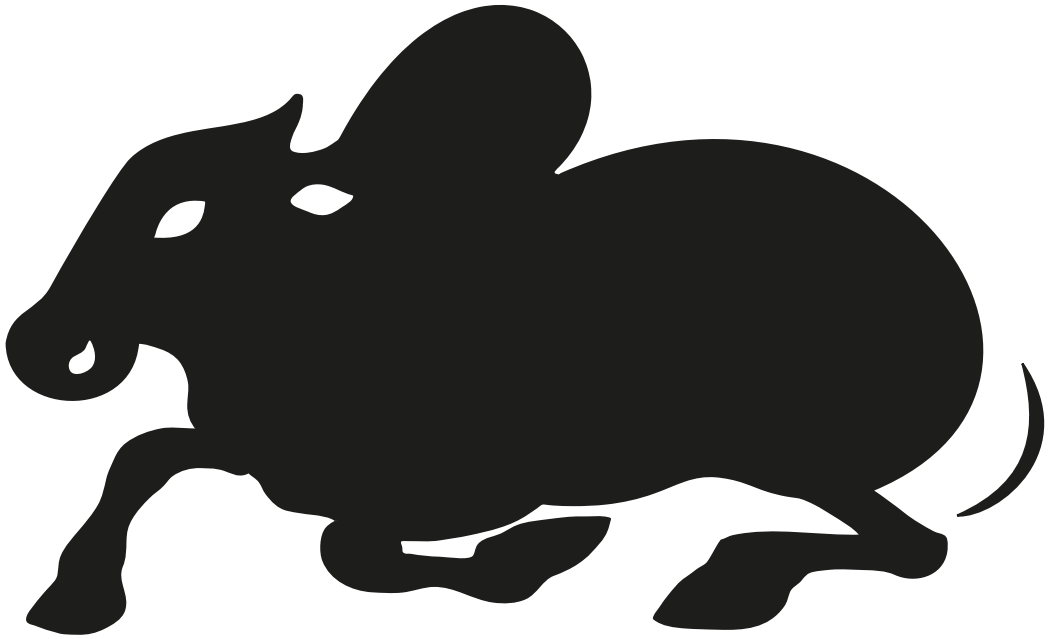


dharma

Transliteration Guide

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1. Introduction

1.1. Version History

Version	Author(s)	Changes	Date
0.1	Balogh, Griffiths	First draft	2019-07
1	Balogh, Griffiths	Expansion and revision for first release	2019-09
2 (1.1)[1]	Balogh	Revision	2019-12
3	Balogh, Griffiths	Revision and expansion	2020-07-05
4	Balogh, with Griffiths	Massive revision and expansion	to 2026-10

This is the final and definitive version of the DHARMA Transliteration Guide. Specific changes to the transliteration scheme since the last release (version 3) are few and minor, but the structure and conceptual background have been thoroughly revised. For these reasons, a detailed change log is not included here.

1.2. Acknowledgements

Over the years, many people have contributed suggestions, questions and critiques for earlier versions of this Guide. The authors wish in particular to thank the contribution of Emmanuel Francis and D. Christian Lammerts.

1.3. Scope

This Guide is essentially intended to cover the scripts relevant to the languages with which the DHARMA project is concerned, i.e., in alphabetical order (omitting the adjective "Old" relevant in several cases): Balinese, Cam, Javanese, Kannada, Khmer, Malay, Prakrit, Sanskrit, Sundanese, Tamil, Telugu. However, the recommendations we give here are certainly intended to be compatible with and extensible to other languages and scripts as and when needed for future projects.

The contents of this Guide are primarily applicable to digital editions of the text of inscriptions and manuscripts for preservation, dissemination and computer-aided research. Digital editions must follow these instructions rigorously. We do, however, hope (and, to some degree, expect) that project members will use the same transliteration method, as far as applicable, in their print publications and other work, and that the solutions recommended here gain currency beyond the scope of the DHARMA project. Section 3.2.5 gives some further pointers on what features of the transliteration system can be ignored outside diplomatic editions. We further encourage all project participants to include a reference to the current release version of the Guide (on HAL-SHS) in their publications, in order to make clear that the conventions DHARMA proposes are a published standard, and to disseminate awareness of this standard.

1.4. Abbreviations

In addition to common abbreviations, this Guide uses:

TG for the DHARMA Transliteration Guide (the present document)
EGD for the DHARMA Encoding Guide for Documentary Editions (version 2.0)

[1] Version 2 bears the internal version number 1.1, but since the automatic versioning in the HAL-SHS repository assigned it number 2, we have chosen to adopt that numbering to eliminate future inconsistencies.

1.5. Brackets for linguistic notation

The concepts indicated by these brackets are introduced in §2.2.

/a/, /ɑː/	**slashes** indicate phonemic entities
[ɑː], [ɐ]	**square brackets** indicate phonetic entities
<a>, <k>	**angle brackets** indicate graphemic entities
|र|, |a|	**vertical bars** indicate graphetic entities

1.6. Terms and definitions

The following list gives cursory definitions of script-related terms, located here for quick reference. Concepts which are vital for working with this guide are underlined in the list. It is strongly recommended that you read the more detailed discussion of grapholinguistic terminology in §2 before reading the contents of this section.

- script and writing (§2.1)
  - **writing** is the graphic representation of language
  - a **script** is an inventory of graphic signs which can be used conventionally for writing
  - a **writing system** is a system of rules governing how certain aspects of a particular language can be recorded by means of a particular script
- typology of writing systems (§2.1.1)
  - a **phonographic** writing system is one which predominantly records language by representing archetypal speech sounds
  - an **alphabetic** writing system is a phonographic system which represents every archetypal speech sound by a visually independent graphic sign
  - an **aksharic** writing system is a phonographic system where the graphic signs representing archetypal speech sounds are often visually dependent on primary signs, so that only their combinations are visually independent; and where primary consonant signs by default also indicate an 'inherent' vowel
- conversion between writing systems (§2.1.2),
  - **transliteration** is the use of a target writing system to represent how something is written in a source writing system, so that target graphemes correspond to source graphemes
  - **transcription** is the use of a target writing system to represent how something sounds in a source language, so that target graphemes correspond to source phonemes or phones
  - **Romanisation** is transliteration or transcription into the Latin writing system
- **graphemes** (§2.3.1) are a finite set of signs belonging to a writing system, comprised of a signifier that is a graphic feature and a signified that is a minimal unit of abstract linguistic information pertinent to analytic interest
  - **graphemics** (or graphematics) is the study of graphemes
- **graphs** (§2.4.2) are an infinite set of graphic signs belonging to a script, which are the concrete graphic instantiations of graphemes, such as |A|, |𐌀|, |अ|, |অ|
  - **graphetics** is the study of graphs
- **graphotactics** is (the study of) the ways in which graphemes and graphs combine into sequences
- **allographs** (§2.5.1) are graphs which instantiate the same grapheme
  - **graphetic allographs** are allographs which do not signify any linguistic information other than that contained in the grapheme, such as Devanagari |अ|, |अ| and |অ|
  - **graphotactic allographs** are allographs of which only one is permitted in a given graphotactic context (in a given sequence of graphemes), such as the graphs corresponding to the grapheme <r> in Devanagari |र्क|=|rka| and |क्र|=|kra|
  - **graphemic allographs** are allographs which potentially signify linguistic information in addition to, and at a different level than, that inherent in the grapheme, such as Devanagari |उ| | versus |ु|
- **homographs** (§2.5.2) are identical or nearly identical graphs which instantiate different graphemes

- **polygraphs** (§2.5.4) are groups of two or more graphemes that together conventionally indicate a particular phoneme
- a **character** (§2.4.1) is the minimal graphetically autonomous unit of a writing system, for example, <A>, <T>, <t·>, <ka> and <rtsnyai> are characters of the Indic writing system
- characters may be comprised of one or more graphemes (§2.4.3)
  - a (graphemically) **simplex character** is a character comprised of a single grapheme, such as such as Indic <A> and <T>
  - a (graphemically) **complex character** is a character comprised of two or more graphemes, such as Indic <t·>, <ka> and <rtsnyai>
- a **glyph** (§2.4.2) is the concrete manifestation of a character, for example, |अ|, |त्|, |क| and |त्स्र्यै| are glyphs of the Devanagari script
- glyphs may be comprised of one or more graphs (§2.4.3)
  - a (graphetically) **simplex glyph** is a glyph comprised of a single graph, such as such as Devanagari |अ| and Bengali |ৎ| as well as Devanagari |त|
  - a (graphetically) **complex glyph** is a glyph comprised of two or more graphs, such as Devanagari |ते| and |क्ल|
  - a **conjunct** or **ligature** in an aksharic writing system is a particular kind of complex glyph involving two or more consonant graphs, such as Devanagari |क्ल|
- the term ***akṣara*** can refer to either the characters or the glyphs of an aksharic writing system
- a **component** (§2.4.3.1) is a visually discernible graph in a complex glyph, such as the graphs |t| and |e| in |ते|
  - a **marker** (§2.6.2) is a graph which can only manifest as a component, and never as an independent glyph, such as the |e| in |ते|
- a (graphic) **element** (§2.4.4) is any salient part of a graph or glyph that is not itself a graph, i.e. which has no graphemic signification of its own
  - a **stroke** is a graphic element which can be written as a single stroke of the writing instrument
  - a **diacritical mark** (§2.4.4.1) is a kind of element which can be added to a graph and thereby alter the grapheme associated with that graph
- graphemes can be classified on the basis of linguistic function as follows
  - a **phonographic grapheme** (§2.3.2) is one that signifies information pertaining to speech sounds in any phonographic writing system
  - **supplementary graphemes** (§2.3.3) are ones that signify information pertaining to other domains of analytic interest, including (according to our particular analytic interest)
    - **ideograms** signify words, morphemes or abstract concepts
      - **numerals** or **ciphers** are ideograms that signify numbers
    - **marks** represent information about the way to convert the adjacent written text to language
      - **punctuation marks** carry information about segmentation into syntactic or metrical units
    - **abstract symbols** are signs we recognise as a grapheme without attributing a particular linguistic function to them
- graphemes can be classified on the basis of how we treat them in transliteration as follows
  - **core graphemes** are those to which we dedicate a unique transliteration equivalent, and this identifies the grapheme precisely enough for our purposes
    - our core graphemes are the Indic writing system's phonographic graphemes (including the inherent vowel and the *virāma*) and its numeral signs, along with a small number of miscellaneous graphemes such as the *avagraha* and the abbreviation mark
  - **peripheral graphemes**, also called simply **symbols**, are those which we transliterate by means of a placeholder, which identifies the grapheme in a vague manner, while more precise identification may be added through encoding
    - our peripheral graphemes include the Indic writing system's non-numeral ideograms, its marks other than the *avagraha* and the abbreviation mark, and its abstract symbols

1.7. Working with Unicode

All text documents in the DHARMA project, transliterated or otherwise, must conform to Unicode. Today, Unicode is the universal default code table for characters, so this requires no special effort. Never use a custom or legacy character encoding, as it will turn into gobbledygook whenever someone attempts to display it in a different font. For disambiguation, we often specify Unicode codepoints in the format U+####, where #### stands for a four-digit hexadecimal code.

1.7.1. Fonts and supported characters

It is up to you to choose your preferred font for working in a word processor or XML editor, so long as it is Unicode-compliant. Some of the widely current fonts lack support for many of the special characters employed in our transliteration scheme, while some others nominally support these characters, but render them awkwardly. When choosing a font, it is advisable to test it for some of the rarer characters relevant to the language you work with (r̥̄, m̆ and ḫ are good candidates).

Some publishers require using a specific font, which may or may not support all the special characters you require. If persuasion does not work, feel free to make any necessary compromises and substitute more widely supported characters for the problematic ones, i.e. to use informal alternatives (§3.2.6).

Of the fonts shipped with current versions of Windows and Mac OS, Times New Roman, Tahoma, Arial and Calibri are fair choices, but we preferentially recommend one of the following free fonts:

- Gentium by SIL, https://software.sil.org/gentium/ (in which the body text of this Guide is typeset)
- Noto Serif and Noto Sans by Google, https://www.google.com/get/noto/

1.7.2. Entering Unicode characters

You probably already have a favourite keyboard layout to access the special characters you need in your work, but some of the characters used in DHARMA transliteration may not be covered. Detailed technical instructions on designing, installing and using keyboard layouts or assigning shortcut keys are beyond the scope of this guide, so we can only offer the following tips.

Out-of-the-box solutions:

- on a Mac, try the layouts Easy Unicode or ABC Extended (formerly US Extended)
- on Windows, there is no suitable keyboard shipped with the system, but you may be able to use and/or adapt John Smith's keyboard layout and Word macros, available at http://bombay.indology.info/software/fonts/induni/index.html

To create your own keyboard for Windows:

- use the Microsoft Keyboard Layout Creator, https://www.microsoft.com/en-us/download/details.aspx?id=102134

To enter special characters as Unicode codes:

- in MS Office you can type the code, then press ALT + x to convert the code into the corresponding character
  - you can enter the code with or without the prefix U+, but using it will make certain the software recognises where the code begins, so the last characters you typed before the code will not interfere with what you want to produce
- on Mac OS (systemwide), you need to enable Unicode Hex Input in Language Preferences
  - once you have done this, whenever you switch to this keyboard layout, you can press and hold Option while you type the character code (without the prefix U+) then release Option

If you can access most of the characters you need via your keyboard, but there are a few that you need occasionally and cannot access, one of the following solutions may help:

- assign a shortcut key or sequence to the inaccessible characters in your editing software
- insert them from a table of available characters
  - in MS Office, use Insert Symbol
  - on Mac OS (systemwide), use the Character Table

– copy and paste the inaccessible characters from this guide (or from a file you keep at your fingertips, listing each of those characters) each time you need one of them
– if all else fails, use informal alternatives (§3.2.6)

1.7.3. Precomposed characters

Unicode caters for Latin letters with diacritical marks in two separate ways. On the one hand, many diacritical characters — such as ü, ā or ṭ — are available in precomposed form, which means that the letter-plus-diacritic is a single Unicode character.[2] On the other hand, diacritical marks — such as ◌̈, ◌̄ or ◌̣ —are themselves available as special combining characters which, when entered after a suitable base character, will be combined with it to render as the base character with the diacritical mark. For example, the character ā may be produced either by entering a (`U+0061` Latin Small Letter A) followed by ◌̄ (`U+0304` Combining Macron), or by entering ā (`U+0101` Latin Small Letter A With Macron). Processing software is generally able to treat the two as equivalent (so, for instance, searching for one will find the other as well), but this cannot be guaranteed in all situations. For this reason we urge you to **always use precomposed characters** whenever they are available, and resort to combining diacritics only when the Unicode codetable does not include a precomposed form. The most common character in our practice for which no precomposed form is available is *r̥*; others will be noted in the body of the Guide where applicable.

The current version of Oxygen (our preferred XML editor) handles combining diacritics as separate entities. As a result, when you wish to delete a combined character or to select it in order to place markup around it, you will need two keypresses instead of one, otherwise stray diacritical marks may be left in unwanted places.

[2] See note 48 about the concept of a Unicode character.

2. Theoretical framework

Throughout this Guide, the word *graphic* is used in a technical sense meaning 'related to writing.' The study of written language will be called *grapholinguistics* (Meletis 2020a, 3). Like many areas of linguistics, grapholinguistics has been approached by theoreticians in various ways, and many of the relevant technical terms refer to (sometimes radically) different concepts depending on approach. Many of its scholars have worked primarily with alphabetic writing, enabling them to ignore features of other writing systems that cannot be readily explained in their terms. We in turn are concerned specifically with the transliteration of Indic writing systems to a Latin one, and are, for pragmatic reasons, at ease to ignore subtle and complicated aspects of written language that are not directly relevant to this. Nonetheless, we must define certain concepts with sufficient accuracy in order to be able to discuss the elements of writing productively and unambiguously. The following discussion, which we recommend you read through at least once, situates the concepts relevant to us in a rough theoretical framework. For future reference, the list of quick-and-dirty definitions §1.6 shall suffice.

2.1. Script and writing

We start out from the premise that **writing** is *glottography*, the graphic representation of language.[3] The representation of meaning as independent from language (called *semasiography*)[4] is thus excluded by default, although we must eventually take it in our stride to some extent (§2.3.3). A **script** is an inventory of graphic signs used conventionally for the representation of linguistic information.[5] A **writing system** is a set of rules governing how certain aspects of a language can be recorded by means of a script.[6] It follows from this that scripts are in themselves independent of language, and that a language may be associated with several writing systems, or none. The concept of a writing system, on the other hand, presupposes both a particular language and a particular script.[7]

We hasten to add that the word "particular" in this statement may be understood at different points on a scale of concreteness. Indeed, to speak in terms like "the system for writing English in the Latin script," as current grapholinguistic theorists often do, is already a generalisation relative to more specific systems.[8] Accordingly, we also use and endorse sweeping generalisations such as **Indic script** for all of the Brāhmī-derived scripts and **Latin script** for all of the scripts derived from that of ancient Rome,[9] and likewise **Indic writing system** and **Latin writing system** understood as the generic set of rules for employing variants of these scripts for any language with which they have been associated historically. In spite of individual differences at lower points of the scale, these rules in the broad sense still share a common systemic framework and prototypical building blocks.

[3] With Meletis (2020a, 20).
[4] As, for example, through the symbols |☢|, |♬| or |👍|.
[5] With e.g. Wellisch (1978, 15) and most theorists.
[6] With Weingarten (2013, 17–18), in agreement with Meletis and Dürscheid (2022, 65–66) and essentially reconcilable with Wellisch (1978, 15, 13), and Coulmas (2003, 35).
[7] With Weingarten (2013, 18) and most recent theorists.
[8] Such as that for writing twenty-first century Bronx English on a screen in Arial, or that for writing Chaucerian English on parchment in insular minuscule.
[9] The terms 'Roman' and 'Latin' are near-synonyms in the context of scripts and writing systems. Coulmas (2006, 285–87, s.v. Latin alphabet) uses 'Latin' specifically for ancient Rome and 'Roman' for the broad family of scripts and writing systems derived from the former (ibid., 438–39, s.v. Roman alphabet). There seems to be no clear rationale for this and Unicode terminology identifies all "Roman" letters as Latin, so for consistency we prefer the term 'Latin' and use 'Roman' only in the word 'Romanisation.'

2.1.1. Writing system typology

Writing systems may be divided into major classes[10] according to the level of language at which they typically establish correspondences between graphic signs and linguistic information. We shall refer to this as the *dominant level of representational mapping* (Meletis and Dürscheid 2022, 212, 216). The writing systems we are concerned with here are **phonographic** or 'sound-writing.' Phonographic writing systems may be further classified as alphabetic, aksharic, abjadic or syllabographic. An **alphabetic** writing system — such as the Latin one — ideally matches each and every phoneme of the language to exactly one freely combinable graphic sign of the script. An **aksharic**[11] system — such as the Indic one — is distinguished from alphabetic systems by the fact that it involves visual segmentation into chunks we shall refer to as characters (§2.4.1), each of which typically consists of a primary sign and zero or more dependent signs. The latter prominently include signs for postconsonantal vowels and secondary forms of consonant signs for use in consonant clusters. Regular characters based on a primary consonantal sign imply an inherent vowel which has no overt graphic representation.[12] The basic principle of aksharic writing does not cater for consonants followed by a hiatus (at the end of a graphic sequence), nor for vowels not preceded by a consonant (at the beginning of a graphic sequence, or after another vowel). Aksharic systems therefore employ separate independent vowel signs, and either independent signs for consonants or a vowel killer sign that can associate with the regular consonants.

Other kinds of phonographic writing systems are only relevant to us inasmuch as they serve as useful contrasts to some aspects of aksharic systems. In a nutshell, the key feature of **abjadic** systems — such as the Arabic one — is that not all of the relevant language's phonemes are represented in writing: typically only the consonants are written. The dominant level of representational mapping for each of the above kinds is that of phonemes. Conversely, **syllabographic** systems — like Japanese kana — are also phonographic, but they employ signs that represent clusters of speech sounds ("syllables"[13]). What distinguishes these from abjadic and aksharic systems, which also operate with signs that correspond to "syllables," is that syllabographic signs cannot be broken down into graphic components that separately represent individual phonemes. Finally, non-phonographic writing systems predominantly represent language at the level of words or concepts and may accordingly be called *logographic* or *ideographic*.[14]

2.1.2. Conversion between writing systems

For the conversion of one writing system to another, the terms 'transliteration' and 'transcription' are sometimes used interchangeably. Moreover, 'transcription' is widely prevalent in epigraphic and manuscript studies for the process of re-recording (using any writing system) the written text witnessed on an artefact. Notably, this usage is present throughout TEI and EpiDoc documentation. In a more general sense, the term is also used in many contexts for that of writing down a text witnessed in a different medium, primarily

[10] There exist various and not entirely compatible typologies of writing systems. We broadly follow Meletis (2020a, 142–51), q.v. for a discussion of other typologies. For an in-depth study, see e.g. Daniels (2018).

[11] In addition to *aksharic* (with varied spellings), technical terms widely used for such writing systems include *abugida* and *alphasyllabary*. The diverse definitions offered for each of these terms are not altogether compatible. See also note 40 below, and see e.g. Gnanadesikan (2017) for a discussion and an attempt at a clearer typology of phonographic writing systems. She endorses the term *āksharik*, which we adopt here apart from the spelling, since it fits the system of related terms better than "akshara script" suggested by Salomon (2003, 78).

[12] The inherent vowel may be absent (or optional) in some specific Indic writing systems, notably early Tamil Brāhmī (Salomon 1998, 36; 2003, 104). The only truly essential feature of an aksharic writing system is that it employs dependent signs for postconsonantal vowels.

[13] The *graphic syllables* predominantly represented by the signs of syllabographic writing systems do not necessarily correspond to speech syllables; for further discussion, see e.g. Coulmas (2003, 62–66); Meletis and Dürscheid (2022, 240–42).

[14] See Coulmas (2003, 40–41). More recently the term *morphographic*, 'morpheme-writing' has gained ground (Meletis and Dürscheid 2022, 243–49). Each of these terms is based on a slightly different interpretation of the nature of linguistic information represented in these systems, but the fine distinctions are irrelevant to us here, so we use these terms by and large synonymously.

audio. In this guide, however, we use and encourage using these terms in specialised senses.[15] **Transcription** in this technical sense means using a target writing system — often but not necessarily an artificial one designed to record speech sounds unequivocally — to represent how a (written or other) text *sounds*. **Transliteration**, in turn, is the use of a target writing system to represent how a text *is written* in a dissimilar source writing system. The modern Indic name written in Devanagari as बलदेव may be transcribed (in loose Anglicisation) as Baldev or even Buldeo and (in IPA) as [bɐldeːɔ], and it may be transliterated (in the DHARMA standard) as *baladeva*. The term **Romanisation** can refer to either transcription or transliteration where the target writing system is (a variety of) the Latin system.

In a nutshell, transcription is essentially concerned with the phones or phonemes of a spoken language, without regard to how they might be written in a source script; conversely, transliteration is essentially concerned with the graphemes of a written language, without regard to pronunciation (Wellisch 1978, 18). That said, transliteration systems in actual practice include a number of transcription-like features, at least for the sake of pronounceability and ease of acquisition.[16] Also, "there is no single 'scientific' system whose principles can be applied uniformly to all scripts and for all purposes ... Rather, there is a plurality of more or less justified but mutually incompatible requirements ... so that a choice must be made among those requirements that are *optimally* needed to make the system work for a particular purpose or task" (ibid., 314; emphasis original).

Our purpose in transliterating original texts is to facilitate their reading, editing and computer processing, and to do so in diplomatic manner, meaning that it allows a reader informed by thorough knowledge of the specific writing system and script employed in the original to produce a reasonably accurate reconstruction of the original from the transliteration. Obviously, transliteration cannot possibly aim to represent each and every graphic detail of the source. What, then, counts as reasonably accurate? According to Wellisch (1978, 322), a transliteration system for philological and epigraphic purposes should strive toward "a one-to-one relationship between each grapheme in a target script and a corresponding [...] grapheme in the source script" in order to be fully reversible. Unfortunately, Wellisch does not tell us what a grapheme is, so with this, we come to muddier waters, and we will have to take a closer look at grapholinguistics.

2.2. Some basic concepts

A distinction between so-called **etic** and **emic** aspects (e.g. Nöth 1990, 300) has been fruitful in several subfields of linguistics and will be important for our discussion of grapholinguistics. The former term concerns variable concrete realisations, while the latter pertains to invariant abstractions on the basis of shared function.[17] Since our readers are presumably familiar with the basic concepts of phonology, we begin by summarising some of these as useful analogies to several grapholinguistic concepts.

The concrete speech sounds constituting spoken language come in a practically infinite continuum. These etic sounds are referred to as **phones**. The study of their characteristics is called **phonetics**. According to widespread convention, we use square brackets to represent phones in written discussion, e.g. [ɐ].[18] The phones of any language can be sorted into a limited number of emic entities termed **phonemes**, on the basis of shared linguistic function.[19] The study of phonemes is called **phonemics**. The term 'phonology' is also

[15] With Wellisch (1978, 18) and Coulmas (2003, 36).

[16] There is no hard reason why we should not transliterate धर्म as *x%it%* (so long as the matching of source signs to target signs is consistent throughout our transliteration scheme), yet most of us prefer more intuitive transliterations such as *dharma*.

[17] This dichotomy is not quite the same as the anthropological distinction of "emic" and "etic" as an insider and outsider perspective; the point of connection is that in both usages, "etic" refers to an objective and impartial description of characteristics, while "emic" is concerned with subjective meaning within a system.

[18] On the few occasions where we mention phones, we represent them in the IPA phonetic alphabet. The accurate pronunciation of the IPA signs is not relevant to our discussion.

[19] The linguistic function of phonemes is generally understood to be the distinction of semantic meaning, demonstrable through minimal pairs in a language's lexicon, i.e. words that have different meanings while differing only in having one or the other phoneme candidate at a certain point. This criterion is not unproblematic, but this need not concern us here.

used in this specific sense, but we prefer to use the latter for the study of speech sounds encompassing both phonetics and phonemics. To represent phonemes in written discussion, we enclose them in slashes, e.g. /a/.[20] The diverse phones which may manifest a given phoneme are known as **allophones**. **Homophony** is the situation where two linguistic units (e.g. words) are identical with regard to speech sounds, yet have different meanings. **Phonotactics** is (the study of) the rules governing how phonemes and phones may combine into sequences in a particular language.

In grapholinguistics, the etic **graphs** are infinitely variable graphic signs that can be used for the representation of linguistic information. The study of graphs is called **graphetics**. In written discussion, graphs are enclosed in vertical bars, e.g. |a|.[21] Their emic counterparts, **graphemes**, are a limited number of discrete abstract entities systematised on the basis of their shared function in representing language.[22] The field of their study is **graphemics**.[23] In discussion, we enclose graphemes in angle brackets, e.g. <a>.[24] Diverse graphic signs which signify the same linguistic information are known as **allographs**. **Homography** is the situation where two graphic entities are visually identical, but have different significations. **Graphotactics** is (the study of) the rules governing how graphemes and graphs may combine into sequences[25] in a particular language.

Thus, the Sanskrit phoneme /a/ may be instantiated phonetically by several allophones such as [a], [ɐ] or [ə]. This phoneme corresponds to the grapheme <a> in Indic writing systems. In the particular writing system that associates modern Devanagari with Sanskrit, this grapheme can be instantiated graphetically by several allographs such as |अ| or |অ|. In the DHARMA transliteration scheme, which is in fact another particular writing system that associates Indic languages with the Latin script, the same grapheme can be instantiated by allographs such as |a|, |A|, |ⷯ| or |𑖀|.

This basic sketch might be all we needed if we worked only with ideal alphabetic writing systems where each and every phoneme of the language exactly corresponds to one grapheme of the writing system and vice versa. This, alas, is not the case, so several of the concepts outlined above, and in particular that of the grapheme, will require elaboration. It is important to keep in mind that entities such as the grapheme and the graph are analytical constructs we employ in order to systematise, and thereby inevitably simplify, a complex and often chaotic reality where smooth spectra are more typical than sharp distinctions. Therefore, in spite of our best efforts to carve nature at its joints, many of our concepts will have fuzzy boundaries (§2.6), and some of the phenomena we must deal with in transliterating Indic writing systems will fit our classification poorly.

2.3. Graphemics

2.3.1. What is a grapheme?

The broader grapholinguistic literature is replete with mutually incompatible, indeed often contradictory, grapheme definitions,[26] so much so that some leading theorists question the usefulness of this concept altogether.[27] Our own working definition is as follows: **a grapheme** is a sign comprised of a signifier that is a

[20] For representing phonemes, we either use the standard spelling (for English) or DHARMA transliteration (for source languages), or use the IPA equivalent of a representative phone (e.g. /ɔ/), as deemed expedient for the phoneme in question.

[21] When the illustration of actual graphic appearance is desired, we use an approximation of the shape concerned. When discussing graphs and graphic units of Indic scripts without needing to (or being able to) illustrate their form, we use transliteration (e.g. |rddhe|, |r|).

[22] The function of graphemes will be addressed in at length in §2.3.1.

[23] The term 'graphematics' is perhaps more prevalent in current scholarship, but the two are largely synonymous (Meletis 2020a, 26 n. 29), and we see no reason to break the analogy with 'phonemics.'

[24] Since graphemes are emic concepts and not etic written signs, the symbols used for their notation are arbitrary (cf. Meletis 2020a, 252–53, n. 9). In this document, we always represent the graphemes of Indic writing systems in transliterated form.

[25] Including characters (§2.4.1) and glyphs (§2.4.2).

[26] See Meletis (2019, 27–34) for an overview.

[27] Notably Daniels (2018, 164–71).

graphic feature and a signified that is a minimal unit of abstract linguistic information pertaining to a particular domain of analytic interest. We find this grapholinguistically tenable,[28] practicable for our purposes, and in harmony with Wellisch's approach to writing and transliteration. We elaborate this definition below, and look at its consequences in the following subsections.

Graphemes are on the one hand **analogous** to phonemes inasmuch as they are abstractions of graphic signs, while phonemes are abstractions of speech sounds. However, we do not extend this analogy to the degree of requiring graphemes to be capable of distinguishing semantic meaning.[29] On the other hand, graphemes, being entities of a writing system, straddle the gap between the disparate domains of language and script. In this respect they are **referential**, establishing a connection from writing to language. Accordingly, we[30] conceive of the grapheme as a sign[31] comprised of a linguistic aspect as its signified and a graphic aspect as its signifier.

The **graphic aspect** of graphemes may be any graphic feature that unequivocally signifies a piece of linguistic information by the conventions of the writing system in question, without requiring interaction with components of the language system, such as the lexicon. As a consequence, we do **not** require the graphic manifestation of a grapheme to be **segmental** (isolable in the linear flow of the script) or **autonomous** (independently combinable).[32] Nor do we require the graphic aspect of a grapheme to be an overt, visually **isolable** graphic sign,[33] meaning that holistic features such as the manner in which graphetic constituents are arranged can also correspond to graphemes if they signify linguistic information without needing input from the language system. Defining the graphic aspect acceptable for a grapheme in this manner is in our opinion necessary in order to be able to deal satisfactorily with some peculiarities of aksharic writing, which will be addressed in more detail in §2.3.1.1 and §2.3.2.1 below.

The **linguistic signification** of graphemes may belong to a broad range of linguistic domains. In order to restrict graphemes to a manageable set, we restrict this to a particular **domain of analytic interest**. Our analytic interest is primarily in speech sounds, so the **primary graphemes** we deal with are those whose linguistic signification pertains to the presence of a phonological unit. We also call these **phonographic graphemes** (discussed further in §2.3.2 below) and distinguish them from supplementary graphemes (§2.3.3), which represent information pertaining to other domains of analytic interest. This is a slight modification of Meletis's approach (e.g. 2019, 36; Meletis and Dürscheid 2022, 127), which defines a 'default grapheme' as one that signifies a linguistic unit at the writing system's dominant level of representational mapping, i.e. a phoneme in alphabetic, abugidic and aksharic systems, a syllable in syllabographic ones, and a morpheme in morphographic ones. In addition to integrating supplementary graphemes into the conceptual framework, defining graphemes in terms of analytic interest also allows us to recognise "honorary graphemes" (§2.7.2), which are pertinent to analysis but are difficult to qualify as graphemes on a purely linguistic basis.[34]

Importantly, the linguistic signification of a primary grapheme is in our view not an actual linguistic unit (such as a phoneme), but rather an **archetypal notion** of a linguistic unit. Grapheme-phoneme correspondence — the manner in which the archetypal phonemes inherent in a grapheme map to the actual pho-

[28] Our approach has much in common with that advocated by Meletis (e.g. 2019), but we have adapted some of his notions in altered form, and reject others. Discussion of where, how and why we agree or disagree with him and other grapholinguists will be kept to a minimum in order to be able to proceed as quickly as possible to our actual subject matter.

[29] With Fedorova (2013, 50) and Weingarten (2013), contra Meletis (e.g. 2019, 35–36).

[30] Broadly in agreement with numerous theorists including Fedorova (2013, 50), Weingarten (2013, 19) and Meletis (e.g. 2019, 35).

[31] In the specific semiotic sense of a dyadic sign, see e.g. Nöth (1990, 59–60).

[32] With Meletis (e.g. 2019, 41) and Weingarten (2013, 18), contra Fedorova (2013, 50).

[33] With Weingarten (2013, 20), largely contra Meletis (e.g. 2019, 35).

[34] For practical purposes we also use a different and overlapping classification based on how various kinds of graphemes are best handled in transliteration. This pragmatic classification will be discussed in §3.1.

nemes of spoken language — is a complex process best kept separate from the study of graphemes themselves,[35] often involving contingency on other graphemic units in the context, as well as reciprocal interaction with higher levels of language processing (such as morphology and the lexicon). Even when correspondence is relatively straightforward, it may be under-specific[36] or over-specific.[37] When for the sake of brevity we speak of graphemes as signifying speech sounds, we always mean archetypal speech sounds, which may not correspond consistently to the spoken phonemes of the language being written.[38]

As a final criterion, we require the grapheme's linguistic signification to be **minimal**,[39] meaning that a grapheme cannot be subdivided into smaller units which themselves represent linguistic information in the same domain of analytic interest. What this entails is that there is no such thing as a complex grapheme, although graphemic units in different domains of analytic interest may be incorporated in or overlap one another.

2.3.1.1. The *akṣara* is not a grapheme

Some discussions of writing systems and Indic palaeography refer to *akṣara*s as graphemes. However, *akṣara*s such as *ke* |के| and *kta* |क्त| fail to satisfy our minimality criterion: their constituents are genuine graphemes, since they signify the same kind of linguistic information as the corresponding independent signs (Meletis and Dürscheid 2022, 130), notwithstanding the fact that they are graphetically dependent on the primary forms. Denying grapheme status to such constituents would give rise to the awkward notion of the 'sub-grapheme' occasionally used for the parts of such *akṣara*s,[40] while awarding grapheme status to *akṣara*s involving such constituents would obscure the pivotal difference between aksharic writing systems and syllabographic ones.[41] We find it preferable instead to recognise that graphemes need **not** coincide with **autonomous graphetic segments**, although they normally do so in alphabetic systems and in some signs of aksharic systems. We shall refer to graphetic segments as characters (§2.4.1) in their emic aspect, and as glyphs (§2.4.2) in their etic instantiations.

2.3.2. Phonographic graphemes

We have defined primary or phonographic graphemes as those whose linguistic signification pertains to the presence of a phonological unit. The scope of our phonographic graphemes thus encompasses the straightforward vowel and consonant graphemes such as <u> and <k> including the inherent vowel of the Indic writing system (§2.3.2.1), as well as the *anusvāra* and *visarga* along with their variants (§2.6.3), and even the *virāma* (§2.6.3).

[35] With Meletis (e.g. 2020a, 65; Meletis and Dürscheid 2022, 121).

[36] Where a grapheme signifies a set of phonemes which typically share some phonological features, e.g. English <s>, which can stand for /s/ in "bits," /z/ in "heads" and /ʒ/ in "vision," as well as for /ʃ/ in certain accents.

[37] Where the grapheme identifies a particular allophone rather than a phoneme proper, as in Sanskrit <ñ> and <ṅ> or standard Hindi <ṣ>, which correspond to phones that are (arguably) context-dependent allophones of /n/ and /ś/ respectively.

[38] We see this as the logical conclusion of Meletis's thoughts about the separateness of grapheme-phoneme correspondence rules. Strangely, he balks from reaching this conclusion, with complicated consequences bordering on the absurd (e.g. Meletis 2019, 36–38).

[39] Contra Weingarten (2013); essentially with Meletis (e.g. 2019, 35), who requires a grapheme to be a minimal unit with both a linguistic value and a capacity to distinguish semantic meaning.

[40] 'Sub-grapheme' is not a widely current grapholinguistic term and is rarely defined. For Fedorova (2013, 50), it includes both diacritical marks (which we discuss under elements, §2.4.4.1) and the components of complex *akṣara*s. In the technical literature on character recognition, it seems to be more or less identical to what we call elements (§2.4.4).

[41] This lack of distinction is implied by the term *alphasyllabary*, which we consider inappropriate.

2.3.2.1. The inherent vowel of aksharic writing systems is a grapheme

The inherent vowel of aksharic systems comprises a genuine grapheme. This position is not normally taken in grapholinguistic circles,[42] but it is a straightforward corollary of our grapheme definition. Linguistically, the inherent vowel is a unit belonging to our domain of analytic interest, while graphically, it is unequivocally signified by a graphic feature (namely an integral consonant-based *akṣara* that has no additional components which would negate that vowel or replace it with a different one), although it does not correspond to any visually isolable graphic sign.

Denying grapheme status to the inherent vowel would, moreover, make it difficult to demarcate aksharic systems from other types of writing systems. On the one hand, doing so would force us to see *akṣara*s involving the inherent vowel as syllabographic graphemes. On the other hand, it would make us lose sight of a key difference between aksharic systems and abjads. A consonantal grapheme in an abjad can signify either the consonant alone or the consonant and an unspecified vowel, so whether a vowel is present (and if yes, then which vowel) can only be determined by referring to other modules of the language system (in particular, the lexicon). Conversely, in an aksharic system, the writing alone is sufficient for determining whether the inherent vowel is present or not.[43]

2.3.3. Supplementary graphemes

Texts written in a phonographic writing system often include graphic signs which signify information at a level other than that of phonography. The most common examples of such signs are punctuation marks, which primarily convey information about the syntactic, semantic and/or prosodic segmentation of the text. In addition, most writing systems are supplemented with graphemes that signify abstract concepts, as well as with sings that pertain to language in an abstract way, like the *avagraha* of the Indic writing system or the apostrophe of the Latin one, which are often used for flagging elision. Yet others may be quite beyond the domain of glottography, such as "auspiciousness" presumably conveyed by many opening and closing symbols used in inscriptions, or "irony" conveyed by a winking smiley in a text message.

Whether such signs are to be considered graphemes — and if yes, exactly how the grapheme ought to be defined to accommodate them — is a vexed question (cf. Meletis and Dürscheid 2022, 132–33). Meletis (2020a, 148; Meletis and Dürscheid 2022, 132) refers to them as non-default graphemes or grapheme-like entities, relegating them to a separate stage of analysis and treating them much less extensively. We feel that the best way to reconcile such signs with the framework presented above is to understand them as graphemes belonging to one or more supplementary writing systems, which are conceptually separate from (though generally associated with) the primary writing system used for a particular language-script pair, and which are not in themselves capable of representing any language in its entirety.

Whatever their exact grapholinguistic status, all such signs need to be reflected somehow in transliteration, and to do so, we treat their signification as additional domains of analytic interest, supplementing the domain of speech sound. Because the linguistic signification of supplementary graphemes is often vague, complex and polyvalent, we only divide them into a small number of broad theoretical classes here.

Ideograms signify words, morphemes or abstract concepts. We prefer not to make a distinction between ideography and morphography (§2.1.1), because this does not seem applicable to the supplementary signs of a primarily phonographic writing system. Ideograms widely used in modern international writing systems include mathematical signs such as |+| and |%| and currency signs like |$| and |€|. Examples from the Indic writing system include signs for *oṁ* (when not in the form of the regular phonographic glyph of the applicable script, as in Devanagari |ॐ|) and *siddham* (when this word is strongly associated with a glyph, as

[42] Meletis (e.g. 2020a, 100) explicitly denies that the inherent vowel of aksharic systems is a grapheme, even though recognising it as one would seem to follow from his approach to graphetically non-segmentable grapheme clusters (q.v. note 62), and would be conceptually analogous to Meletis and Dürscheid's (2022, 235–36) view that the vowels of the Cree writing system (represented by the orientation of the consonant graph) are graphemes.

[43] The grapheme corresponding to the inherent vowel of an aksharic system is of course present even when — in particular contexts in particular languages — the inherent vowel is not pronounced, or pronounced differently than in other contexts. That is a matter of grapheme to phoneme correspondence, not essentially different from cases such as the grapheme <e> of English or French, which may be silent or may be pronounced in various ways.

in Śāradā |𑆭|), as well as a number of special signs in the Tamil, Telugu and Burmese writing systems. Numeral signs such as |0|, |3|, |¼| and |৪| are likewise ideographic, but are something of a special case, since they are quite universally used in conjunction with all modern writing systems and usually have a definite graphemic signification.

Marks (not to be confused with markers, §2.6.2) are graphic signs which signify neither linguistic units (such as speech sounds or morphemes) nor ideas, but which indicate something about the manner in which the reader should convert the adjacent written text to language. This motley category includes, but is not limited to, punctuation marks such as the international |.|, |,| and |?|, which indicate syntactic or prosodic units in a continuously written text. Another example of a mark is the modern end-of-line hyphen, whose information content is that the end of a written line is not the end of a word.

Abstract symbols is our catch-all category for any glyphs that we at least provisionally recognise as graphemes, but do not explicitly classify as phonographic signs, ideographic signs, or marks. Examples of abstract symbols in modern international typography include signs such as |→|, |✓| and |☺|. All of the above categories have permeable boundaries, so any graphic sign that qualifies as an ideogram or mark in a particular text may be unclassifiable in another text, or even in another part of the same text.

2.3.4. Other entities with a graphemic function

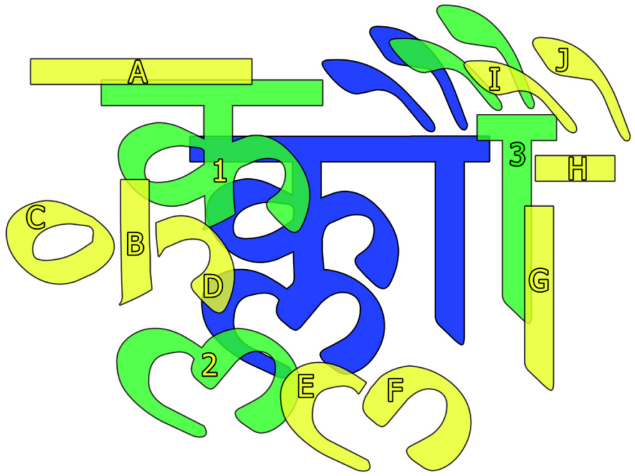
In addition to the graphemes discussed above, there are features of writing which are both less concrete in terms of graphic signs and less concretely pertinent to language, such as paragraph indentation, underlining, italicisation and so on. All of these, and more, are arguably graphemic functions, but such a profusion of multifarious and often overlapping graphemes would be all but impossible to systematise and certainly impractical to transliterate. In working with texts, we remain aware of these graphemic functions, but exclude them from the domain of analytic interest for transliteration.[44]

2.4. Graphetics

Having arrived at a concept of the grapheme that appears to be feasible in general and practicable for our purposes, we now proceed to explore salient graphic entities which are not graphemes by our definition and investigate how they interact with graphemes. As illustrated in Figure 2.4.A, a glyph is a discrete and autonomous graphetic unit, whose graphemic counterpart we call a character. Glyphs may consist of one or more graphs, and characters may consist of one or more graphemes; in the latter case, we refer to the individual entities as components. On the purely graphetic level, each graph is made up of one or more graphic elements, which are in themselves "meaningless," i.e. they do not have a graphemic counterpart. The following subsections elaborate these concepts.

[44] Largely with Meletis (2020a, 49–54; 2020b, 256–57).

Figure 2.4.A. Graphetic entities in the Devanagari glyph |klau|

graphemic comp.		graphetic composition		
grapheme	character	glyph |klau|	graph	element
<k>			1 |k|	A
				B
				C
				D
<l>	<klau>		2 |l|	E
				F
<au>			3 |au|	G
				H
				I
				J

2.4.1. Characters

We define a **character** as a minimal graphetically autonomous unit of a writing system, meaning that the character cannot be broken down into smaller graphetically autonomous parts that, together, have the same graphemic signification as the character. In alphabetic writing systems, characters are normally simplex (§2.4.3), meaning that they coincide with graphemes. This is not necessarily so in other types of writing systems.[45] In aksharic systems, some characters coincide with graphemes, as in the case of independent (typically initial) vowel signs such as Devanagari |अ| (which represents the character and grapheme <A>), and independent (typically final) consonant signs such as Bengali |ৎ| (representing <T>). In most cases, however, *akṣara*s are complex characters consisting of clusters of graphemes. Thus, in Figure 2.4.A above, <klau>[46] is a single complex character in an Indic writing system, which consists of the graphemes <k>, <l> and <au>. The topic of character complexity will be discussed further in §2.4.3 below.

The term 'character' is widely used in a vague non-technical sense, and is rarely defined in a rigorous manner;[47] instead, much of the grapholinguistic literature tends to deal with the character — as defined by us — under the blanket term 'segment.' Importantly, a 'character' in information technology, specifically in the Unicode standard,[48] is *not* what we define as a character.

[45] In true syllabographic systems, graphemes and characters again coincide. Our notion of the character seems difficult to reconcile with cursive abjadic systems such as Arabic, but it could be applied productively to non-cursive abjads such as Hebrew, to typologically complex writing systems such as Hangul and Mayan hieroglyphics, and probably also to morphographic systems.

[46] Characters are emic units of a writing system in spite of being graphetically determined, and therefore we use angle brackets for representing them.

[47] Ollett and Taylor (draft) do define a character in the same way as we do: "an element of the writing system that can be used independently according to the logic of that writing system". Wellisch's (1978, 16) definition is somewhat vague but appears compatible with ours. Neef's (2015, 711) treatment of the character seems to encompass both our grapheme (a term Neef avoids) and our character. Iyengar (2024) rejects the notion of the grapheme, but his understanding of the *akṣara* corresponds straightforwardly to our etic glyph and emic character.

[48] In Unicode, a character is an abstract element of the script defined as a "member of a set of elements used for the organization, control, or representation of textual data" (ISO/IEC 2020, 2, §3.5). That is to say, a Unicode character can be either a "graphic character" (which is much like the grapheme as we define it; ibid. 5, §3.28), a "control character," or a "format character;" both of the latter influence the processing and rendition of adjacent charac-

2.4.2. Graphs and glyphs

We have cursorily defined graphs in §2.2 above as the concrete graphic signs that can be used for the representation of linguistic information. More rigorously, we can now state that a **graph** is a graphic sign that corresponds to one grapheme. Having differentiated the grapheme from the character, we introduce the term **glyph** for the graphic manifestation of a character. Where the distinction between glyphs and graphs is irrelevant, we use the umbrella term **graphic sign**.[49] Thus, in our terms, Devanagari |अ| and Latin |A| (corresponding to the character <A> in their respective writing systems) are simultaneously glyphs and graphs, while the Devanagari glyph |क्लौ| (corresponding to the character <klau>) in Figure 2.4.A is constituted of three graphs (|k|, |l| and |au|, which individually correspond to the graphemes <k>, <l> and <au>). As is the case with 'character,' the term 'glyph' is rarely used in a technical sense and, when used, is rarely distinguished from 'graph.'[50]

As also noted in §2.2, we endorse referring to glyphs and graphs with the transliteration of the corresponding graphemes, especially when the visual appearance of the glyph or graph is irrelevant to the discussion or difficult to represent in Unicode. Thus, it is fully acceptable to use phrases such as "the graph |r| in the glyph |rddhe|."

2.4.3. Character and glyph complexity

Characters (graphemic entities) may be **simplex** when they consist of a single grapheme, and **complex** when they consist of more than one grapheme. Likewise, **glyphs** (graphetic entities) may be **simplex** when they consist of a single graph, and **complex** when they consist of more than one graph. Devanagari |अ| and Latin |A| are simplex graphs corresponding to simplex characters, while Devanagari |क्लौ| in Figure 2.4.A is a complex graph corresponding to the complex character <klau>.

Graphemic and graphetic complexity do not fully overlap in an aksharic writing system, chiefly because the inherent vowel has no isolable graphetic counterpart (§2.3.2.1). In Figure 2.4.B, rows 1 and 2 show graphemically and graphetically simplex units of Latin and Devanagari script. Rows 3 and 4 show graphemically complex units of Devanagari, of which 3 is graphetically simplex, while 4 is graphetically complex. Row 5 shows a special case in Devanagari, with a graphetically simplex graphotactic allograph (§2.5.1.2) of a graphemically complex glyph where none of the three graphemes present in the character have an isolable graphetic counterpart.

ters. Accordingly, many specimens of what we call a character are composed of several Unicode characters. Devanagari क्त्र <ktra>, for instance, consists of one each for the graphemes <k>, <t>, <r> and <a>, plus two instances of the Unicode *virāma* control character (one each after the first two consonants).

[49] The Unicode definition of a 'graphic symbol' as the "visual representation of a graphic character or of a composite sequence" (ISO/IEC 2020, 5, §3.29) appears to be identical to our 'graphic sign.'

[50] Iyengar (2024, 420) defines a graph in a way we find on the whole compatible with ours. According to Meletis (2020a, 46, n. 57), 'graph' and 'glyph' can be considered synonyms. Neef (2015, 711) uses only 'glyph' and does so in a sense that seems to cover both our graph and our glyph, while Coulmas (2006, 173–74, s.v.) defines a graph much as we define a glyph, and all he has to say about 'glyph' (2006, 168, s.v.) is that the term has been "used in the description of writing systems whose units were not well understood", such as Mayan, where it is a "collective designation that can refer to a logogram, a phonetic sign or a compound sign".

Figure 2.4.B. Character and glyph complexity

	grapheme	character	glyph	graph	comment
1	<A>	<A>	|A|	|A|	simplex character, simplex glyph
2	<A>	<A>	|अ|	|अ|	simplex character, simplex glyph
3	<k> <a>	<ka>	|क|	|क| —	complex character, simplex glyph (vowel grapheme represented holistically)
4	<k> <u>	<ku>	|कु|	|क| |ु|	complex character, complex glyph (vowel grapheme represented discretely)
5	<k> <ṣ> <a>	<kṣa>	|क्ष|	— — —	graphetically simplex graphotactic allograph for a graphemically complex glyph (vowel grapheme represented holistically)

2.4.3.1. Glyph components

We use the term **component** (or, for explicitness, **glyph component**) for a graph that is part of a graphetically complex glyph. Latin |A|, Devanagari |अ|=|a| and Devanagari |क|=|ka|, as well as |क्ष|=|kṣa|, have no components because they are graphetically simplex, while Devanagari |क्लौ| consists of three components (1 |k|, 2 |l| and 3 |au|). In Devanagari |क्ल|=|kla|, only |k| and |l| are components, because <a> is represented holistically rather than by a visually isolable graph.

2.4.3.2. Conjunct consonants

A **conjunct** in an aksharic writing system is an *akṣara* that involves two or more consonant components. The term **ligature** is also widely used for them, and we approve of this usage, but it must be kept in mind that the same term in the context of Latin typography refers to typographic ligatures, which we prefer to see as graphetically modified sequences of independent characters (§2.5.1.2).

Conjuncts are therefore always graphemically complex, but a conjunct is not the same as a complex character: the latter also includes characters comprised of a single consonant combined with any grapheme other than a dependent consonant (i.e. with one or more of the following: the inherent vowel, a vowel manifesting as a marker, a *virāma*, an *anusvāra* and a *visarga*). Nor is a conjunct identical to a complex glyph. On the one hand, graphetically complex glyphs also include those comprised of a single consonant combined with any graph other than a dependent consonant (i.e. one or more of the following: a vowel manifesting as a marker, a *virāma*, an *anusvāra* and a *visarga*). On the other hand, some conjuncts are graphetically simplex, being expressed by a graphetically simplex graphotactic allograph such as |क्ष|=|kṣa|.

2.4.4. Graphic elements

Even graphetically simplex graphs may be quite complex shapes, for instance Devanagari |औ|=|Au| and Tamil |ஒ◌ா|=|o|. Graphic complexity thus does not necessarily entail graphetic complexity. Palaeographic as well as grapholinguistic discussion nonetheless often needs to refer to parts of graphs, so we need a term for such parts that is distinct from 'component,' which is restricted to graphs (i.e. manifestations of graphemes) in a complex glyph. We therefore recommend the term **element** — or, for extra clarity, graphic element — for any visually discrete constituent parts of individual graphs, i.e. for salient shapes which are not themselves graphs. Elements can be distinctive features of graphs — as the horizontal strokes that distinguish Brāhmī |+| = |ka| from |ı| = |ra| or Latin |E| from |F| — but they do not in themselves carry linguistic information.

Graphically simple graphs, such as Devanagari |े|=|e|, cannot be broken down into elements.[51] When a graph is made up of several elements, these may be contiguous (as in Devanagari |ो|=|o|) or disjoined (as in Tamil |ொ|=|o|). Elements of graphs can sometimes look like individual graphs, as elaborated in §2.4.4.1 below.

A graphically complex element may itself be spoken of as consisting of multiple simpler elements. When discussing the parts that make up Devanagari |क्लौ| in Figure 2.4.A, it may, for instance, make sense to refer to the body of |k| (shown as elements B, C and D in the figure) as a single element. At the simplest level, we recommend the term **stroke** for elements which are uninterrupted lines and may have been written as a single stroke of the writing instrument. All the lettered elements in Figure 2.4.A are thus in fact strokes.

In referring to the details of a graph, chiefly in palaeographic description or a discussion of reading difficulties, we encourage DHARMA collaborators to use the terms 'element' and 'stroke' constructively. For pointing to particular graphic elements, we recommend intuitive biological and architectural analogues such as arm, leg, wing, tail, stem, lobe, arch, base, etc.

2.4.4.1. Dual patterning

Some morphemes (i.e. semantically meaningful linguistic units) consist of a single phoneme (e.g. English "I" or "o!"), but the same phonemes (in themselves meaningless) can be constituents of other morphemes (e.g. "hi" or "go"). This feature of language is called dual patterning (Nöth 1990, 237), and it is also applicable to grapholinguistics, where it is a particular case of homography (§2.5.2). Most sign inventories utilise some recurring graphic shapes that can occur as (linguistically significant) graphs on their own, but also as (linguistically insignificant) graphic elements in various combinations with other shapes. Importantly, even if such a shape can comprise a graph (i.e. manifest a grapheme) on its own (such as Devanagari |आ| = |Ā|, |े| = |e| or |ा| = |ā|), that shape is still merely an element when it combines with other elements to constitute a different graph (such as Devanagari |ओ| = |O| or |ो| = |o|).[52] Although graphically complex, Devanagari |ओ| is simplex both graphetically and graphemically. It is not a combination of the graphs representing the graphemes <Ā> and <e>, merely of elements which, incidentally, manifest those graphemes when they are on their own.[53]

2.4.4.2. *Diacritical marks*

Diacritical marks (or, in short, diacritics) are, in our conceptual scheme, a special kind of element which can combine with certain graphs to alter their graphemic signification.[54] Most diacritical marks can be added to a range of graphs and often (though not always) alter the signification of those graphs in a somewhat consistent manner. For instance, in the DHARMA transliteration, the macron always changes the unmarked short state of vowels to long, and the underdot usually changes the unmarked dental state of certain consonants to retroflex. Diacritical marks thus arguably have a graphemic function. However, what they signify is only a feature of a phonological segment, while the presence of that segment is already signified

[51] From an alternative viewpoint it might be said that they are themselves elements, much in the way that simplex glyphs are themselves graphs.

[52] For contrast, compare the digraphs mentioned in note 65 below.

[53] We thus emphatically disagree with Meletis (e.g. 2020a, 101), in whose view Tamil |ொ| = |o| corresponds to two separate graphemes. His reasoning is that the graphs |ெ| = |e| and |ா| = |ā| can occur separately on their own, hence they correspond to separate graphemes, hence their combination is not minimal, and the association of |ொ| with the phoneme /o/ is a matter of higher-level grapheme-phoneme correspondence. We assert that it is simply the strokes comprising Tamil |ொ| that happen to be homographic to those comprising the graphs |ெ| and |ா| = |ā|, which is irrelevant to the grapheme status of Tamil <o>. A parallel case in point is Latin |Y|, which looks like a combination of |V| and |I| (and is indeed thought to have been deliberately created from these primary graphs, originally to represent a vowel phoneme "in between" /u/ and /i/), but this is irrelevant to the grapheme status of <y>.

[54] This definition is more rigorous than, but entirely compatible with, Coulmas's "A mark added to a basic letter to alter its pronunciation" (Coulmas 2006, 126, s.v. diacritic).

by a grapheme. Features of phonological units fall outside our domain of analytic interest, so from our perspective, a graph with a diacritical mark corresponds to a single grapheme, and is also graphetically simplex in spite of its graphic complexity.

As such, diacritical marks per se have no relevance to transliteration: like any distinctive element, they only matter inasmuch as they distinguish one graph from another.[55] If a source grapheme is distinguished from another source grapheme by a diacritical mark, then the distinction itself must be preserved in transliteration, but it need not involve a diacritical mark, just as diacritical marks in the transliteration must indicate distinctions in the source, but not actual diacritical marks in the source.[56] However, we see no objection to recognising certain diacritical marks as graphemes in their own right when this serves analytic interest, as discussed in §2.7.2.

On the other hand, we emphasise that the vocalisation signs of the Indic writing system (which we call markers, §2.6.2) are *not* diacritical marks by our definition (even though secondary literature often refers to them as such), since they do represent graphemes at the same analytic level as the base signs. These signs could only be called diacritics if the *akṣara* was equated to the grapheme, a notion that we emphatically reject (§2.3.1.1).

2.5. Complications

2.5.1. Allography

Allography literally means something like 'other writing.' We have briefly defined allographs above (§2.2) as diverse graphic signs which signify the same linguistic information. With the stricter definition of the graph as the etic counterpart of a grapheme (§2.4.2), this primarily means that **allographs** are graphs which manifest the same grapheme. However, even more so than with allophones in phonology, there are several different levels at which allography can occur, from the most superficial variation between one handwritten |a| and another, all the way to the striking variation found for instance in the graphs expressing the archetypal phoneme /u/ in Devanagari |उ| (<U>), |कु| (<ku>) and |रु| (<ru>). A qualitative distinction between the levels of allography might be grasped in terms of supplementary linguistic information carried by the graphs in question, as well as in terms of the conditions that determine the choice of one form or another. Accordingly, we propose distinguishing three kinds of allography: graphetic, graphotactic and graphemic.[57]

2.5.1.1. Graphetic allography

Graphetic allographs are alternative graphs which do not signify any linguistic information other than that contained in the corresponding grapheme, as in Latin |a| - |ạ| - |*a*| and in Devanagari |अ| - |अ| - |अ| or |झ| -

[55] For instance, the diaeresis (Umlaut) distinguishes |ö| from |o| in the Latin writing system for German, the macron distinguishes |ū| from |u| in the DHARMA transliteration system, and the nuqta (dot) distinguishes |ड़| from |ड| in Devanagari for Hindi. For our purposes, these elements are not conceptually different from the horizontal stroke that distinguishes Brāhmī |+| = |ka| from |ᅵ| = |ra|.

[56] Thus, Devanagari |ड| is transliterated *ḍa*, while |ड़| is transliterated *ṛa*. In the source graphemes, a diacritical mark in |ड़| indicates that the grapheme stands for a flap allophone of the stop phoneme represented by the base graph |ड|; an incidentally identical-looking diacritical mark in the transliteration of both indicates the retroflex quality of both phonemes, but the distinction of the graphemes takes place in the base graph.

[57] Our distinction between graphemic and graphetic allography is inspired by Meletis (e.g. 2020b), but he defines these concepts in a very different way. For him (e.g. Meletis and Dürscheid 2022, 64), the crux is the physical template ("basic shape"), identified strictly on the basis of phenomenal criteria such as the number and topological arrangement of graphic elements. He further classifies graphetic and graphematic allography with regard to whether the alternation takes place within or between inventories (scripts), and whether the choice between any given set of forms is paradigmatic or syntagmatic. Iyengar (2024, 427–28) uses the label 'homophonous heterography' for some cases of what we call graphemic allography, but is concerned only with the phonological signification of alternate graphic representations, and ignores their potential graphemic role.

|झ| - |झ|.[58] The scope of our term 'graphetic allography' also includes the trivial variation between instances of a particular graph written in the same hand or printed in the same font, although in other contexts of research it may be useful to treat this most superficial kind of allography separately. The choice between graphetic allographs is *paradigmatic*, meaning that any one of a set can be freely swapped for any other.[59] Their alternation may be essentially random, or it may be governed by aesthetic considerations.

To be sure, practically any graphic feature can be a *potential* carrier of additional linguistic (or quasi-linguistic) information, and thus have a graphemic role. This includes suprasegmental graphetic allography (the use of a different graph inventory for words or other chunks of text), such as italics for emphasis in a text written in the Latin script, Grantha for a Sanskrit word in a text written in the Tamil script, or a more ornate script for the royal signature on an Indic inscription. This kind of graphemic function is outside our domain of analytic interest (§2.3.4).

2.5.1.2. Graphotactic allography

Graphotactic allographs are alternative graphs whose occurrence is governed by graphotactic context, as in the |r| of Devanagari |र|=|ra|, |र्क|=|rka| and |क्र|=|kra| or the |u| of |कु|=|ku| and |रु|=|ru|.[60] In the Latin script, such allographs occur mainly in cursive handwriting where, for instance, the letter |n| may look different in an initial position, when following an |o|, and when following an |a|. The choice between graphotactic allographs is *syntagmatic*, meaning that it is determined by the position of the graph in a sequence of other graphs and by the nature of the other graphs in that sequence. The dependence of graphotactic allographs on context means that in damaged source texts, they can inadvertently imply something about their lost context, but this implication is not in the nature of a graphemic sign.

The concept of **graphotactic allography** can be usefully extended to cases where graphotactic context requires or favours certain allographic formations for **clusters of graphs**. In aksharic writing systems, this phenomenon can explain the existence of special glyphs such as Devanagari |क्ष|=|kṣa| and |ज्ञ|=|jña|, which are graphetically simplex (i.e. they cannot be analysed into components) while being graphemically complex (more so than consonant glyphs with an inherent vowel, discussed in §2.4.3 above). Our approach also appears applicable to situations in alphabetic writing systems written in cursive handwriting or with typographic ligatures such as |fi| and |ffi|, where certain clusters of graphs combine into a visual whole involving special forms of the individual graphs. It would be theoretically possible instead to perceive cluster-level graphotactic allographs like |क्ष| as syllabographic graphemes in an otherwise aksharic systems,[61] and to perceive cluster-level graphotactic allographs like |fi| as complex characters in an otherwise alphabetic system. However, we prefer to interpret features of any writing system in accordance with, rather than as exceptions to, the prevailing representational mode of that writing system (§2.7.1).

Cluster-level graphotactic allography does not necessarily produce graphetic simplicity. In Devanagari |श्र|=|śra|, for instance, the graphs for |ś| and |r| are isolable even though neither is identical to the base graph for these graphemes. The graphetically simplex cluster-level graphotactic allographs are palaeographically derivable from graphetically complex forms, and are thus just an unusual combination of the standard rules and processes of the writing system, not a special case requiring separate rules. We therefore advance that the relevant graphemes are individually present in such graphetically simplex *akṣara*s,[62] and that, analogously, the cursive or ligated units of alphabetic writing systems are series of simplex characters rather than complex characters.

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[58] Meletis (2020b, 257–59) would call pairs like |अ| and |ञ| graphematic allographs because they involve different "basic shapes," and recognise only the allography of |अ| and |अ| as graphetic (Meletis 2020b, 255 Fig. 3). Our concept of graphetic allography thus includes Meletis's graphetic allography and his free graphematic allography.

[59] Graphetic variation is of course constrained by the choice of sign inventory or "font", as in our examples above. But in handwriting and some traditional typography, truly paradigmatic alternative forms do exist.

[60] In Meletis's terms (2020b, 259–60), our graphotactic allographs are positional graphematic allographs, but that category also includes many cases of what we consider true graphematic allography.

[61] Indeed, they are sometimes appended to traditional alphabetic lists of Indic scripts.

[62] Largely in agreement with Meletis (2019, 41, 45–46 n. 32), who recognises such "graphetically non-segmentable clusters" as a challenge, but, in spite of his blanket requirement of visual isolability (ibid., 35), his last word on

2.5.1.3. Graphemic allography

Finally, **graphemic allographs** are alternative graphs which typically signify linguistic information in addition to, and at a different level than, that inherent in the grapheme, as in Devanagari |उ|=|U| versus |u| in |कु|=|ku| and |रु|=|ru|, and in Bengali |t| in |ত|=|ta| versus |ৎ|=|T| (final |t|).[63] The distinction between upper and lower case (e.g. Latin |a| versus |A|) is also a case of graphemic allography in our view, although the additional linguistic information carried in these allographs is of course different from that represented by our Indic examples. Examples from non-aksharic writing systems might include Greek |σ| and |ς| (allographs for the grapheme <s>, the former being used in medial positions, the latter in final positions), or the isolated, initial, medial and final forms of Arabic script. The choice between graphemic allographs is neither free and random, nor mechanically determined on the basis of graphotactic context. Instead, it is driven by linguistic context at a level other than that represented in the graphemes themselves. Consider the Sanskrit words *kṛtam etat* written in early Brāhmī as |𑀓𑀽𑀢𑀁 𑀏𑀢𑀁| (involving the final allograph of <m> and the initial allograph of <e>) and as |𑀓𑀽𑀢𑀫𑁆 𑀏𑀢𑀁| (involving the *akṣara* <me>). Both are legitimate alternatives in the writing system, but the former carries additional linguistic information: it implies a pause (and hence, a syntactical or semantic boundary analogous to that expressed by punctuation and/or spacing) between the two words. Likewise, the use of a Latin capital letter instead of a lowercase one can, among other things, indicate the beginning of a sentence or the beginning of a proper name.

2.5.2. Heterography

Heterography, literally 'different writing' is a term for situations where a certain linguistic entity can be represented in writing by two or more different graphemes or grapheme clusters. Our concept of graphemic allography could arguably be considered a form of heterography (cf. note 57 in §2.5.1.2 above). More commonly, heterography designates cases where the same sequence of phonemes can be represented with different graphemes (e.g. English 'right' and 'write'), which is an issue of grapheme to phoneme mapping; or where the same morpheme can be represented with different graphemes (e.g. English 'color' and 'colour'), which is a question of orthography.

2.5.3. Homography

In our quick terminological sketch (§2.2) we have defined **homography** (literally, 'same writing') as a situation where two graphic entities are visually identical, but signify different things. While for situations where the same "thing" is written in different ways, a distinction is usually made between allography (different graphs for the same grapheme) and heterography (different graphemes for the same linguistic entity), homography is not distinguished with regard to the level at which it takes place. At the level of words, for instance, English 'read' (present tense) and 'read' (past tense) are homographs, as they consist of the same graphs, but have different meanings. This kind of homography has no relevance to transliteration, because it also involves the same sequence of graphemes, and the difference becomes manifest only in the mapping of graphemes to language.

However, full or partial homography can also apply to graphs and glyphs, so that an identical or near-identical graphic entity corresponds to different graphemes or grapheme sequences. Scripts tend to employ a relatively limited repertoire of graphic elements. Each combination of these elements into graphs, graph clusters, glyphs and glyph clusters sits in a fuzzy cloud of allographs, and some of these clouds overlap. Examples include Latin |l| - |I| - |1| (lowercase L, uppercase I and the numeral one) or |O| - |0| (the letter O and the digit zero) at the level of graphs, and (especially handwritten) Latin |*k*| - |*lc*| or |*nu*| - |*un*|, or Devanagari

them is that they correspond to sequences of individual graphemes in the same way as the more transparent complex *akṣara*s. The problem apparently baffles Iyengar (2024, 432), according to whom Tamil |ஸ்ரீ| = |śrī| is "effectively logographic in nature".

[63] Meletis seems to consider our graphemic allography no different from his positional graphematic allography (Meletis 2020b, 257–60), except possibly for the matter of upper and lower case in Latin, about which he is undecided (2020b, 260–61).

|ख|=|kha| - |रव|=|rava| at the cluster level.[64] As a result, the graphemic interpretation of graphic signs sometimes requires input from context.

2.5.4. Polygraphy

Many writing systems employ established clusters of signs with a conventionally associated graphemic function, such as English |sh|, which corresponds to the phoneme /ʃ/ with great consistency. This phenomenon is polygraphy, 'multiple writing,' and such combinations are technically known as **polygraphs** (and as **digraphs** when comprised of two members). Polygraphs are common in alphabetic writing systems, though rare in other phonographic systems,[65] and have been recognised as graphemes in some approaches.[66] Our definition of the grapheme (which must be a minimal unit but is not required to represent an actual phoneme) precludes this: polygraphs can always be broken down into individual graphemes (such as <s> and <h>) belonging to the same domain of analytic interest. The phenomenon that these graphemes together correspond to a single phoneme is purely a matter of complicated grapheme to phoneme mapping.

2.6. Revisiting the *akṣara*

Having set up a framework for interpreting the graphetic constituents of phonographic writing systems and the graphetic constituents of their scripts, we proceed to investigate some details of the Indic *akṣara* that have so far been left in the twilight.

2.6.1. Demarcating the *akṣara*

It is not always obvious whether a given cluster of graphemes is to be conceived of as a string of characters (each grapheme being graphetically autonomous) or one complex character (with all but one grapheme being graphetically dependent). The matter essentially depends on the graphotactic rules of the applicable writing system, but these rules are not written, much less formalised in terms of our grapholinguistic framework. Our general approach is that character demarcation should be based on the prevailing representational strategy of the writing system. Alphabetic systems typically employ a separate character for each grapheme, while aksharic systems typically combine graphemes into *akṣara*s. Thus, signs in either system that seem to behave differently should be reinterpreted as conforming to the applicable system's prevailing strategy unless there is a strong reason to the contrary.[67]

The demarcation of *akṣara*s in the Indic system is irrelevant to Romanised transliteration, because the target writing system is not equipped to handle complex characters. A reader familiar with the graphotactic rules of the source writing system will usually be able to reconstruct the source's *akṣara* boundaries on the basis of the transliterated graphemes. However, because these graphotactic rules are not completely uniform across our scope, it is sometimes expedient to represent the manner in which graphemes combine into characters, but this can only be done by means other than transliteration.

For the sake of completeness, we summarise here the kinds of grapheme clusters that can congregate into an *akṣara* in the generic Indic system. The simplest *akṣara*s are independent vowels and the special final forms of consonants (§2.5.1.3), comprised of a single grapheme. In the next simplest setup, a primary consonant standing on its own consists of the consonant grapheme and the holistically represented inherent

[64] See also §2.4.4.1 for a separate discussion of graphic elements homographic to graphs.

[65] Examples can, however, be found even in the Indic system, such as the use of |ळ्ह| for [ḷh], a Vedic allophone of /ḍh/; the use of the combination <ys> in some North Indian scripts of the first millennium CE to represent the phoneme /z/, which does not occur in Sanskrit; or, arguably, the concurrent use of the dependent vowels |ui| in Khmer, Burmese and Mon to represent a vowel phoneme alien to Sanskrit.

[66] Coulmas (2006, 129, s.v. digraph 1) asserts that polygraphs are graphemes in their own right. Meletis and Dürscheid (2022, 128–29) classify certain polygraphs as "complex graphemes" (cf. note 38), but deny grapheme status to most.

[67] See also §2.5.1.2 about typographic ligatures in alphabetic systems and apparently syllabographic signs in aksharic systems, and §2.7.1 for more cases of relying on the prevailing strategy.

vowel (§2.3.2.1). All other *akṣara*s follow the same basic pattern, where a primary consonant grapheme serving as the "head" is supplemented with one or more graphemes expressed through dependent graphs, which we shall call markers (§2.6.2). In the broad sense, this category includes dependent consonants, dependent vowels, the *virāma* (§2.6.3), as well as the *anusvāra* and *visarga* (§2.6.4). When such a complex *akṣara* lacks an explicit vocalisation (in the form of a vowel marker of the *virāma*), the inherent vowel grapheme is implicitly present.

A consonant or conjunct[68] combined with a *virāma* is thus in our view[69] an autonomous and integral character even though it does not correspond to a syllable, and even if it is employed in a non-final position as an alternative to part of a ligature. That is to say, combinations such as Devanagari |द्म| = |d·ma| and Tamil |க்க| = |k·ka|) are not *akṣara*s but strings of two *akṣara*s each.

The *anusvāra* and *visarga* could arguably be viewed as autonomous characters in their own right. They are always graphetically distinct, and can in many specific writing systems occupy a separate segmental space of their own, be physically separated from their parent *akṣara* by a physical feature, or even be graphetically associated with the following character. On the other hand, they can never occur in an initial position, so we prefer to align with the traditional view, which considers them to be dependent on *akṣara*s.

2.6.2. Markers

'**Marker**' is not an established grapholinguistic concept[70] but our term for a special set of components (§2.4.3.1) that can only appear in a graphetically dependent position, i.e. which can never manifest as autonomous glyphs, only as glyph components. In actual practice we prefer to use the term 'marker' primarily in a more limited sense for markers of vocalisation, meaning the dependent vowel signs and the *virāma* (§2.6.3). For other kinds of markers in the broad sense, instead of a class name we prefer specific designations: *anusvāra*, *visarga* and dependent consonant. Note also that a marker is different from a mark (§2.3.3).

2.6.3. *Virāma*: graph or diacritical mark?

The *virāma*, the "vowel killer" sign of aksharic writing systems, negates the inherent vowel of an *akṣara*, but does not signify any actual phonological unit, so it is difficult to recognise it as a traditionally defined grapheme. Probably for this reason, it is sometimes called a diacritical mark, but this is only tenable in a conceptual framework where *akṣara*s are equated to graphemes.[71]

All things considered, we deem it best to recognise the *virāma* as a primary grapheme.[72] Defining a phonographic grapheme (§2.3.1) as one signifying information pertaining to the presence of a phonological unit (rather than signifying the presence of a phonological unit) allows us to include the *virāma* in this category. For a rationale, we emphasise that functionally the *virāma* is fully analogous to the dependent vowel signs (markers, §2.6.2), which are definitely graphemes in their own right. A basic *akṣara* is comprised of one or

[68] Although combinations of the *virāma* with conjunct consonants are rare, there is no inherent rule in the Indic writing system that would forbid them, and practical examples do exist (e.g. Devanagari |र्द्व| = |rd|, Tamil |க்ஷ்ய| = |kṣ·|).

[69] With Iyengar (2024, 430–31).

[70] Meletis (2020a, 100) simply describes the corresponding graphemes as graphetically subsegmental and dependent or bound, while Weingarten (2013, 18) calls them affigated graphemes.

[71] For details, see our discussion of markers in §2.4.4.2.

[72] We are not aware of any publication that discusses the status of the *virāma* vis-à-vis graphemes. Theoretical frameworks which equate *akṣara*s to graphemes can unproblematically handle the *virāma* as a diacritical mark which changes one grapheme to another, and this tacit understanding may be in the background of the fact that the *virāma* is often referred to as a diacritic (e.g. Gnanadesikan 2017, 18). However, such approaches implicitly treat the Indic script as syllabographic rather than aksharic (cf. note 40), which we find unacceptable. Weingarten (2013, 22) implies that he considers the *virāma* to be a grapheme, but since his broad grapheme concept includes diacritical marks, this does not expressly support our stance. Iyengar (2024, 427–28) treats the *virāma* as a graph (defined much as we define it, cf. note 50), which would imply that on the emic level (which Iyengar dismisses) it is a grapheme. Meletis does not address the status of the *virāma* as far as we are aware, but Meletis and Dürscheid (2022, 233) state that it "appears as the diacritic <◌्>" which, unless the angle brackets (indicating graphemes) are a mistake for vertical bars (denoting graphs), implies that they are willing to recognise it as a grapheme at least in some sense.

more consonant graphemes and the holistically represented inherent vowel grapheme. The addition of a vowel marker to such an *akṣara* signifies the presence of another vowel in place of that inherent vowel, while the addition of a *virāma* signifies a zero-vowel in place of that inherent vowel.

A further practical consideration leading to this decision is that it is essential for a transliteration scheme used in philology and palaeography to make a distinction between a vowelless consonant expressed using a *virāma* and one expressed as a special final (*halanta*) form of a regular consonant grapheme. Since we accept any graphic feature as the graphic aspect of a grapheme (§2.3.1), these final forms, used in many early variants of the Indic writing system, could be fitted into our scheme as signs which represent the *virāma* (in addition to a consonant phoneme) holistically, just as we have conceived of basic *akṣara*s as holistically representing the inherent vowel in addition to a consonant phoneme (§2.3.2.1). However, it is part of our analytic interest to make a distinction between these two strategies for representing vowelless consonants, and therefore we deliberately choose to recognise a *virāma* grapheme only when it is represented by an isolable graph, while special final forms of consonants are given a different treatment in transliteration (§2.5.1.3).

2.6.4. *Anusvāra* and *visarga*

The *anusvāra* and *visarga* normally represent a nasal consonant and an aspiration, but are sometimes understood to signify the nasal or breathy quality of the preceding vowel. We have already chosen to treat them as dependent graphs rather than independent characters (§2.6.1), but in the light of this usage they might even be classified as diacritical marks (§2.4.4.2) rather than graphs in their own right. In some writing systems they also have alternative shapes (such as the *candrabindu* or *anunāsika* for the *anusvāra*, and the *jihvāmūlīya* and *upadhmānīya* for the *visarga*), which may be arguably considered either graphs in their own right or allographs for a single grapheme.

Given the temporal, spatial and linguistic scope of our undertaking, we prefer to treat the *anusvāra* and the *visarga*, as well as their variants, as phonographic graphemes in their own right. The rationale for this is that primary and more universal signification (§2.7.1) of the *anusvāra* and *visarga* is a consonantal speech sound. When the basic signs for these graphs express only a feature of an adjacent vowel in some particular writing systems, this can be understood as a question of grapheme to phoneme correspondence. As for the variant signs, some of these may be used mechanically as the shape appropriate to a specific writing system or to a given graphotactic context in a specific writing system, but the original reason for the creation of these signs can be presumed in most cases to have been a desire to distinguish their linguistic function from that of the basic *anusvāra* and *visarga*. Recognising a number of these signs as distinct graphemes and transliterating them with dedicated Latin letters helps maintain homogeneity across the corpus of transliterated texts and avoids the unfounded assertion of a specific function while still allowing for a distinction of potential linguistic function.

That said, for specific writing systems of modern languages with a well-known orthography and pronunciation, we have no objection to treating the *anusvāra* as a diacritical mark indicating vowel nasality and transliterating it accordingly, for example Hindi हाँ as *hã̄* (as opposed to *hāṁ*, with the *anunāsika* transliterated as a distinct grapheme).[73]

2.7. Fuzzy categorisation

When analysing a specific writing system whose script, language and orthographic conventions are stable and well-known, it is usually unproblematic to distinguish graphemes from other entities, such as clusters of graphemes, allographs of the same grapheme, and elements which may be distinctive features of graphemes but are not themselves graphemes. For instance, in the writing system for modern German written in the Latin script, the glyphs |e|, |o| and |ö| each correspond to a grapheme; |ö| is not an allograph of |o| but a separate graph which represents a different archetypal phoneme; the diaeresis |◌̈| is not a graph, only an

[73] However, it could be argued that even in such systems the *anusvāra* or *anunāsika* represents an archetypal nasal consonant, and the fact that it manifests as the nasal quality of an adjacent vowel is simply an issue of grapheme to phoneme correspondence.

element (specifically a diacritical mark), while |oe| is a cluster of two graphs corresponding to two discrete graphemes, notwithstanding the fact that these graphemes <o> and <e> can be mapped together to the same phoneme as that represented by |ö|. The distinction of these graphs from others which may be homographic or near-homographic to them is also largely self-evident: |0| is not the letter for the phoneme /o/ but the numeral zero, while |€| is not the letter for /e/ but the ideogram for the Euro.

However, writing systems change as they evolve, co-evolve with their languages, or are adapted to other languages. Hand in hand with changes in how graphemes are mapped to speech sounds, pre-existing graphemes may be endowed with a new function, either in addition to or in lieu of their older function. Novel graphemes (for example to represent sounds which had been allophones in the earlier or original language but are phonemic in a later evolutionary stage or a different language) may develop out of allographs of earlier graphs, be created through the addition of distinctive elements, or arise from graphetic combinations of graph clusters already mapped to the desired phoneme(s). In the latter case, the graphs originally constituting such a cluster become demoted to mere elements in a new graph that represents the new grapheme. Since such changes usually occur gradually, we have no choice but to live with the fact that our analytical constructs are not always an ideal fit for the ground truth.

How then are we to handle spectra such as |oe| - |œ| - |o͡e| - |ö| (from cluster to simplex glyph involving a diacritical mark), |sz| - |ʃʒ| - |ʩ| - |ß| (from cluster to simplex glyph), and |et| - |ɘr| - |&| (from cluster to ideographic sign)? What is the graphemic interpretation of the English sentence "2 b or not 2 b," where the numeral sign |2| represents the phonemes /tʊ/, while the letter |b| represents the phonemes /bɪ:/? Are we still dealing with the numeral grapheme <2> and the phonographic grapheme <b> in such a case?

We do not pretend to solve such questions, but when analysing the broad and ramified Indic writing system for the purpose of transliteration, we are obligated to establish correspondence between graphic entities and our analytical constructs. In particular, we need decisions on the following matters for all graphic entities. First, is it merely an element (§2.4.4), or is it a graph (§2.4.2), or even a graphemically complex glyph (§2.4.3) that may be manifested as a cluster-level graphotactic allograph (§2.5.1.2)? In the first case, transliteration must distinguish the graphemes distinguished by this element in the source, but not represent the element directly. In the second case, it requires direct representation in transliteration, which prompts a second question: does that graph correspond to a single grapheme (with potentially very different linguistic correspondence contingent on context), or to either of two separate graphemes (which happen to manifest homographically, §2.5.3)? Finally, in the third case, the entity itself gets no representation in transliteration, but the graphemes corresponding to its constituent graphs do. To arrive at practical solutions to these problems, we offer two basic approaches to fuzzy cases: sticking with a default interpretation so long as doing so is tenable and analytically acceptable (§2.7.1), and recognising entities of special interest as honorary graphemes (§2.7.2).

2.7.1. Defaults

As already noted in the context of character demarcation (§2.6.1), writing systems have prevailing representational strategies which orchestrate most of the things that happen in a given system. According to the prevailing strategy of alphabetic writing systems, graphemes represent archetypal speech sounds and characters are comprised of single, individual graphemes. A broad view of related writing systems usually reveals which forms, and thus which kind of graphemic analysis, is earlier in an evolutionary view and/or more universal. Thus, progressions toward apparent simplicity such as |oe| - |œ| - |o͡e| and |sz| - |ʃʒ| - |ʩ| can be plausibly regarded as clusters of graphs, each represented by a separate glyph, with various cluster-level graphotactic allographs. We have made the same argument for glyphs like Devanagari |क्ष| (§2.5.1.2), based on the prevailing strategy of aksharic writing systems, according to which graphemes represent archetypal speech sounds and cluster into complex characters.

Indeed, even |ö| and |ß| could be approached by the same logic, and doing so might serve analytic interest in a broad diachronic study. Conversely, recognising |ö| and |ß| (and even |œ| and |o͡e|) as single graphs would also be in accordance with the prevailing strategy of alphabetic systems, and in many such spectra it is best to break with the evolutionary analysis at some point. Exactly where the breaking point shall be located in a continuum (and how many breaking points there ought to be) depends on analytic interest and

the writing systems involved. As a rule of thumb for such cases, if in any of the writing systems relevant to analysis a particular graphetic shape that is clearly distinguishable from its alternatives is consistently associated with a particular graphemic function, it is worth considering it as a candidate for graph status even if by default it is but a cluster of already known graphs.

As regards the choice between two homographic graphemes and a single grapheme with different context-dependent significations, the principle of defaults means that the evolutionarily primary and/or more universal graphemic function should be prioritised. In other words, in the sentence "2 b or not 2 b," we prefer to see the numeral grapheme <2> and the phonographic grapheme <b> (in special functions) rather than two novel graphemes. The same approach might be applied to the rebus principle commonly exploited in logographic writing systems, where a given sign's primary signification is a particular word, but the same sign may secondarily signify the phonological sequence constituting that word, which can be used to build other words for which there is no separate ideogram. A somewhat forced example of the rebus principle in a modern writing system might be the English word /start/ written as |*t| with the expectation that the reader would map the ideogram |*| to the phonological value /star/. However, in spite of an overall preference for the defaults, secondary significations can be recognised as separate graphemes (which may manifest homographically to the grapheme with the original signification), provided that a sign's association with the secondary function is sufficiently prominent and consistent in a particular writing system.

2.7.2. Honorary graphemes

In some cases, analytic interest may be best served by promoting an entity to the status of a graph (and thus, of a grapheme) even though from an evolutionary point of view it is only an allograph of a known graph, and/or if in a particular writing system it is only a diacritical mark distinguishing one known grapheme from another. For instance, some or all manifestations of the "dependent |e|" in the forms |œ| - |ů| (and even |ö|) might be awarded graph status for an analysis focussing on such transitional forms.

As regards transliteration, doing so can be expedient on a prosaically practical basis. If diacritical marks occur widely in the source writing system, it would be impracticable to dedicate a distinct target grapheme to each diacritically distinguished source grapheme (cf. note 56 in §2.4.4.2), while recognising certain diacritical marks as graphemes requires only one dedicated target grapheme for the diacritical mark.

Since the linguistic function we require for a grapheme is contingent on analytic interest (§2.3.1), we are not bound to make grapheme status exclusive to signifiers of linguistic units at the writing system's dominant level of representational mapping. This sets the stage for recognising as graphemes signs signifying virtually anything, simply by declaring a sign's signification (even when it is polyvalent or vaguely known) to be within our analytic interest. Obviously, honorary graphemes must be managed sparingly and considerately, and may best be completely avoided in scientific grapholinguistic analysis, but they are certainly useful in the practice of transliteration.

3. General principles of the DHARMA transliteration scheme

3.1. Representation of graphemes

The essence of transliteration is to establish a one-to-one relationship between each grapheme in a target script and a corresponding entity in the source script (§2.1.2). In order to achieve this, we must be mindful of two distinct objectives. **Demarcation** is establishing the extent of a grapheme as distinct from adjacent graphemes as well as from anything that is not a grapheme. By our definition (§2.3.1), a grapheme is a graphic signifier of a minimal unit of abstract linguistic information pertaining to a particular domain of analytic interest. Graphemes are thus demarcated in the source on the basis of their signification: each packet of linguistic information that is represented in writing and is not divisible into smaller packets at the current level of analytic interest constitutes one grapheme. The most straightforward way to preserve demarcation in transliteration is for each source grapheme to be represented by exactly one target grapheme. Failing that, a cluster of target graphemes corresponding to a single source grapheme must be unambiguously identifiable as such.[74] **Individuation** is establishing the specific identity of each grapheme as distinct from any other grapheme. By the same definition, one grapheme is different from another grapheme if the packet of linguistic information represented by it is different at the current level of analytic interest from the packet represented by any other grapheme. To preserve individuation in transliteration, we need as many different target entities as there are distinct source graphemes, and maintain a one-to-one correspondence between these.

In the course of diplomatic transliteration, the scope of our analytic interest is broader than that of phonography (§2.3.1). As a result, the inventory of individual graphemes that we must deal with is potentially tremendous. Moreover, the linguistic function of many relevant graphemes can be difficult to identify and may lack a consistent association with a recognisable graphic shape. In order to take this complexity and vagueness in our stride, we divide graphemes into two broad classes from the pragmatic perspective of transliteration: the class of core graphemes, which are precisely individuated in transliteration through the use of dedicated target graphemes; and the class of peripheral graphemes, whose graphemic individuation is represented in transliteration only in broad lines.

3.1.1. Core graphemes

Core graphemes, from the viewpoint of transliteration, are those which are fully demarcated and individuated through transliteration, by means of conventionally dedicated target graphemes (or digraphs, §3.2.3). Core graphemes are individuated solely on the basis of their graphemic function, and such individuation is considered sufficient for our analytic interest. In other words, the graphetic appearance of these graphemes is irrelevant to transliteration, because the shape of the source graph or glyph is implicit in the identification of the grapheme. The reader familiar with the applicable writing system will have a good idea of how the transliterated text would have appeared in the source, while idiosyncratic variation (graphetic and graphotactic allography, §2.5.1) may be annotated (§3.3.4) outside the transliteration.

3.1.1.1. Phonographic graphemes

All **phonographic graphemes** of the Indic writing systems are core graphemes of the DHARMA transliteration and most of our core graphemes are phonographic. The obvious vowel and consonant graphemes, including the inherent vowel of the Indic writing system as well as the *anusvāra* and *visarga* along with their variants, are transliterated according to the ISO 15919 conventions (see also §3.4) with a number of additions for writing systems not covered in that standard. Phonographic graphemes (§2.3.2) also include the *virāma*,

[74] See §3.2.2 about digraphs in the DHARMA transliteration and §3.1.2 about arbitrary placeholders.

which we transliterate explicitly with the · (middle dot) character.[75] In individuating phonographic graphemes, we grant full grapheme status to their independent forms (§3.1.1.2).

3.1.1.2. The independent forms of phonographic graphemes

Phonographic graphemes of the Indic writing system can manifest in dependent (in-*akṣara*) or independent (in-hiatus) forms. We have called this alternation of forms graphemic allography (§2.5.1.3), and noted that although it is to some degree deterministic, it can signify additional linguistic information at a level other than that of phonology, namely the absence or presence of a pause or a hiatus. The advantage of distinguishing graphemic allographs in transliteration is not only that a potentially graphemic feature of the original written text is preserved in the transliterated text, but also that in the transliteration of poorly legible and lacunose text it tells the reader whether a specific legible grapheme appears as a dependent or independent graph, thereby facilitating conjectural restoration. We therefore include this kind of information in our analytic interest and accordingly treat these allographs in transliteration as separate graphemes. Since the alternation of Indic independent graphs and in-*akṣara* graphs is superficially parallel to the alternation of uppercase and lowercase Latin letters, we dedicate uppercase Latin letters to the independent allographs of Indic graphemes (e.g. Devanagari |अ|=|A| *a* in |क|=|ka|, and Bengali |ৎ|=|T| as opposed to *t* in |ত|=|ta|).[76] This makes DHARMA transliteration case sensitive (§3.2.3).

Some of DHARMA's predecessor projects have used a the character ° (degree sign) before transliterated vowels and/or after transliterated consonants to denote their special independent forms. DHARMA has decided on a projectwide level to discontinue using that notation, and to endorse and propagate the use of uppercase Latin letters instead. Considerations in favour of doing so include the following:

- unlike independent consonants written with a vowel killer graph, the special independent forms do not involve an additional grapheme, so it is better to use a single character for their transliteration
- uppercase Latin letters are (as pointed out above) superficially analogous to Indic independent allographs, and are moreover easy to enter on any keyboard, so their inclusion in the transliteration scheme helps productivity
- text-based search algorithms can easily be switched to case-insensitive mode and thus retrieve, for instance, *taM Eva* when searching for *tam eva*, whereas in order to retrieve *tam° °eva* for the same search string, special programming would be required
- using uppercase letters for the independent forms eliminates ambiguity with other uses of the ° sign

3.1.1.3. Numerals

Numeral signs, also called **ciphers** or simply **numerals**, are ideographic in nature (cf. §2.3.3), but in practice we treat them as core graphemes because they are widely prevalent, have a clearly definable and recognisable graphemic function, and can be readily represented and fully individuated in transliteration through Arabic numerals.[77] For decimal digits (the numerals 0 to 9) and for numbers represented in a decimal place-value notation,[78] demarcation is also fully preserved in transliteration. The case of non-digital numerals is less straightforward, because they require clusters of target graphemes to represent a single source grapheme. A human reader is expected to be able to resolve the resulting ambiguity intuitively, while in digital editions, encoding is added for demarcation and semantic interpretation.

[75] This practice will be endorsed in the forthcoming revised version of ISO 15919 (Jan Kučera, personal communication, 21 January 2026).

[76] Note that vowelless consonants written with a *virāma* are not allographs of the consonant graphemes but complex glyphs (§2.6.3).

[77] Throughout this guide, the term 'Arabic numeral' refers to the modern international numeral signs.

[78] Place-value notation, also called positional notation, is where a digit's position in the sequence of numbers functions as a multiplier to the digit's default value. This is distinguished from sign-value notation, where each numeral sign has a fixed value regardless of its position.

3.1.1.4. Additional core graphemes

Our core graphemes also include a small number of signs which are neither primary graphemes nor numerals, but are nonetheless individuated in transliteration with dedicated target graphemes. Some of these, namely the *avagraha* and the abbreviation mark, have a definite graphemic function by which they can be individuated in texts where they occur, and already have widely used conventional transliteration equivalents (namely the right apostrophe ’ and the degree sign °). Others, found in specific varieties of the Indic system, are in fact diacritical marks, but we choose to recognise them as honorary graphemes (§2.7.2) and transliterate them with dedicated characters (§4.3.3). Doing so helps ensure diplomatic accuracy in the transliteration while preventing the proliferation of target graphemes that the dedication of a target grapheme to each diacritically modified source grapheme would entail.

3.1.2. Peripheral graphemes

From our pragmatic perspective, **peripheral graphemes** are those that are not fully individuated in our transliteration. We use the term ‘**symbol**’ in a broad sense as a friendlier synonym for ‘peripheral grapheme,’ but, depending on context, the same term in a narrower sense can also mean symbols par excellence, the smaller subset of abstract symbols (§2.3.3). Many non-phonographic graphemes across the gamut of Indic writing systems have a linguistic function that resists precise definition. Even more often, there is no consistent association between a particular graphetic shape and a particular linguistic signification. On the other hand, precisely because their linguistic signification cannot be determined accurately, it can be useful to individuate non-phonographic graphemes on the basis of graphetic shape, so that different-shaped signs in the source remain distinguishable in the target even if their graphemic function is deemed to be identical, or if no assertion is made regarding their graphemic function. In transliteration, we subdivide peripheral graphemes into four functional classes: those of ideograms (excluding numerals, which are treated as core graphemes, §3.1.1.3), punctuation marks, connectors and abstract symbols. These classes will be discussed at more length in §4.4.7 in the practical part of this Guide.

We endorse three approaches to representing peripheral graphemes in transliteration. As an essential minimum of representation, we recommend a narrow range of **default placeholders.** One default placeholder is dedicated to each functional class of peripheral grapheme: @ (at sign) to ideograms, . (full stop) to punctuation marks, § (section sign) to connectors, and * (asterisk) to abstract symbols. These placeholders indicate the presence of a peripheral grapheme and demarcate it precisely, but do not individuate it beyond a simple functional classification.

Alternatively, **arbitrary placeholders** for peripheral graphemes can be any Unicode-supported character or string of characters. These may be chosen on the basis of graphic resemblance to the original glyph (e.g. |, ~, ❋), they may be concise labels describing the glyph in etic terms (e.g. “floret,” “bar-with-hook”), or labels based on an emic name for the glyph in question (e.g. “kaḷañcu”). The former two provide a measure of vague individuation on the basis of graphetic appearance, while the latter usually indicates both the graphetic shape and the graphemic function (or range of functions) associated with the glyph in a given specific writing system.

Finally, either kind of placeholder may be supplemented with XML markup (§3.3) where applicable, in order to clarify demarcation and to provide an additional measure of formalised individuation. In the practical part of this Guide, §5.1 discusses the nature and interaction of these strategies in more detail.

3.1.3. Space

Space is not an explicit graphic sign, yet the presence or absence of space between glyphs deserves recognition as the graphic aspect of a grapheme (§2.3.1), because it often has a linguistic signification. In our classification of supplementary graphemes, space would technically qualify as a punctuation mark (§2.3.3), since it indicates the boundary of a unit of linguistic structure.

Because most of the texts in our scope do not employ word spacing, we declare all spaces in transliterated text to be editorial and use them for morphological segmentation (§7.3.4). Accordingly, when a space is present in the source and is deemed significant, it needs to be explicitly transliterated with a dedicated placeholder grapheme. Unlike punctuation marks, cannot be individuated more precisely on the basis of

graphetic appearance, but XML markup provides methods for encoding the size and, where applicable, nature of spaces.

3.1.4. Entities other than graphemes

Since our aim in transliteration is to represent the graphemes present in the source, we must briefly examine salient entities that may be present in the source but are not qualified as graphemes. One such entity, the *akṣara*, will be discussed separately (§3.2.4) because of its importance in working with texts transliterated from an aksharic writing system. This section briefly addresses other entities, which are by and large ignored in transliteration.

3.1.4.1. Graphic elements

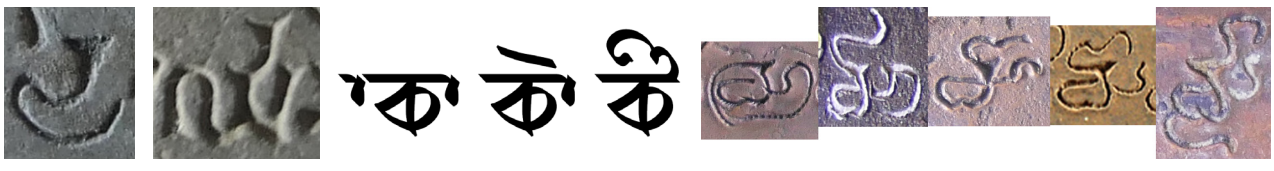
Graphic elements are visually distinct parts of graphs which are not themselves graphs, meaning that they do not manifest a grapheme (§2.4.4). Graphic elements are therefore not represented separately in transliteration unless their promotion to honorary graphemes (§2.7.2) serves analytic interest. Nonetheless, elements may be relevant to diplomatic transliteration inasmuch as they can occasionally be separated by a physical feature from their mother glyphs. To allow representing this phenomenon in transliterated text, we provide dedicated placeholders for such "orphaned" elements (§3.3.1.2).

3.1.4.2. Non-graphemic allographs of core graphemes

The Indic writing system is normally deterministic as regards the selection of **graphotactic allographs** (§2.5.1.2), which depends on position within a complex glyph. Nonetheless, when an unexpected graphotactic allograph occurs in a source text, this can be of palaeographic interest, while in damaged text where only part of an *akṣara* is legible, the kind of graphotactic allograph present can carry clues about the illegible context within the same *akṣara*. XML markup can reflect situations where the graphotactic allograph employed is not the one expected in a given position (as in Figure 3.1.A/1, where <rya> is written with a body <r> and a subscript <y> instead of a superscript <r> and a body <y>), and where the spatial configuration of a ligature is unusual (as in Figure 3.1.A/2, where a prescript <g> has been joined to the left of a body <gh> instead of joining a subscript <gh> to a body <g>). However, attempting to reflect this level of detail in transliteration would greatly complicate the transliteration scheme without much tangible gain in return. Therefore, we have deliberately chosen to ignore graphotactic allographs in our transliteration.

The same consideration applies all the more to **graphetic allography** (§2.5.1.1), to which transliteration shall remain blind. This includes some conspicuous variation, such as the use of two alternative glyphs within the same inscription for the same simplex glyph, or different forms of a vowel marker which may be attached to a consonant in different ways (as in Figure 3.1.A/3, 4). When deemed interesting or potentially relevant to scholarly study, such allographic variation can be noted in your palaeographic description, but will not be directly represented in the transliteration or the markup.

Figure 3.1.A. Allographs ignored in transliteration



3.1.4.3. Features ancillary to the text

Source texts are often accompanied by written features which are not graphemic because they do not represent language. The two primary kinds of such features are metamarks (scribal marks) and imagery. As a rule, neither is to be represented in transliteration, but the former may be represented in XML encoding and

both may be described in annotation (§3.3.4). Since peripheral graphemes are not always obviously different from ancillary features, we establish some guidelines for their distinction below.

Normally, the graphemes of a text are fairly uniform in dimensions and are arranged in a neat linear sequence, while ancillary features are of a different size and/or shape, and are not integrated into the linear sequence of graphemes. Instead, they are inserted between characters, between lines, in the extramarginal space, or in insets within the inscribed surface which the sequence of graphemes skips. Occasionally, graphemes may also disrupt the neat sequence (mostly when they are of a larger size than the rest of the text), but in this case they are usually recognisable as particular shapes associated with a known graphemic function. An abstract symbol grapheme inscribed outside the normal sequence of graphemes cannot always be distinguished on a purely objective basis from a decorative image. As a rule of thumb, complexity and iconicity point towards images, while simplicity and abstraction point toward symbol graphemes. However, ornamental elaborations of shapes which are known to also occur as symbol graphemes can be transliterated as graphemes.

3.1.4.4. Layout and materiality

On the whole, the characteristics of a source text's layout, including the ways in which the materiality of the support may affect the physical arrangement of the writing, fall outside the domain of transliteration. Such characteristics can be annotated (§3.3.4), and some of them can be specifically encoded in XML markup (§3.3).

3.2. Accuracy and interpretability

3.2.1. Strict transliteration

The DHARMA system is by default a scheme of **strict transliteration**, meaning that its analytic interest is diplomatic: a reader conversant with the source writing system should be able to reconstruct from the transliteration a reasonably accurate reproduction of the text as written in a particular original source document. To this end, it prioritises the transliteration of graphemes over the transcription of speech sounds (§2.1.2), and recognises a gamut of graphemes beyond those that represent phonological units (§2.3.3). To increase the accuracy of representation, the independent and in-*akṣara* allographs of phonographic graphemes are distinguished by means of upper case (§3.2.3).

As noted in §2.1.2, there is no such thing as a perfect transliteration system, and the DHARMA scheme is no exception, falling short of ideal in the following respects. On the one hand, in order to preserve compliance with established conventions and ease of keyboard input, we use digraphs as the transliteration equivalents of certain source graphemes (§3.2.3), for which one-to-one correspondence does not take place at the level of individual graphemes. On the other hand, we diverge from the maxim "the truth, the whole truth, and nothing but the truth" in order to permit certain editorial additions and interpretative alterations to our transliterated text with the tacit understanding that these are *always* editorial. These additions and alterations are integrated into the transliterated text, but are in fact a form of markup (§3.3) that is essentially extraneous to the transliteration itself.

The adaptation of the Indic writing system to languages whose phonology and phonotactics differed considerably from Old and Middle Indo-Aryan languages entailed a number of innovations in specific varieties of the writing system. The DHARMA transliteration scheme strives, on the whole, to represent any given source grapheme consistently with one and the same target grapheme, regardless of the precise role played by the source grapheme in any specific writing system, and especially regardless of the variations in grapheme to phoneme mapping. Exceptions are only made for cases when the graphemic function of a sign in a particular writing system is conspicuously different from that of related and ancestral signs in other particular varieties of the Indic writing system (§2.7.1). In the reverse direction, we also strive to dedicate each target grapheme to one and only one source grapheme, although this constraint has to be relaxed for peripheral graphemes represented by placeholders (§3.1.2) and has been deliberately relaxed for a small number of niche graphemes in order to keep the range of target graphemes manageable.

Depending on the context of use, the strictures of the transliteration system may be loosened either by a shift toward transcription when the representation of language, rather than of a particular written source, is desired (§3.2.5), or by using simplified and potentially ambiguous transliteration solutions when a given printing or display environment does not permit "proper" transliteration (§3.2.6).

3.2.2. Digraphs in the transliteration

In accordance with long-established and widely followed transliteration conventions, our scheme dedicates digraphs (§2.5.4) to diphthongs (e.g. *au*) and to aspirated consonants (e.g. *kh*). In this respect, it fails to establish a one-to-one relationship between source and target graphemes: the transliterated word *ratha* has five (target) graphemes, but there are only four (source) graphemes in the same word written in an Indic script, where, <th> is a single grapheme which, in Sanskrit, corresponds to the phoneme [t^h]. It is therefore very important to keep in mind that in any discussion of Indic graphemes, *source* graphemes are meant, but some of these may be represented by digraphs (i.e. two graphemes) in transliteration. The target digraphs must, in all respects, be treated as if they were a single, atomic entity. This leads to some complications in computer processing, but all of these are surmountable in current software environments. Nonetheless, it may be necessary to disambiguate target graphemes which can appear in digraphs but are present on rare occasions in their individual roles. The method for doing so is introduced in §3.3.1.1 below.

3.2.3. Case sensitivity

As explained in §3.1.1.2, the strict DHARMA system employs the uppercase forms of Latin letters to represent the independent forms of Indic vowels and consonants. As a result, the transliteration is case sensitive, while transliteration systems currently in widespread use, including the ISO 15919 scheme (ISO 2001, 16, §8.1.1), are case insensitive. However, case sensitivity in accordance with the DHARMA norm will be an explicitly recognised option in the next release of the ISO standard, foreseeably forthcoming in 2027.[79] Incidentally, this distinction also reduces the range of situations in which disambiguation (§3.3.1.1) becomes necessary for the digraphs of the transliteration system. The inevitable drawback is that in strict transliteration, uppercase cannot and must never be used for the purposes to which capitalisation is traditionally put in the Latin writing system, such as marking the initials of proper names, sentences and metrical units.

3.2.4. Implicit graphotactics

In Romanised transliteration, graphemes follow one another in a strictly linear sequence. In an aksharic system, however, this strict linearity is present only at the level of *akṣara*s, which may be coterminous with graphemes (when they consist of an independent vowel or an independent consonant), but are not graphemes per se (§2.3.1.1). The individual graphs constituting a complex glyph can appear in a complicated topology without any intuitively obvious correspondence between their spatial arrangement and the order in which they must be read.

As a consequence, Romanised transliteration is by nature blind to the segmentation of the source text into *akṣara*s and to the spatial arrangement of graphs within an *akṣara*. Since the phonographic graphemes, their independent graphemic allographs, and the *virāma* are all represented individually, a reader familiar with the graphotactic rules of the source writing system will almost always be able to reconstruct the source's *akṣara* segmentation from the transliterated text. Thus, knowledge of the source writing system's graphotactic rules is implicitly assumed for anyone who approaches the transliterated texts from a palaeographic perspective.

Some specific writing systems deviate from the default graphotactic rules of the generic Indic writing system and construct some kinds of *akṣara*s in their own way. This can complicate the reconstruction of the original written form from the transliteration. In order to take systematic graphotactic variation in our stride, for certain specific writing systems we recognise **alternative graphotactic modes** that are different from the default mode applicable to the Indic system at large (§4.3.3). Conversely, idiosyncratic deviations

[79] Jan Kučera, personal communication (21 January 2026).

from the graphotactic mode characteristic of the writing system in question (which may be the default mode or an alternative mode) cannot be indicated in the transliteration (§3.1.4.2).

3.2.5. Loose transliteration

Unlike strict transliteration, which is predominantly diplomatic (§3.2.1), **loose transliteration** is predominantly linguistic in nature. It is applicable to contexts where words, passages or complete texts in a source language are treated in the abstract, rather than as a particular written instance. Accordingly, some constraints of the strict system are relinquished, resulting in a shift toward the transcription end of the spectrum. Loose transliteration is not a rigorously defined system, and we expect that in contexts not requiring diplomatic accuracy, users of the DHARMA system will transliterate source languages in ways that are either intuitive or conformant to the general practice of the linguistic sub-field (or both). The degree of "looseness" depends on context, and may be for instance smaller in a critical edition and more extensive in isolated words or names mentioned in modern-language discussion. For these reasons, we only outline the basic features of loose transliteration here and give some non-exhaustive specific suggestions in the body of this Guide.

- loose transliteration does not distinguish graphemic allographs from the base graphemes
  - this frees up uppercase target graphemes for traditional purposes, such as the capitalisation of the initials of loosely transliterated proper names
  - hence, whenever a proper name in a source language is mentioned in the course of discussion in an international language, we recommend using capital initials
- when the independent forms of vowels and consonants are not distinguished in transliteration from their in-*akṣara* forms, it becomes necessary to disambiguate the diphthongs <ai> and <au> from the corresponding vowel clusters (although they are unambiguous in strict transliteration, e.g. *praUga*, *caUttho* and *daIĀ*)
  - the ISO 15919 disambiguation colon (§3.3.1.1) can be used for this purpose (e.g. *pra:uga*, *ca:uttho* and *da:iā*)
  - but our preference is to follow the convention of using a diaeresis (pair of dots) above the second vowel (e.g. *praüga*, *caüttho* and *daïā*)
  - disambiguation (of any form) should be avoided when there is no ambiguity, for example
    - because one of the vowels is long, e.g.
      - *āicca* (strict transliteration: *āIcca*), never *āïcca* or *ā:icca*
    - because a space or other editorial insertion separates the vowels, e.g.
      - *mṛta iva* (strict transliteration: *mṛta Iva*), never *mṛta ïva* or *mṛta: iva*
      - *mana-icchā* (strict transliteration: *mana-Icchā*); never *mana-ïcchā* or *mana:-icchā*
- loose transliteration permits the silent normalisation of orthographic peculiarities (heterography, §2.5.2), including but not limited to
  - the alternation of double and single consonants in certain phonological contexts, particularly in Sanskrit
  - the alternation of *anusvāra* and *visarga* variants
  - the alternation of *anusvāra* with the class nasal or a generic nasal such as *n*
- loose transliteration permits (vaguely) phonological transcription where this is preferred to the accurate transliteration of the source graphemes, including but not limited to
  - distinguishing short *e/o* from long *ē/ō* even if the original writing does not do so
  - transcribing *h* for *visarga* in languages where it represents a final /h/ rather than the Sanskrit *visarga*
  - transcribing the actual phoneme or phoneme sequence represented by graphemes repurposed from their traditional signification in certain writing systems
  - transcribing single consonants for consonants doubled at morpheme boundaries in Old Javanese (e.g., *lavann ika* simplified to *lavan ika*, *muvaḥhakan* to *muvahakan*, *at thana* simplified to *at hana*)

3.2.6. Informal alternatives

In order to achieve a one-to-one correspondence between source and target, the DHARMA transliteration scheme assigns one specific target grapheme (or digraph) to each recognised source grapheme. The arbitrary placeholders endorsed for peripheral graphemes (§3.1.1.2) are a partial exception to this, since they allow the use of alternative representations for any particular source grapheme as well as the use of character strings for a single source grapheme. In some situations it may be necessary or beneficial to relax the strictures of the transliteration scheme in other ways and use informal alternatives for certain target characters. Informal alternatives suggested in the body of this Guide will be highlighted as shown.

Certain characters, especially diacritical combinations, may be **difficult or impossible to enter** in your hardware and software environment. The recommended solution for such cases is to copy and paste these characters from this Guide (or a document you keep at your fingertips, where you have collected the special characters you need). An alternative solution for work in progress that may suit your working style better and improve productivity is to use informal alternatives that you can enter easily. Whenever you do this, it is your responsibility to ensure that your alternative notation is unique (so a batch search-and-replace will not accidentally change something else too), and that you replace it with the proper notation when finalising your work.

Some characters and diacritical combinations may also be **impossible to reproduce** in a given (print or screen) publication environment and/or rejected by style requirements imposed on a publication. In such a case, use informal alternatives that suit the platform in question. If the text you are working on is not destined solely for that publication, then we strongly recommend using strict DHARMA transliteration in your master copy of the text and only replacing the problematic characters with their informal alternatives in the copy meant for publication in such an environment.

3.3. Synergy with markup

While creating a digital edition, we strive to keep the transliteration *representing* original content (i.e. what is written on a certain support) separate, or at least separable, from our annotations *describing* and *analysing* various aspects of the source (for instance how it is written and laid out, how clearly it is readable, or what sort of additional information it carries) and the transliteration (how it is to be interpreted). Content is to be transliterated according to the methods covered in this Guide, while annotation is to be added in the form of formalised markup and free annotation. Ideally, this separation of concerns should be complete: no issues pertaining to description and analysis should be recorded in the transliterated text itself; and likewise, no issues pertaining to the content should be omitted from the transliterated text and recorded only in annotation. In practice, there are a number of borderline cases that could arguably belong to either of these domains, with a certain degree of inevitable fuzziness and permeability. For instance, the distinction of independent vowel and consonant forms from their in-*akṣara* forms (§3.1.1.2) could also be achieved by using the same transliteration (based solely on the phonological unit represented by them) and adding markup to record their form.

Markup in our practice can be of two basic kinds: electronic markup (namely TEI-compliant encoding in XML as described in the Encoding Guides), or text-based markup such as the various brackets employed in printed epigraphic editions. Strict DHARMA transliteration is primarily for use in XML-encoded diplomatic and critical editions of source texts.[80] The full arsenal of XML encoding is, however, not available in other working environments, and not entirely necessary in XML documents outside the edition proper (for example in a translation or commentary, where transliterated source-language text is still necessary).

In order to facilitate markup in a non-XML environment and to reduce code clutter in an XML environment, our transliteration scheme is supplemented with some elements of text-based markup comprised of characters at the level of the transliterated text rather than on the separate tier of XML encoding. For simplicity's sake, we refer to such markup as **shorthand**, implying that it is a simplified notation which could

[80] XML encoding also allows the separation of a third concern, the way in which the transliterated text will be ultimately presented to users on screen and in print.

also be expressed more fully in the form of XML code. Shorthand solutions are divided into the categories of essential, optional and private shorthand depending on the context of their use and their relationship to XML markup in our practice, as discussed in the following subsections. Where the main body of this Guide suggests a shorthand solution, it will always be assigned to one of these categories and labelled as shown below. All of the shorthand solutions recommended in this Guide, as well as any other private shorthand, may be used freely in a non-XML context as and when deemed expedient.

3.3.1. Essential shorthand

Essential shorthand is shorthand notation for features that could feasibly be represented via XML encoding, but we choose not to do so in the DHARMA encoding scheme, because using XML for these features would hinder productivity by making the encoded text less human-readable and more prone to human error without providing a substantive advantage. Essential shorthand can therefore be used as and when required in both XML and non-XML contexts and need not be replaced or complemented with XML markup in the latter, although future projects may decide to do so.

In addition to the solutions described below, our sandhi analysis (§7.2) and editorial segmentation (§7.3) are also in principle essential shorthand.

3.3.1.1. Colon for disambiguation

Because our transliteration scheme involves some digraphs (§3.2.2), it is on rare occasions necessary to disambiguate these from their constituent target graphemes used in their individual roles. The ISO 15919 scheme (ISO 2001, 17, §8.1.15) employs : (the colon) as a disambiguation sign in such situations, as well as in sundry contexts where the graphetic composition of an original *akṣara* is unusual. The latter is in our perception a very different kind of situation, for which we prefer a different notation (§3.3.2.1). As for the former, the need to distinguish vowels in hiatus from diphthongs is eliminated by our use of uppercase for independent vowels (§3.1.1.2). Thus, the words written in Devanagari as प्रउग, चउत्थो and दइआ are to be transliterated *praUga*, *caUttho* and *daIĀ* rather than *pra:uga*, *ca:uttho* and *da:iā* as in default ISO 15919. In loose transliteration, either the colon or a diaeresis may be used for the disambiguation of diphthongs (§3.2.3).

We do endorse the use of the disambiguation colon in the exceedingly rare cases where an <h> follows a stop consonant grapheme, in order to indicate that it is this combination, and not the aspirated consonant grapheme, that is present in the original. Thus, the cluster |प्ह| is to be transliterated *p:ha*, distinguishing it from |फ| = |pha|.

Since the disambiguation colon hardly ever occurs in our texts, we have chosen to also employ the colon for an honorary grapheme, the *ā* marker as an indicator of vowel length or consonant doubling in Indonesian writing systems (§4.1.7.2)

3.3.1.2. “Ceilings” as placeholders for orphaned elements

An *akṣara* is normally an atomic unit of the Indic writing system. However, *akṣara*s interacting with a material feature of the support (such as a line break, or a defect or other feature on the writing surface) are sometimes graphetically separated into their constituent parts. A constituent separated from its parent *akṣara* may be a component, i.e. a graph with its own graphemic signification, but it may also be an element (§2.4.4), without graphemic value in itself. In this latter situation, we dedicate the target characters ⌈ (left ceiling, U+2308) and ⌉ (right ceiling, U+2309) as placeholders for representing orphaned elements (located respectively before or after the parent *akṣara*). In this way, the transliterated text can indicate the fact of such separation, show the location of the separated element, and allow the addition of XML markup specifically targeting the separated element when needed.

3.3.2. Optional shorthand

Optional shorthand is text-based markup for features that can also be represented through XML markup in our practice. It is therefore intended primarily for contexts where encoding is not applicable. In an XML document, the use of optional shorthand is acceptable in our practice, but it is strongly recommended that the shorthand be supplemented with or replaced by encoding, as indicated in the following subsections. Our

transliteration system's dedicated placeholders for peripheral graphemes (§3.1.1.2) and space (§3.1.3) are in fact also a form of optional shorthand, which can and normally should be supplemented by XML markup when applicable.

In addition to the solutions described below, our supplied *avagraha*s (§4.3.1.1) and editorial interpretive alterations (§7.1) are also in principle optional shorthand.

3.3.2.1. The equals sign as character joiner

We use the = (equals) sign as optional shorthand in the role of a character joiner, for use between two target graphemes that are manifested in the source as parts of a single complex character (§2.4.3), when these would normally manifest as (parts of) separate characters according to the general rules of the Indic writing system.[81] This shorthand notation is inspired by the classical Indological convention of using an = sign (or a double hyphen) to indicate that the end of one transliterated word and the beginning of the next belong to a single source *akṣara*. However, because we transliterate the independent forms of source graphemes differently (§3.1.1.2), the = sign is not needed (and must be avoided) in the situations where it is normally employed in classical editions, namely at word boundaries.[82]

The corresponding XML markup indicates the boundaries of the *akṣara* rather than joining any two adjacent graphemes, and is therefore more flexible and less equivocal. Nonetheless, because the shorthand is simpler to produce and to read, we approve of employing it in XML editions, especially for cases of systematic variation that the informed reader is expected to be able to interpret. XML markup can either supplement or replace the shorthand, at the encoder's discretion.

3.3.2.2. V and C as vowel and consonant wildcards

According to widespread convention, we endorse the use of the characters V (uppercase v) and C (uppercase c) as wildcards representing "any vowel" and "any consonant" respectively. These are mainly for use in morphological and phonological discussion, but are also suitable for the informal representation of the illegible vowel or consonant components of an *akṣara*.

This use of uppercase V and C is systemically questionable given that in general we dedicate uppercase Latin letters to independent allographs. However, we are not aware of any source text with final |V| or |C| glyphs, and these consonants cannot appear in a final position in the phonotactics of Sanskrit and related languages, so the possibility that one will someday be found is remote. There is, therefore, no conflict in practice between the two solutions.

3.3.3. Private shorthand

Private shorthand means any kind of text-based markup used in the production stage to speed up the preparation of electronic texts, including collaborative text preparation. Private shorthand must always be eventually replaced with acceptable shorthand and/or XML encoding, and it is the responsibility of the person using the shorthand to do so before the work is made public.

Any kind of notation may be used as private shorthand, but the following cautions are strongly recommended. Private shorthand should always consist of characters or strings that otherwise never occur in the text and the subcorpus. Private shorthand solutions should be kept track of and documented at least cursorily. When using batch search-and-replace to convert private shorthand into a publishable form, care must be taken to avoid inadvertently replacing anything else.

[81] A similar use of the = sign will be recognised in the forthcoming revised version of ISO 15919 (Jan Kučera, personal communication, 21 January 2026).

[82] The classical rendition *tam eva* with a space corresponds to *taM Eva* (with a final *m* and an initial *e*) or *tam· Eva* (with a *virāma* and an initial *e*) in the DHARMA scheme, while the classical *tam=eva* corresponds simply to *tam eva*, where the lowercase letters make it clear that *m* and *e* appear in their in-*akṣara* forms.

3.3.4. Free annotation

For all features of the source that cannot be represented accurately through markup, human-readable annotation is always an option. In XML editions, general annotation may be placed in the palaeographic description or the commentary as applicable, while specific spots of text are best annotated in apparatus notes.

3.4. Compatibility with other transliteration systems

Aside from increasingly obsolete ASCII-based transliteration schemes optimised for early computer processing, the only internationally prevalent schemes for the Romanisation of Indic scripts are ISO 15919 and IAST (the International Alphabet of Sanskrit Transliteration).[83] The schemes used in many traditional Indic epigraphic publications and those still current in India have much in common with both of these, but are less standardised, and generally employ more polygraphs in order to approximate English phonological transcription better. The transliteration in the Madras *Tamil Lexicon* is identical to ISO 15919 on all fundamental points. IAST is the most widespread among Sanskritists, but does not cater for some graphemes associated with other languages written in an Indic script and is prone to become ambiguous when extended for this purpose. For this reason, the DHARMA transliteration scheme is based on the ISO 15919 scheme, extending it in order to handle graphemes of specific writing systems not covered in the standard, and modifying it by incorporating case sensitivity (§3.2.3) and text-based editorial markup (§3.2.3). Among writing systems not or not wholly covered by ISO 15919, our transliteration scheme is by and large compatible with current recommendations for the transliteration of Kannada (Ollett and Taylor draft) and Burmese (Lammerts and Griffiths 2016).

3.4.1. Extensibility

The inventory of dedicated transliteration equivalents and shorthand solutions discussed in this guide comprises those that have been found relevant to the DHARMA project. This inventory may be expanded freely according to the needs of future projects for any linguistic or regional context that we have not yet catered for. For instance, our transliteration scheme has no provision for the Gurmukhi *addak*, a sign whose graphemic function is difficult to describe.[84] Should it be desired to represent it in a text transliterated according to our conventions, it would probably be best recognised as an honorary grapheme (§2.7.2) with a dedicated transliteration.

While adding new signs to the inventory, the following rules of thumb should be considered.

- **compatibility**: unless there are strong reasons to the contrary, use the signs prescribed by ISO 15919 when applicable,[85] and signs already in conventional use in the relevant subdiscipline when an ISO standard is not applicable
- **simplicity**: as far as feasible, avoid polygraphs and give preference to single target graphemes
- **uniqueness**: as far as possible, the Latin grapheme (or polygraph) selected for any particular source grapheme should not be identical to any target grapheme already in use for another source grapheme in the DHARMA scheme
  - deviate from this rule only when it is reasonable to assume that the two different source graphemes transliterated by an identical target grapheme or digraph will never co-occur in any source text

[83] The ISO standard is described in a proprietary publication (ISO 2001), but a summary is freely available on Wikipedia (2026b). There is no definitive standard for IAST, but the convention is also described on Wikipedia (2026a).

[84] The *addak* |ੱ| usually indicates the gemination of the following consonant, but may also function as a stress marker for the preceding vowel.

[85] For example, the ISO 15919 provisions for Bengali |য়| → *ẏ* and Devanagari |ज़| → *z* could be added straightforwardly to the DHARMA system.

4. Transliterating core graphemes

Core graphemes (§3.1.1) are graphemes which we represent in transliteration by means of a dedicated target grapheme. The various target graphemes indicate the graphemic function of the source grapheme, most often by identifying the archetypal speech sound (§2.3.1) that it signifies. The graphetic appearance of core graphemes is on the whole not represented in transliteration, although in some cases distinctions are made on a graphetic basis. The reader familiar with the source script is expected to be able to reconstruct the original appearance of the graphs with sufficient precision.

4.1. Phonographic graphemes

4.1.1. The universal graphemes of the Indic writing system

The DHARMA transliteration for the basic graphemes of the Indic writing system is fully identical to ISO 15919. We assume all readers to be familiar with this system, but for the sake of completeness we list the target graphemes in Figure 4.1.A below. Readers primarily used to working with IAST should keep in mind a key difference between that scheme and ISO 15919 (and the DHARMA scheme), namely that the former employs underdotted *ṃ* for the *anusvāra* and underdotted *ṛ*, *ṝ* and *ḷ* for vocals, while in the latter, the underdot is primarily for retroflex consonants, so *anusvāra* is transliterated *ṁ* with a dot above, and the vocals *r̥*, *r̥̄* and *l̥* with an undercircle.

Figure 4.1.A. The basic inventory of Indic graphemes

a	ā	i	ī	u	ū	r̥, r̥̄	l̥, l̥̄	e, (ē)	ai	o, (ō)	au
k	kh	g	gh	ṅ		c	ch	j	jh	ñ	
ṭ	ṭh	ḍ	ḍh	ṇ		t	th	d	dh	n	
p	ph	b	bh	m		y	r	l	v		
ś	ṣ	s	h			ṁ	ḥ				

4.1.1.1. Transliteration of vocalic *r* and *l*

The diacritical letters corresponding to the ISO 15919 and DHARMA transliteration of vocalic *r* and *l* are not available in Unicode as pre-composed characters and must be composed by entering an *r* or *l* as applicable, followed by ◌̥ (U+0325 Combining Ring Below) and, if needed, by ◌̄ (U+0304 Combining Macron), in this order. Alternatively, copy and paste the composed characters from Figure 4.1.A above, or use informal alternatives as a provisional solution.

- straightforward informal alternatives involve underdots instead of undercircles, i.e.
  - *ṛ* instead of *r̥*
  - *ṝ* instead of *r̥̄*
  - *ḷ* instead of *l̥*
  - *ḹ* instead of *l̥̄*
  - but note that in some languages, *ḷ* and/or *ṛ* may be required for the transliteration of retroflex consonants (§4.1.5.1)

4.1.1.2. Transliteration of *e* and *o*

In transliterating languages that do not make a phonemic distinction between /e/ vs. /ē/ and /o/ vs. /ō/, the ISO 15919 standard (ISO 2001, 18, §9.1) provides the option of using use the letters *e* and *o* without a macron to transliterate the corresponding graphemes, regardless of whether they are normally pronounced long in the source language. DHARMA transliteration makes use of this option and, importantly, applies it to any **writing systems** that do not make a graphemic distinction, which includes many historic Dravidian

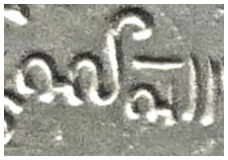
writing systems where a length-based contrast exists in the language, but is not reflected in writing. Since the specific signs for the long variants of these vowels (such as those in Tamil |கே| *kē* and |கோ| *kō*) are usually palaeographically derived from the original signs for the generic — and, in Sanskrit, long — vowel (such as those in Tamil |க| *ke* and |கா| *ko*), the default transliteration in our scheme is always the unmarked Latin character *e* or *o*.

– see also §7.1.1 on the editorial distinction of the long phonemes /ē/ and /ō/ when written with the basic graphemes <e> and <o>

4.1.2. Vowelless consonants

Aksharic writing systems need special provisions for representing consonants not followed by a vowel (§2.1.1). Many historic variants of the Indic writing system employ special final (*halanta*) forms of the regular consonant graphemes for this purpose (Figure 4.1.B). We consider these to be graphemic allographs (§2.7) of the consonant graphemes and deem it important to distinguish them from the default in-*akṣara* allographs in diplomatically accurate transliteration (§3.1.1.2). Conversely, many Indic writing systems (especially more recent ones) employ an explicit zero vowel marker (*virāma*, *puḷḷi, patén/pangkon*, etc.; also referred to as a vowel killer) in combination with the regular consonant graphs in order to represent vowelless consonants. In our view, the vowel killer is a genuine phonographic grapheme (§2.6.3), which must thus be transliterated with a corresponding target grapheme. Representing zero vowel markers by a separate character in the transliteration has the added advantage of allowing the application of targeted XML markup to that sign. Guidelines for distinguishing a final (simplex) form from (a complex) one involving a vowel killer are given in §4.1.2.1. In some writing traditions, a vowelless consonant may also be represented by the regular consonant graphs, as discussed in §4.4.1.

Figure 4.1.B. Independent consonant graphs

1	2	3	4	5
				
yaN	*dattaM*	*ṇāM||*	*yaT*	*dṛK*

4.1.2.1. Distinguishing independent forms from characters with a vowel killer

Final consonant forms may differ from the regular *akṣara*-forming graphs in any combination of size reduction, subscript positioning, stroke simplification and the absence of a headmark. Independent consonant forms often also involve an additional "diacritical" stroke — such as a horizontal dash, a vertical squiggle above the primary graph, or a curl below it — which may be interpreted as a proto-*virāma* (as in Figure 4.1.B/4,5). These transitional forms constitute a fuzzy boundary between simplex and complex glyphs.

The primary **criterion** for assigning a particular written instance to one or the other of these classes is whether the body of the consonant is graphically identical to the form seen in regular *akṣara*s. If it is, and a stroke that may be interpreted as a zero-vowel marker is present, then transliterate it as a complex glyph with a *virāma* (§4.1.2.3). If, on the other hand, the body is graphically altered in one or more of the ways mentioned above, then transliterate it as a simplex final character (§4.1.2.2), even if a stroke that might be interpreted as a zero-vowel marker is present. Being fairly objective, this is a practicable criterion, but it does not necessarily do justice to all occurrences. In ambiguous cases, base your decision on your knowledge of the practice of the source text in question and of the writing system of the time and region. A relevant consideration is the productivity of these nascent vowel killer signs. If they appear only in final positions, with a small range of consonants, and especially if their form varies depending on the consonant, then it is better not to recognise them as *virāma* graphemes. Conversely, if they can appear word-internally (where

ligatures would be a feasible alternative), are combined with a variety of consonants, and they have the same form regardless of the consonant, then it is more prudent to interpret them as actual *virāma*s.

4.1.2.2. Simplex glyphs for independent consonants

Simplex independent consonant forms shall be transliterated as the corresponding uppercase Latin consonant, e.g. *N*, *M*, etc.
- should you encounter simplex final forms of consonants for which our transliteration uses digraphs (which should not be possible in Indo-Aryan languages and is unlikely in other languages written in an Indic script), use uppercase only for the first letter of the digraph, e.g. *Th*
- guidelines for deciding what constitutes a special final form are given in §4.1.2.1 above
- the same transliteration applies to the graphetically dependent final consonants of Pyu, for which see also §4.4.6

4.1.2.3. Vowelless consonants as complex glyphs involving a vowel killer

Since we recognise the *virāma* as a phonographic grapheme (§2.6.3), we always transliterate it explicitly. Complex glyphs involving a regular consonant form and any explicit zero vowel marker shall be transliterated as follows.
- transliterate the consonant component normally (with the lowercase equivalent)
- transliterate the vowel killer as ·
  - `U+00B7` Middle Dot
- e.g. Devanagari |त्| → *t·*
- this applies to all vowel killers by any name, including for instance the Tamil *puḷḷi*
  - e.g. |த்த| → *t·ta*
  - see also §3.3.2.1 for the "implicit *puḷḷi*"

4.1.3. Independent vowels as simplex glyphs

Aksharic writing systems employ independent vowel graphs for writing vowels in an initial position and in hiatus (§2.1.1). As is the case with special vowelless consonant forms (§4.1.2), these independent vowels are in our view graphemic allographs (§2.7) of the vowel graphemes, and we deem it important to distinguish them from the default in-*akṣara* allographs in diplomatically accurate transliteration (§3.1.1.2).
- independent vowel forms shall be transliterated as the corresponding uppercase Latin vowel, e.g. *A*, *I*, etc.
- for the independent forms of the diphthongs *ai* and *au*, capitalise only the first character of the digraph in your transliteration, i.e. use *Ai* and *Au* (whereas *AI* and *AU* would transliterate अइ and अउ)
- independent vowels represented in some Southeast Asian systems by complex glyphs involving a vowel support grapheme are discussed in §4.1.7.2

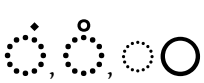
4.1.4. *Anusvāra* and its relatives

As per §2.6.3, we always recognise the *anusvāra* as a grapheme, regardless of whether a particular instance signifies a consonant phoneme or allophone, or a nasal quality to an adjacent vowel phoneme. However, because various forms based on the plain *anusvāra* can, in some specific writing systems, fulfil more or less differentiated graphemic roles, we distinguish *anusvāra* variants in transliteration as follows.

The basic *anusvāra* graph of Indic script, typically a dot or a circle, is always transliterated as *ṁ*, with a dot above as in ISO-15959 rather than with a dot below as in IAST. Conversely, *ṃ* with a dot below is used for a different function (§4.3.3.1) in the DHARMA scheme. In the linear sequence of transliterated characters, the *anusvāra* shall be positioned where it logically falls in language, regardless of whether it is graphically above an *akṣara*, in a separate inline space, or even above the next *akṣara*. The graph representing the *anusvāra* may vary somewhat within a source (e.g. dots alternating with small circles), but unless this variation is likely to correspond to different linguistic information (i.e. if the various graphs are not allographs but separate graphemes), the transliteration should remain identical.

Conspicuously different alternative graphs may carry a different linguistic signification, for example some writing systems employ the *bindu* |◌̣| graph for the consonantal *anusvāra* and the *candrabindu* |◌̐| graph to signify the nasal quality of the preceding vowel. In some other cases, the linguistic signification of the alternative graphs is less clear and/or less consistent. In the diplomatic transliteration of primary texts, we prefer to err on the side of caution and therefore not only do we recognise all *anusvāra* forms as individual graphemes, but also use a different target grapheme for conspicuously different *anusvāra* graphs, even if it is not certain that their linguistic value is different from that of the basic *anusvāra* dot. *Anusvāra* variants are thus differentiated in transliteration on a primarily graphetic basis, as shown in Figure 4.1.C.

Figure 4.1.C. Anusvāra *variants*

1	2	3	4	5
anusvāra, bindu	*anunāsika, candrabindu*	*anusvāra-candra*	*ulu ricem*	Bengali *anusvāra*
				
				
saṁ, vaṁ	*Oṃ̆*	*tad-vaṃ̆śe*	*vṛttaṁ*	*ṁ*

Anusvāra and its variants shall be transliterated with the following dedicated equivalents.

- *ṁ* for the basic *anusvāra*, which takes the shape of a *bindu* (a dot, a circle or occasionally a small dash), as in Figure 4.1.C/1
  - this is the character `U+1E41` Latin Small Letter M with Dot Above
  - use this transliteration regardless of size and position in the source, which may be above the preceding *akṣara*, above the following *akṣara* or in its own segmental space between two *akṣara*s at any height
    - but see §4.3.3.1 for dots below *akṣara*s, which are transliterated with *ṃ*
  - you can opt to use this transliteration for shapes other than a typical *bindu* if the shape is the primary form of *anusvāra* in your text or subcorpus and you are reasonably certain that the shape is not associated with a different graphemic function than the dot
  - the same transliteration shall be used for dots above *akṣara*s even if their function is not clear or not quite the same as that of the classical *anusvāra*, as in Mon and Pyu, for which see also §4.4.6
- for *anusvāra* variants which in your opinion might be graphemically distinguished in your source text from the basic *anusvāra*, use the following dedicated equivalents, distinguished on the basis of shape regardless of size, position and orientation in the source
  - *ṁ̆* for the classical *candrabindu* graph comprised of an *anusvāra* dot (*bindu*) and a semicircle (*candra*), as in Figure 4.1.C/2
    - no precomposed glyph available: enter a regular *m* followed by ◌̆ (`U+0310` Combining Candrabindu)
    - do not use the *candrabindu* Unicode diacritic in combination with any other Latin letter, thus *tāṁ̆l lakṣmīm* rather than *tāl̆ lakṣmīm*
    - inverted *candrabindu* signs (with the semicircle above the dot) may be transliterated as *ṁ* (see below) when it is expedient to distinguish them in transliteration
  - *ṁ̆* for a semicircle (*candra*) without a dot (*bindu*), in any orientation
    - no precomposed glyph available: enter a regular *m* followed by ◌̆ (`U+0306` Combining Breve)
    - in the DHARMA corpus, such signs are exemplified by the Cam *anusvāra-candra* (Figure 4.1.C/3)
    - the same transliteration is recommended for similar signs not occurring in our corpus, such as
      - the Telugu “half-*anusvāra*” |◌ఁ| called *arasunna* or sometimes, misleadingly, *candrabindu*

  - the same transliteration is recommended for the Gurmukhi *ṭippī* |◌̐|, an inverted semicircular shape[86]
- *ṁ* for any combination of an *anusvāra* dot (*bindu*) with an additional (straight, curved or semicircular) stroke, including *candra* + *bindu* combinations when these differ graphetically from the standard *candrabindu* and distinguishing them is desirable
  - this is the character U+1E3F Latin Small Letter M with Acute
  - in the DHARMA corpus, such signs are exemplified by the Javanese/Balinese *ulu ricem* (Figure 4.1.C/4) and the Bengali *anusvāra* (Figure 4.1.C/5)

4.1.5. *Visarga* and its relatives

As per §2.6.3, we always recognise the *visarga* as a grapheme, regardless of whether a particular instance signifies a consonant phoneme or allophone, or a breathy quality to an adjacent vowel phoneme. Some varieties of the Indic writing system make a graphic distinction corresponding (with more or less regularity) to allophones of the *visarga*, using the sign called *upadhmānīya* before palatals and another sign called *jihvāmūlīya* before velars (Figure 4.1.D). We treat these as graphemes distinct from the *visarga*, so when such a distinction is made in the text you are transliterating, preserve this distinction in transliteration regardless of whether the *visarga* alternatives are used systematically and correctly or not.

Figure 4.1.D. Visarga relatives

1	2	3	4	5
visarga	combining *upadhmānīya*	combining *jihvāmūlīya*	inline *upadhmānīya*	inline *jihvāmūlīya*
				
yaḥ	*ḫ pu*	*traḫ ka*	*ḫ*	*yoḫ ka*

Visarga and its variants shall be transliterated with the following dedicated equivalents.

- ***visarga*** → *ḥ*
  - U+1E25 Latin Small Letter H with Dot Below
  - the same transliteration shall be used for *visarga*-like double dots whose graphemic function is uncertain, as in Pyu
- ***upadhmānīya*** → *ḫ*
  - U+1E2B Latin Small Letter H with Breve Below
  - the informal alternative *f* may, when necessary, be used instead
- ***jihvāmūlīya*** → *ẖ*
  - U+1E96 Latin Small Letter H with Line Below
  - the informal alternative *x* may, when necessary, be used instead

4.1.5.1. Combining and inline *upadhmānīya* and *jihvāmūlīya*

While the basic *visarga* is always written inline as a semi-independent graph (§2.6.1), the *jihvāmūlīya* or *upādhmānīya* can, depending on the specific writing system, either appear independently (Figure 4.1.D/4,5) or behave like a consonant graph and form a conjunct with the following consonant (Figure 4.1.D/2,3).

[86] This treatment, although not an established convention, seems preferable to the haphazard approach of ISO 15919, whereby *ṭippī* is to be transliterated in modern language with the class nasal when it occurs before a stop or a class nasal, with *ṁ* (no distinction from *bindī*) when it occurs in another position, and in "older language" as *ṃ*.

Whether a source grapheme manifests as an autonomous glyph or as a dependent graph cannot be represented through transliteration (§2.6.1). If the two behaviours alternate within a single text, or if all instances of these signs in a text behave otherwise than expected on the basis of related texts (in the same language, from the same region and time), then note the default behaviour in your palaeographic description and annotate the deviations from it. The character joiner sign = (§3.3.2.1) may in this case be used as optional shorthand to indicate the idiosyncratic association of the *visarga* variants.

- if the default behaviour is *visarga*-like, then idiosyncratic combining *upadhmānīya* and *jihvāmūlīya* may be transliterated *ḫ̲=* and *h̲=*, indicating that it combines with the following consonant
  - e.g. Figure 4.1.D/2: *ḫ̲= pu*; Figure 4.1.D/3: *traḫ̲= ka*
- if the default behaviour is consonant-like, then idiosyncratic inline *upadhmānīya* and *jihvāmūlīya* may be transliterated *=ḫ̲* and *=h̲*, indicating that it, like a *visarga*, is associated more closely with the preceding *akṣara*
  - e.g. Figure 4.1.D/5: *yo=ḫ̲ ka*
- when the character joiner sign is placed between two graphemes which are separated by an editorial space or hyphen (§7.3), always put the = sign first, and the space or hyphen second

4.1.6. Additional graphemes in other Indian writing systems

The transliteration equivalences listed in this section are conformant with ISO 15919.

- variants of the *anusvāra* are discussed in §4.1.4
- variants of the *visarga* are discussed in §4.1.5
- the "double *kāl*" in Tamil script is not a grapheme but a graphetic combination of two graphemes, to be treated as per §6.4
- **Dravidian (and other) short *e* and *o***, as distinct from long *ē* and *ō*, are discussed in §4.1.1.2 above
- **Tamil *āytam*** |ஃ| → *k̲*
  - U+1E35 Latin Small Letter K with Line Below
- **Dravidian alveolar nasal**, e.g. Tamil |ன| → *n̲*
  - U+1E49 Latin Small Letter N with Line Below
- **Dravidian, Vedic and New Indo-Aryan retroflex lateral**, e.g. Tamil |ள|, Kannada/Telugu |ళ|, Vedic/NIA |ळ| → *l̤*
  - U+1E37 Latin Small Letter L with Dot Below
  - **Vedic aspirated retroflex lateral** |ळ्ह| → *l̤h*
- **Dravidian retroflex approximant / frictionless continuant**, Tamil |ழ|, Kannada/Telugu |ఴ| → *l̲*
  - U+1E3B Latin Small Letter L with Line Below
- **Dravidian alveolar trill/stop**, Tamil |ற|, Kannada/Telugu |ఱ| → *r̲*
  - U+1E5F Latin Small Letter R with Line Below
- **New Indo-Aryan retroflex flap**, Hindi |ड़| → *ṛ*
  - U+1E5B Latin Small Letter R with Dot Below
  - **New Indo-Aryan aspirated retroflex flap**, Hindi |ढ़| → *ṛh*

4.1.7. Additional graphemes in Southeast Asian writing systems

The transliteration equivalences listed in this section are not covered by ISO 15919 (which is targeted at Indian writing systems).

- variants of the *anusvāra* are discussed in §4.1.4
- the glottal stop grapheme, which can also function as a vowel support, is discussed in detail in §4.1.7.2
- the combination of |u| and |i| to represent a particular phoneme is addressed in §4.1.7.3.
- the Indonesian use of |ā| as a mark of length is addressed in §4.3.3.1
- the underdot in Pyu and Burmese is to be treated as per §4.3.3.2
- shorthand for short vowels in lieu of long vowels in Sanskrit loanwords is described in §7.1.2
- **barred/dotted variant of *b* in Mon and Pyu** → *ḅ*
  - U+1E05 Latin Small Letter B with Dot Below

4.1.7.1. The mid-central vowel, “schwa”

- short form (Javanese/Balinese *pepet* |◌ꦼ|)
  - lowercase → *ə*
    - U+0259 Latin Small Letter Schwa
  - uppercase (for an independent vowel form) → Ə
    - U+018F Latin Capital Letter Schwa
- long form (Javanese/Balinese *pepet tedung* |◌ꦼꦴ|)
  - lowercase → *ə:*
    - U+0259 followed by a colon[87]
  - uppercase (for an independent vowel form) → Ə:
    - U+018F followed by a colon
  - in loose transliteration, *ə̄* and Ə̄ may be used instead
    - these are not available as a precomposed glyph: enter the base character and ◌̄ (U+0304 Combining Macron)
- if *ə* / Ə is difficult to access on your system, the informal alternative *ĕ* / *Ĕ* may be used instead
- be aware that there exist several Unicode codepoints for characters with an identical or similar appearance, which should be avoided
  - if creating a custom keyboard assignment, make sure it produces the desired Unicode character
  - if entering these characters on a keyboard that does not support them, copy and paste from this document or enter the exact Unicode code (§1.7.2)
  - in particular, make take care to avoid the following characters / codepoints
    - *ə* (U+01DD Latin Small Letter Turned E) and its uppercase equivalent *Ǝ* (U+018E Latin Capital Letter Reversed E)
    - *ə* (U+04D9 Cyrillic Small Letter Schwa) and its uppercase equivalent *ə* (U+04D8 Cyrillic Capital Letter Schwa)

4.1.7.2. The vowel support grapheme

In specific Southeast Asian systems, the original graph for the independent vowel <A> has evolved into a sign representing a “zero consonant” or a “minimal consonant” (a glottal stop [ʔ] or a [h]-like glottal fricative).[88] In this role, it can function as a *vowel support*, taking on vowel markers in combinations which replace the original independent vowel graphs. Moreover, in some writing systems this graph can even form conjuncts with consonant graphs, indicating that it was perceived by the users of those systems as a fully fledged consonantal grapheme.[89]

When this graph functions only as a vowel support, it is questionable whether it can be properly called a graph (i.e. the representation of a grapheme), or merely a graphic element. However, in a consonantal role it is definitely a graph, representing a grapheme. For this reason, we prefer to always recognise it as a graph regardless of the role it plays, and always transliterate it explicitly with the dedicated letter *q*. The following

[87] In the Javanese/Balinese writing system, this phoneme is always written by adding a length mark to the basic grapheme <ə>, so for the sake of consistency we transliterate it as per §4.3.3.1.

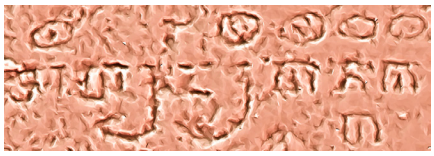
[88] We suspect two evolutionary processes here, which may have worked synergistically. On the graphetic side, certain independent vowel graphs in Indic writing systems serve as bases from which graphs for related vowels are derived by the addition of a stroke similar or identical to the corresponding dependent vowel graph. Thus, in many specific systems, the graph |Ā| is derived from |A|, |Ai| from |E|, and |Au| from |O|; in some, such as modern Devanagari, |O| is also derived from |A|. In the case under discussion, this generalisation has been carried to the extreme, whereby all independent vowel graphs are derived in this manner from the basic graph for the neutral vowel. On the linguistic side, some languages which adopted an Indic writing system had a minimal consonant phoneme for which the original writing system had no provision, and therefore adapted the writing system to accommodate that phoneme.

[89] This systemic change is complete in Khmer, where |ᩉ| functions fully as a consonantal graph, but has been carried through to varying degrees in the writing traditions of Java, Bali and Lombok. According to Ida Bagus Komang Sudarma (personal communication, 16 Aug. 2019), in Sasak writing ᬘ can be combined with a *pasangan* consonant, e.g. ᬘᭀᬶ *qhi* and ᬘᭀᬸ *qhu*, but cannot itself become a *pasangan*, while in Balinese writing neither possibility exists.

instructions for the strict and loose transliteration (§3.2.3) of vowels composed with a vowel support are summarised and illustrated for the Khmer and Balinese writing systems in Figure 4.1.F below.

In **all writing systems in which a vowel support graph exists**, regardless of whether it corresponds to a consonantal phoneme or is used only as a vowel support, apply **strict transliteration** as follows.

Figure 4.1.E. Vowel support



qət ṛ̀ŋəkən tikiṁ

- the vowel support shall be transliterated with the Latin letter *q*, as in Figure 4.1.E.
  - however, when the same graph is used on its own as independent <A>, transliterate it as *A* (not *qa*)
  - independent vowel graphs which do not involve a vowel support shall be transliterated in the regular manner with the corresponding uppercase vowel
    - e.g. Khmer អេត → *qeta* but ឯត្ត → *Etta*
- the same transliteration is to be used when the graph representing the glottal stop combines into a ligature with other consonantal graphs
  - in a syllable-initial position, e.g. |អ្នក| → *qnaka*
  - in post-consonantal position, e.g.
    - |ផ្អែក| → *phqaka*
    - |ល្អិត| → *lqita* (as opposed to |ល្ត| → *lIta*)
    - |ប្អូន| → *pqūna* (as opposed to |ផ្ទួន| → *phUna*)
    - |ក្អែក| → *kqeka* (as opposed to |ក្ត| → *kEka* and |ក្អែក| → *kqaeka*[90])

In the **loose transliteration** of characters involving a vowel support, **transcription** of the language has priority over transliteration of the script, therefore

- in languages **where the vowel support is silent** (i.e. it stands for a zero consonant), as in Balinese in Figure 4.1.F, ignore it in loose transliteration and simply transcribe the relevant vowels
  - e.g. Balinese |ᬍᬾ| = |qe| → *e*
- in languages **where the vowel support signifies a consonant**, as in Khmer in Figure 4.1.F, always transcribe it explicitly
  - in loose transliteration, either *q* or *’* (the right single quote) may be used to represent the glottal stop
  - note that this sound should be transcribed even when the vowel support is on its own (as the independent vowel <A>)
    - e.g. Khmer |អ| = |A| → *qa* or *’a*
  - except for Sanskrit names and loanwords occurring in an Old Khmer text, whose loose transliteration should transcribe the Sanskrit and not involve the glottal stop (e.g. *Amoghapura* rather than *Qamoghapura* or *’Amoghapura*)
    - even though Khmer speakers would have pronounced a glottal stop in these names and words

[90] The example is taken from Modern Khmer. The digraph *ae* is not included in the DHARMA system because it does not exist in Old Khmer, but can function analogously to *ui* (§4.3.3.1).

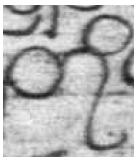
Figure 4.1.F. Independent vowels composed with and without a vowel support

language and writing system	IPA transcription of the phoneme	simplex glyph			complex glyph				
			transliteration				transliteration		
								loose	
		glyph	strict	loose	glyph	components	strict	lowercase	uppercase
Khmer	/ʔa/	អ				|A|	A	qa / ’a	qA / ’A
Balinese	/a/	ᬅ						a	A
Khmer	/ʔi/	ឥ	I	i	អិ	|A| + |i|	qi	qi / ’i	qI / ’I
Balinese	/i/	ᬇ			ᬅᬶ			i	I
Khmer	/ʔi:/	ឦ	Ī	ī	អី	|A| + |ī|	qī	qī / ’ī	qĪ / ’Ī
Khmer	/ʔu/	ឧ	U	u	អុ	|A| + |u|	qu	qu / ’u	qU / ’U
Balinese	/u/	ᬉ			ᬅᬸ			u	U
Khmer	/ʔu:/	ឩ	Ū	ū	អូ	|A| + |ū|	qū	qū / ’ū	qŪ / ’Ū
Khmer	/ʔe/	ឯ	E	e	អេ	|A| + |e|	qe	qe / ’e	qE / ’E
Balinese	/e/	ᬌ			ᬅᬾ			e	E
Khmer	/ʔaj/	ឰ	Ai	ai	ឱ្អែ	|A| + |ai|	qai	qai / ’ai	qAi / ’Ai
Khmer	/ʔo/	ឱ	O	o	អោ	|A| + |o|[91]	qo	qo / ’o	qO / ’o
Balinese	/o/	ᬍ			ᬅᬾᬅ			o	O
Khmer	/ʔao/	ឳ	Au	au	អៅ	|A| + |au|	qau	qau / ’au	qAu / ’Au
Balinese	/ə/				ᬆ	|A| + |ə|	qə	ə	ə
Balinese	/ə:/				ᬆᬉ	|A| + |ə| + |ā|[92]	qə:	ā	Ə̄

[91] The graph |o| is of course composed of elements which, in themselves, correspond to the graphs |e| and |ā|; cf. §2.4.4.1.

[92] Unlike the case of |o|, the |ā| marker is used here to indicate length, treated as per §4.1.7.2.

4.1.7.3. Graphemic combination of the vowel markers |u| and |i|

Figure 4.1.G. Combined ui	
1	2
	
gui	*tui*

In Khmer, Burmese and Mon, the vowel markers which independently signify <u> or <ū> and <i> may be combined so that together they represent a particular phoneme. Since this is an established feature of the writing system, in principle the combination of these graphic elements is a single graph (cf. §2.4.4.1[93]). Nonetheless, in accordance with the conventions of the field and in order to foreground the palaeographic precedent of this graph, we transliterate them individually. In practice, this is largely equivalent to using a digraph for this source grapheme, but since in this case the two transliteration graphemes severally correspond to source graphs, they can be treated separately in markup (e.g. unclear reading, editorial correction).

- the vowel marker |u| or |ū| used in combination with the marker |i| to represent a phoneme shall be transliterated as *ui* and *ūi*, for example:
  - the Khmer glyph in Figure 4.1.G/1 is transliterated *gui*
  - the Burmese glyph in Figure 4.1.G/2 is transliterated *tui*
- the character joiner sign = (suggested for instances of multiple vowel markers in a single *akṣara*, §4.4.5) must not be used
- note that in certain writing traditions, a combination of the markers |u| and |i| may signify scribal deletion
  - scribal marks are not phonographic graphemes and must not be transliterated as such (§3.1.4.3), only represented in XML markup

4.2. Numeral signs

As core graphemes of our transliteration system, numerals (in spite of being technically ideographic) are rendered with Arabic numerals in transliteration, which individuates them fully with respect to their graphemic function. However, as noted in §3.1.1, this transliteration method requires more than one target grapheme to correspond to a single non-digital numeral in the source, which can make the demarcation of graphemes ambiguous. This can and must be disambiguated rigorously in XML markup. In contexts where XML is not involved (including the preparatory stage of XML editions, if desired), the appropriate use of editorial spacing (§7.3.4.2) may be supplemented with private shorthand, for which we recommend the + (plus) sign, as described in the following subsections.

In XML editions, but nowhere else, all numbers written in numeral signs must, and all numbers written in words may, be supplemented with semantic encoding, which records the value of the number as a whole. This latter encoding is not applicable to (originally) numeral signs which are used in a function other than to denote numbers, such as the cipher <2> for the phonemes /ro/ (§6.3) and the cipher <1> used as an auspicious symbol (§6.2).

[93] See in particular the remark about the Latin graph |Y| in note 53 to that section.

4.2.1. The digits 0 to 9

Any numeral sign which denotes an integer between 0 and 9 is to be transliterated plainly with the equivalent Arabic digit. These numeral signs require no markup for demarcation.

Figure 4.2.A. Numeral signs 0-9

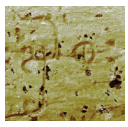
1	2
	
1218	8

- standalone single-digit numbers (Figure 4.2.A/2) shall be transliterated with a corresponding Arabic numeral, with spaces separating the number from any adjacent grapheme on either side
- multi-digit numbers written in place-value notation (Figure 4.2.A/1) shall be transliterated with a corresponding cluster of Arabic numerals, without intervening spaces, but with spaces separating the number from any adjacent grapheme on either side
- single-digit numbers additively combined with other numeral signs (Figure 4.2.B/2,3,5) shall be transliterated with a corresponding Arabic numeral, with spaces separating the number from any adjacent grapheme on either side (including other numerals)

4.2.2. Signs for numbers greater than 9

Historic Indic writing systems often use simplex glyphs for multiples of 10 and 100, as in Figure 4.2.B. Some of these glyphs are (or may arguably viewed as being) composed of two numeric graphs (for the respective power of 10 and a multiplier). For our purposes, this is irrelevant, and these glyphs shall be transliterated according to their final value.[94] Each source glyph representing an integer greater than 9 shall be transliterated with the required number of Arabic digits, separated with spaces from any adjacent grapheme on either side (including other numerals).

Figure 4.2.B. Numbers greater than 9

1	2	3	4	5
				
10+	80+ 2	200+ 80+ 2	100+ 20+	1000+ 4 100+ 4 10+
10	82	282	120	1440

- the fact that these digits together represent a single source grapheme must be indicated in XML markup
- outside XML contexts, the + sign may be added after the last Arabic numeral as private shorthand to make the demarcation clearer
  - although the use of the + sign is inspired by the conventional transliteration of sign-value ciphers in additive notation, in this notation it does not signify addition, and its use is not identical to that conventional transliteration, as the examples below illustrate
  - in Figure 4.2.B/1, |10|, i.e. the number ten written with a single glyph, such as Brahmi |𑁒| → 10+
    - note that the + is present even though nothing is added to this 10
  - in Figure 4.2.B/2, |80| |2|, i.e. the number eighty-two written with a glyph denoting eighty and one denoting two, such as Brahmi |𑁓𑁦| → 80+ 2
  - in Figure 4.2.B/3, |200| |80| |2|, i.e. the number two hundred and eighty-two written with glyphs respectively denoting two hundred, eighty, two → 200+ 80+ 2
  - in Figure 4.2.B/4, |100| |20|, i.e. the number one hundred and twenty written with a glyph denoting one hundred and one denoting twenty → 100+ 20+

[94] This does not apply to Tamil number notation, where multiplier glyphs can be placed before glyphs for powers of 10, as in Figure 4.2.B/5, but these are visually separate.

  - note that the + is present after the 20 as well as after the 100
- |80| |10|, i.e. the number ninety written (unusually) with a glyph denoting eighty and one denoting ten, such as Brahmi |𑀪𑁂| → 80+ 10+
  - note that the + must be present after the 10 as well as after the 80
- in Figure 4.2.B/5, |1000| |4| |100| |4| |10|, i.e. the number one thousand four hundred and forty written in Tamil notation with glyphs respectively denoting one thousand, four [times], one hundred, four [times], ten → 1000+ 4 100+ 4 10+
  - note that the + is present after 100, 100 and 10, but not after the instances of 4

4.2.3. Numbers denoted by bars

Cambodian inscriptions may denote numbers by groups of vertical bars. This notation shall be transliterated by an identical number of *I* (uppercase i) characters.
- XML markup must be added to disambiguate this transliteration from the phonographic grapheme <I> and to demarcate groups of target graphemes as a single source grapheme
  - this applies even when there is only a single bar for the number 1
- outside XML contexts, the + sign may be added after the last *I* as private shorthand to make the demarcation clearer
  - in Figure 4.2.C, three vertical bars denoting the number three → III+

Figure 4.2.C. Number bars



III+
3

4.2.4. Fraction signs

Source glyphs denoting fractional numbers shall be transliterated as common fractions, i.e. with the numerator and denominator represented in Arabic digits and separated by a fraction slash, without any intervening spaces.
- because a single source grapheme is represented by more than one transliterated grapheme, XML markup must be added to demarcate the source graphemes
- outside XML contexts, the + sign may be added after the last Arabic numeral as private shorthand to make the demarcation clearer, for example
  - one half → 1/2+, as in Figure 4.2.D/1
    - double-barred allographs of the cross-shaped Khmer fraction sign, as in Figure 4.2.D/2, are not to be distinguished in transliteration from the single-barred form
  - one eighth → 1/8+
  - one fortieth → 1/40+ as in the Tamil fraction sign of Figure 4.2.D/1

Figure 4.2.D. Numbers greater than 9

1	2	3
	 	
1/2+	1/2+	1/40+

4.2.4.1. Vulgar fraction signs

The Unicode codetable provides for a number of dedicated “vulgar fraction signs,” some of which enjoy wide font support, while others are absent from most fonts. At your discretion, these signs may be used as informal alternatives to multicharacter fractions, especially in texts prepared for print publication.
- the following vulgar fraction signs are quite widely supported and are probably safe for most ecosystems
  - one half → ½ (U+00BD Vulgar Fraction One Half)
  - one third → ⅓ (U+2153 Vulgar Fraction One Third)
  - two thirds → ⅔ (U+2154 Vulgar Fraction Two Thirds)
  - one fourth → ¼ (U+00BC Vulgar Fraction One Quarter)
  - three fourths → ¾ (U+00BE Vulgar Fraction Three Quarters)

4.3. Miscellaneous core graphemes

For convenience's sake, our core graphemes (§3.1.1) include some sundry signs, namely the honorary graphemes (§2.7.2) recognised in transliteration for practical reasons, and a small set of marks (§2.3.3), with clearly established graphemic functions. The graphetic appearance of these signs may vary from script to script, but since their graphemic function is unequivocal, in transliteration we represent only that function, by a dedicated target character. Their graphetic details, when of palaeographic interest, can be described for human readers outside the edition.

4.3.1. *Avagraha*

The ***avagraha*** (§4.3.1) is a special grapheme present in some varieties of the Indic writing system, whose function is to help the identification of morphemes that would be otherwise homographic (§2.5.2). A sign identified in the source as an *avagraha* must always be transliterated. *Avagraha*s not present in the source but expected in modern standard spelling may be supplied by the editor, marked up in XML as editorial (§4.3.1.1).

- use the transliteration ’ (U+2019 Right Single Quotation Mark) to represent original *avagraha*
  - e.g. Devanagari |ऽ| → ’
  - if necessary, ' (U+0027 Apostrophe), which is accessible on most keyboards, may be used as an informal alternative
- original *avagraha*s must be transliterated regardless of whether their usage conforms to modern conventions
  - when it does not, XML markup may be used to normalise the spelling

4.3.1.1. Editorial *avagraha*

According to modern spelling convention, the *avagraha* is used in Sanskrit when an initial short /a/ is elided due to sandhi after a final /e/ or /o/. Where an *avagraha* is expected according to this convention, but none is present in the source, it may be supplied by the editor in order to clarify the interpretation of the edited text. Editorial *avagraha*s in principle require XML markup, but since the texts in our scope hardly ever contain an original *avagraha*, we choose to recognise all apostrophes in the transliterated text as editorial. Thus, in essence, the apostrophe is optional shorthand for encoding as a supplied *avagraha*. The use of the apostrophe for certain kinds of sandhi analysis (§7.2.2) is superficially similar to the supplying of *avagraha*, but essentially different because the *avagraha* is an actual grapheme of the Indic writing system that may or may not be present in the source and may be supplied by the editor when it is not (so the corresponding apostrophe in the transliteration is a grapheme that implies markup), while sandhi analysis is purely interpretive (so the corresponding apostrophe in the transliteration is only markup).

- when to supply *avagraha*
  - supply *avagraha* only to indicate the elision of initial <a> after <e> or <o> in sandhi
    - never supply *avagraha* to indicate the merging of initial <a> or <ā> with a preceding <a> or <ā>, as is occasionally done in Devanagari texts
  - supplying *avagraha* is recommended in general, and especially in cases where the received text would be meaningful (and often contradictory in meaning) both with and without an *avagraha*
    - for example, the inscribed sequence *sohataḥ* may stand for *so hataḥ* (“he was killed”) or *so ’hataḥ* (i.e. *saḥ ahataḥ*, “he was not killed”), so if you interpret the text as the latter, then supply an *avagraha* to make this clear
    - if you feel that such ambiguity is a deliberate poetic device (bitextuality, *śleṣa*), then the decision whether or not to supply an *avagraha* should be based on what you consider to be the prima facie meaning of the text
  - the recommendation of supplying *avagraha* applies likewise to words in compound, whether hyphenated (§7.3.5) or not
    - e.g. *yaśo’mṛtam* or *yaśo-’mṛtam* and *saro’nte* or *saro-’nte* are preferable to *yaśomṛtam* or *yaśo-mṛtam* and *saronte* or *saro-nte*
- how to distinguish supplied *avagraha*s from original ones

  - in XML editions, proper encoding is strongly recommended for all supplied *avagraha*s and mandatory when a single text contains both supplied and original *avagraha*s
  - when working outside an XML context, distinction between original and supplied *avagraha* may (always) or must (when both occur within a text) be made through private shorthand as follows
    - either +' for editorial *avagraha*s
    - or '! for original *avagraha*s
    - or both
- if you are converting a previously prepared text to XML markup and want to use batch search and replace to add encoding to editorial *avagraha*s, take care not to do the same for apostrophes in any other function, such as Dravidian elision sandhi (§7.2.2)
  - the elision apostrophe is always followed by a space, while the *avagrha* never is

4.3.2. Abbreviation marks

The **abbreviation mark** found in some varieties of the Indic writing system (such as Devanagari |∘|, also called the *lāghava* sign) is used in conjunction with a string of phonographic graphemes (which may or may not be otherwise meaningful) to indicate that they comprise an abbreviation standing for a word that is not fully written. The mark on its own does not constitute an abbreviation, so it is not an ideogram. We are not aware of any occurrence of such a sign in our corpus so far, but in order to provide for the eventuality, we propose the conventional transliteration with the degree sign, even though this results in some ambiguity with the use of the same sign for editorial truncation (§7.4).

- a sign confidently identified as an abbreviation mark, regardless of its graphetic appearance, shall be transliterated as ° (U+00B0 Degree Sign)
  - when a sign identification as an abbreviation mark is doubtful, it is preferable to transliterate or encode it as an abstract symbol (§5.5) and to annotate its possible function separately
- note that we use the same target grapheme as an indicator of editorial truncation (§7.4) in fragments of words extracted verbatim from their context
  - the possibility of conflict between these two practices is meagre because original abbreviation marks do not occur in our texts, and because editorial truncation is never used inside an edition (but only in excerpts, such as apparatus lemmas)
  - however, should original abbreviation marks occur in a future text, any instances of such a mark occurring outside the edition proper (e.g. in a lemma) will need to be disambiguated
    - we recommend adding XML encoding for original graphemes to the abbreviation mark

4.3.3. Honorary graphemes

In order to provide an accurate rendition of certain features of specific source writing systems without requiring an inordinate number of dedicated target characters, some items which, from a theoretical perspective, best qualify as diacritical marks (§2.4.4.1) are recognised in transliteration as honorary graphemes (§2.7.2).

4.3.3.1. The |ā| graph as a signifier of length in maritime Southeast Asia

In some Southeast Asian writing systems, we see the evolution of the original vowel marker for the grapheme <ā> (Javanese *tarung* |◌ꦴ|, Balinese *tedong* |◌ᬵ|) into a diacritical mark signifying length in association with any vowel,[95] and even with consonants, as shown in Figure 4.3.A. By default, we normally transliterate graphemes according to their evolutionarily earlier and/or more universal signification (§2.7.1), but since an innovative graphemic function is evidently present in these writing systems, we prefer to differentiate this from the generic <ā>. Since we also wish transliteration to distinguish long vowel signs involving the length

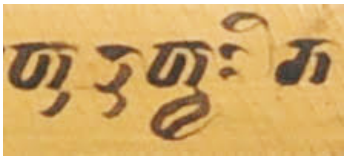
[95] This appears to be a straightforward generalisation of the phenomenon that in most Indic writing systems the independent |Ā| graph is distinguished from independent |A| by the addition of a stroke similar or identical to dependent |ā|, as in Devanagari |आ| comprised of the elements |अ| and |◌ा|.

mark from traditional simplex long vowel signs, we recognise the length mark as an honorary grapheme (§2.7.2).

Taking advantage of the facts that the colon is widely used in transcription as an indicator of length and that the disambiguation colon of ISO 15919 (§3.3.1.1) is hardly ever needed in our transliteration scheme, we dedicate the colon to the transliteration of this honorary grapheme. This is a slight inconsistency in the transliteration system, which we deem acceptable when weighed against the gain of providing easy keyboard access and intuitive interpretability for the length mark. If in any text or corpus an ambiguity should arise (i.e. both the disambiguation colon and the length mark colon need to be present), use the IPA triangular colon : (U+02D0 Modifier Letter Triangular Colon) for the length mark.

In loose transliteration, transcribe the pronunciation instead of using a colon.

Figure 4.3.A. The |ā| graph as a signifier of length

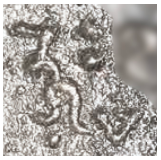
1	2	3
		
qə:bni pilaṁ	*gnəp:ipitu*	*turut:vaḥna*

- when the graph originally denoting <ā> is associated with another vowel marker to transform the latter into a long vowel, add a colon (:) after the transliterated short vowel to represent the length mark
  - e.g. Figure 4.3.A/1 → *qə:bni pilaṁ* (see also §4.1.7.2 about the transliteration of the vowel support)
  - for the sake of consistency, in writing systems which use the <ā> graph systematically as a length marker, this graph shall be transliterated as *a:* when used without another vowel marker to represent the phoneme /ā/ (see Figure 4.4.G for an illustration)
- when the graph originally denoting <ā> is attached to an *akṣara* to indicate the lengthening (doubling, gemination) of the consonant component of that *akṣara*, add a colon (:) after the transliterated consonant to which the doubling pertains
  - e.g. Figure 4.3.A/2 → *gnəp:ipitu* (loose transliteration *gənəp pipitu*, Old Sundanese "fully seven")
  - the colon shall be next to the transliterated consonant even if it is not adjacent in the original
    - e.g. Figure 4.3.A/3 → *turut:vaḥna* (loose transliteration *turut tvaḥna*, "you should follow his behaviour")

4.3.3.2. Dots below *akṣara*s

The Mon, Burmese and Pyu writing systems use a dot placed below an *akṣara* as a diacritical mark, as in Figure 4.3.B. In our transliteration practice we treat this mark, with a vague graphemic function, as an honorary grapheme (§2.7.2) and dedicate a distinct transliteration equivalent to it.

- underdot in Mon, Burmese and Pyu → *ṃ*
  - U+1E43 Latin Small Letter M with Dot Below
  - see also §4.4.6 about the transliteration sequence of Pyu syllabic *akṣara*s
- should it be desired to recognise the Devanagari *nuqtā* and related diacritical marks (e.g. the Gurmukhi *pairī bindī*) as honorary graphemes, the same transliteration equivalent may be used for these
  - but since these marks affect a consonant, the transliteration character should be placed immediately after the relevant transliterated consonant

Figure 4.3.B. Underdots in mainland Southeast Asia		
1	2	3
		
Burmese *v·ṃ*	Pyu *gauṃ*	Pyu *tgaMṃḥ*

4.4. Graphotactic peculiarities

As noted in §2.6.1, transliteration is blind to *akṣara* composition, but a reader informed by knowledge of the graphotactic rules of the Indic writing system will be able to reconstruct the boundaries of source *akṣara*s on the basis of the transliterated graphemes. However, some specific varieties of the Indic writing system modify the default graphotactic rules. For certain systematic modifications, we recognise alternative graphotactic modes (§3.2.4), which affect the transliteration and the reconstruction of the source from the transliterated text.

Additionally, in all situations where *akṣara*s are composed idiosyncratically, i.e. in a manner that would not be expected (on the basis of the default mode or the alternative mode where applicable), the = (equals) sign may be used as optional shorthand to indicate that the two adjacent target graphemes belong to the same *akṣara* in the source. The use of this shorthand is not mandatory, but recommended in many situations. In XML editions it can, but does not have to, be either replaced or supplemented with markup indicating *akṣara* boundaries. When the character joiner sign is placed between two graphemes which are separated by an editorial space or hyphen (§7.3), always put the = sign first, and the space or hyphen second.

4.4.1. Vowelless consonants in Tamil: the "implicit *puḷḷi*"

Tamil inscriptions often use the regular consonant forms (which would normally imply the presence of the inherent vowel) for writing the non-final members of consonant clusters, as in Figure 4.4.A/1. Because this practice is widely prevalent while conjunct *akṣara*s (ligatures) are rare in Tamil, we declare it to be the graphotactic mode for Tamil written in Tamil script (excluding Grantha).[96] That is to say, whenever a consonant cluster is present in a DHARMA-transliterated Tamil text, it is to be understood that the non-final consonants of that cluster were written in the source with the regular (*akṣara*-constituting) consonant glyphs. As a consequence of this decision, any cases where a consonant cluster is written differently in the source must be made explicit in the transliteration. The possibilities for writing a consonant cluster in Tamil are illustrated in Figure 4.4.A. In the transliteration of consonant clusters written with a ligature, it is recommended that you use the optional character joiner sign = to indicate that the graphemes in question belong to a single *akṣara*.

Figure 4.4.A. Consonant clusters in Tamil		
1	2	3
நந கக	ந்ந க்க	ஶ ஶ
nna, kka	*n·na, k·ka*	*n=na, k=ka*

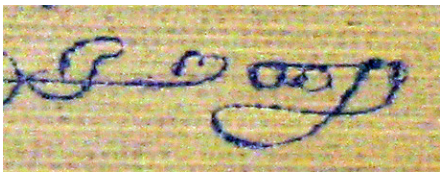
[96] In future projects, similar alternative defaults may be declared for other languages and writing systems as deemed necessary, such as the writing system of the Bhaṭṭiprōḷu inscriptions (Salomon 1998, 35).

– where *puḷḷi* is not present in the source but is to be understood implicitly, as in Figure 4.4.A/1, simply transliterate the consonant cluster without any additional characters
  – e.g. |ததை| (understood as த்த /tta/) → *tta*
– when *puḷḷi* is present in the source, as in Figure 4.4.A/2, always transliterate it explicitly like any other *virāma* (§4.1.2.3)
– where a Tamil text written in Tamil script employs a ligature, as in Figure 4.4.A/3
  – use the = sign as optional shorthand between the corresponding transliterated consonants to distinguish the ligature from the script's default method of writing conjunct consonants as two glyphs with an explicit or implicit zero vowel marker, e.g.
    – *n=na* (as distinct from |ந்ந| *n·na* and |நந| *nna*)
    – *k=ka* (as distinct from |க்க| *k·ka* and |ககை| *kka*)
  – Tamil ligatures should be made explicit in this manner whenever feasible
  – however, the = sign must never be used in ligatures of a writing system where conjoining is the default method of representing consonant clusters, and this includes Tamil written in Grantha

4.4.2. Tamil ligatures (*kūṭṭeḻuttu*) with subjoined M

Tamil manuscripts sometimes join a graph for final M (based on Grantha) to the lower part of the preceding *akṣara*. The resulting complex glyph is more in the nature of a typographic ligature (§2.5.1.2) than an *akṣara* proper as we define it in §2.6.1, although it is graphotactically analogous to the latter (as well as to the Pyu syllabic *akṣara*, §4.4.6).

Figure 4.4.B. Tamil subjoined |M|



caraṇa=M

– Tamil final |M| subjoined to the preceding *akṣara* as in Figure 4.4.B may be transliterated normally, ignoring the graphotactic peculiarity, e.g. *caraṇaM*
– at your discretion, you may use the = sign as optional shorthand before the subjoined M to indicate that it forms a graphotactic unit with the preceding character, e.g. *caraṇa=M*

4.4.3. Syllable-final *r* in Indonesian writing systems

In most varieties of the Indic writing system, the superscript |r| (e.g. in Devanagari |र्क| = |rka|) is a graphotactic allograph (§2.7) used when the grapheme <r> is to be read before the body consonant of a complex *akṣara*. Indonesian writing systems, however, often employ the superscript graph (called *layar* or *surang*) in an alternate mode in order to cater for closed syllables, where the /r/ phoneme represented by it is to be pronounced at the end of the sequence corresponding to the *akṣara* with the superscript r, as in |ஸர்| = |samar| and in Figure 4.4.B. We recognise this as an alternative graphotactic mode applicable to Indonesian texts, which can, however, also employ the Indian mode where the subscript |r| is to be interpreted in the traditional way, as in |ஸர்ப| = |sarva|.

Figure 4.4.C. Indonesian superscript |r|



Ina=rpaṇakan·

If a given text uses only one of the two modes, then it is sufficient to note this in the text's palaeographic description. If both modes are present within a single text, then you must declare one of the modes (the dominant one) to be the default for the text and note the exceptions which use the other mode. In addition, you can use the character joiner sign = as optional shorthand before or after the transliterated <r> to indicate that it is graphetically dependent on the preceding or the following *akṣara*. This is recommended for the non-dominant mode in a text that uses both modes, and may (for the sake of explicitness) be used with every instance of a superscript |r| in any text where alternates modes might be applicable.

– if the Indonesian mode is dominant in a text,
  – preferably add = to instances of the Indian mode
    – e.g. ஸர்ப → *sar=va*

  - for maximum precision, optionally also add = to instances of the Indonesian mode
    - e.g. ᨾᨿᩮᩬ → *sama=r*
- if the Indian mode is dominant in a text,
  - preferably add = to instances of the Indonesian mode
    - e.g. ᨾᨿᩮᩬ → *sama=r*
    - e.g. Figure 4.4.B → *Ina=rpaṇakan·*
  - for maximum precision, optionally also add = to instances of the Indian mode
    - e.g. ᨾᨻᩮᩬ → *sar=va*

4.4.4. Independent vowel graphs forming conjuncts

In some Southeast Asian varieties of the Indic writing system, some of the graphs which traditionally represent independent vowels can combine into conjuncts (§2.4.3.2) with consonantal graphs. Situations where complex graphs representing independent vowels (i.e. combinations involving a vowel support) form ligatures with consonants have already been discussed under §4.1.7.2 and are not relevant here, since the vowel support graph in that case represents a consonantal grapheme. This section is concerned with simplex vowel graphs which are normally standalone glyphs, but which form ligatures in certain specific writing systems, as in Figure 4.4.D.

Figure 4.4.D. Independent vowel graphs in ligatures

1	2
	
maṟk= ṟmpva	*Umiṅsor= I*

Our transliteration scheme represents independent vowel glyphs with uppercase Latin characters (§4.1.3). The same transliteration is to be used for the corresponding graphs when they enter into ligatures. However, since this would imply that the graphs transliterated in this manner are independent glyphs (§2.4.1), it is preferable in this case to use character joiner sign = as optional shorthand, indicating that they form a complex glyph with an adjacent grapheme.

- where an independent vowel graph is combined with a regular consonant graph or a superscript |r| graph to form a complex glyph
  - transliterate the vowel in uppercase as usual
  - preferably add the = sign between the consonant and the vowel sign to indicate that the two belong to the same *akṣara*
  - e.g. Figure 4.4.D/1 → *maṟk= ṟmpva* (pronounced /marək rəmpva/)
  - e.g. Figure 4.4.D/2 → *Umiṅsor= I*

4.4.5. Multiple vowel markers within a complex glyph

Some specific writing systems use a combination of two vowel markers in a systematic manner, as in the grapheme <ui> (§4.1.7.2) and where the graph for |ā| is used in conjunction with another vowel marker to signify vowel length (§4.3.3.1). Multiple vowel markers, typically also |u| and |i|, are also sometimes used for marking characters as deleted, i.e. as a metamark which shall not be transliterated (§3.1.4.3).

Any other instance of more than one vowel marker in an *akṣara* is likely to be a result of non-standard orthography or simple scribal error (e.g. intended scribal correction from one marker to the other, without deletion of the former). XML markup is available for normalising non-standard usage, correcting scribal error, and encoding scribal correction. Since our transliteration system employs lowercase Latin vowels for source vowels manifesting as markers (and uppercase for those materialising as independent vowel signs), the transliteration of the received text in itself indicates unambiguously when more than one vowel marker is present in a source *akṣara*. Therefore, simply transliterate both vowel markers, in an arbitrary order that seems most plausible. Nonetheless, when deemed desirable, such occurrences in the source can be made explicit by adding joiner sign = as optional shorthand between the two transliterated vowels. For example:

– Figure 4.4.E/1 → *du=ā*
  – this instance is probably a scribal mistake for an intended *ddhā*, which can be encoded as editorial correction
– Figure 4.4.E/1 → *mr̥=i*
  – the simultaneous use of the vowel markers |r̥| and |i| (as well as of subscript |r| and the |i| marker) to represent the sonant /r̥/ is not uncommon, so we prefer to perceive it as non-standard orthography (rather than a scribal mistake), and encode its normalisation

Figure 4.4.E. Multiple vowel markers

1	2
	
du=ā	*mr̥=i*

4.4.6. The syllabic *akṣara* in Pyu

The *akṣara* in the Pyu writing system often consists of a larger number of components than in the generic Indic system. The function of some of these components is not confidently understood and some may be properly classified as elements (§2.4.4) rather than components. Supralinear dots, resembling an *anusvāra* (which may also be to the right of the body, aligned vertically with the double dot as in Figure 4.4.F/2), are transliterated as *ṁ* (§4.1.4); sublinear dots as *ṃ* (§4.3.3.2); double dots resembling a *visarga* as *ḥ* (§4.1.5); and diminutive sublinear consonants as uppercase consonant letters (§4.1.2.2). All of these, including the "independent" consonant graphs, are considered to belong to a single *akṣara* in the graphotactic strategy of Pyu, which incorporates innovations for the representation of closed syllables. In transliteration, Pyu *akṣara*s shall normally be separated by spaces, so the character joiner sign = is not needed for any of them. However, because of the complexity of these characters, care must be taken always to transliterate the components in a fixed order, in accordance with the practice of Griffiths et al. (2017).

Figure 4.4.F. The Pyu *akṣara*

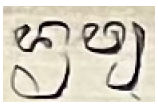
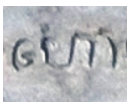
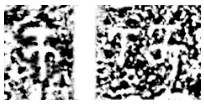
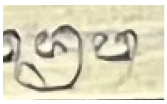
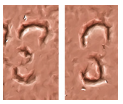
1	2
	
tvaNṃṁḥ	*tgiMṃṁḥ*

– the components of a Pyu *akṣara* shall be transliterated in the following sequence: consonant(s) – vowel marker – sublinear consonant – underdot – overdot – visarga, e.g.
  – Figure 4.4.F/1 → *tvaNṃṁḥ*
  – Figure 4.4.F/2 → *tgiMṃṁḥ*

4.4.7. Glyphs or graphs split by an intervening feature

As discussed in §3.3.1.2, we dedicate special placeholder characters to orphaned glyph elements which are not in themselves graphemic but may be graphetically separated from their parent *akṣara* in some circumstances. The use of these placeholder characters is optional, but strongly recommended in all editions created with access to the original or a surrogate. Some such instances are illustrated in Figure 4.4.G, and the rules for using these placeholders are given below. The upper rows of the figure demonstrate integral glyphs, while the lower rows show similar or identical glyphs interrupted by a feature. In the figure and the instructions, <> represents any kind of interruption, which will be represented by encoding in an XML edition.

Figure 4.4.G. Split glyphs and graphs

1	2	3	4	5	6
					
malaṁka	*kācu*	*dṛvya*	*sa*	*ke*	*ho*
					
mala<>ṁka	*k<>ācu*	*dṛv<>ya:*	*sa<>⌉*	*⌈<>ke*	*A⌈<>ho*

- the placeholders for orphaned elements and element groups are as follows
  - ⌈ (left ceiling, U+2308) to represent a prescript element split off from the following glyph
  - ⌉ (right ceiling, U+2309) to represent a postscript element split off from the preceding glyph
  - if you have difficulty entering these characters, you may use private shorthand such as [[and]]
- the placeholders are not necessary and **should not be used** when the orphaned part is a component rather than an element, i.e. it corresponds to a grapheme which can be represented separately in the transliteration, and the place of this component in the graphetic sequence corresponds to the place of the grapheme in the sequence of graphemes, for example
  - when an *anusvāra* is split off from its parent *akṣara*, as in Figure 4.4.G/1, *mala<>ṁka*
  - when a complete vowel marker is split off from its parent *akṣara*, as in Figure 4.4.G/2, *k<>ācu*
    - similarly: ಕ<>◌ಾ → *k<>ā* (ignore the Unicode dotted circle)
  - when a complete dependent consonant graph is split off from its parent *akṣara*, as in Figure 4.4.G/3, *dṛv<>ya:* (see also §4.3.3.1 about the transliteration *a:*)
  - when the splitting of an *akṣara* is not due to the deliberate action of the scribe but to a fracture in the support (in which case the entire *akṣara* should be transliterated arbitrarily on one side of the fracture)
- the placeholders **should be used** in the following circumstances
  - when a single consonant graph is separated into two parts, as in Figure 4.4.G/4, *sa<>⌉*
  - when an orphaned vowel marker precedes the rest of the *akṣara* in the graphetic sequence but follows it in the sequence of graphemes, as in Figure 4.4.G/5, *⌈<>ke*
    - similarly: ◌ೊ<>ಕ → *⌈<>ke* (ignore the Unicode dotted circle)
  - when only part of a vowel marker is orphaned, as in Figure 4.4.G/6, *A⌈<>ho*
    - similarly: ◌ೊಕ<>◌ಾ → *ko<>⌉*; ◌ೊ◌<>ಕಾ → *⌈<>ko* (ignore the Unicode dotted circle)
- rules of thumb for determining **which target graphemes should appear on which side** of the interruption are as follows
  - by default, put all of the transliterated characters belonging to the split original character on that side of the interruption where the consonant body is located, and add only the applicable placeholder character on the other side of the interruption
  - if the split entity is a consonant graph without a vowel marker (Figure 4.4.G/4), then the consonant and its inherent vowel shall be transliterated before the interruption
  - in more complex cases (e.g. if a split consonant graph is accompanied by a vowel marker and/or a dependent consonant), put all transliterated graphemes on the side where there is more of the original glyph, and put only the applicable placeholder on the other side of the interruption (even if there is more than a single element on that side of the interruption in the source)

5. Transliterating peripheral graphemes

Peripheral graphemes, informally also called symbols, are graphemes which we represent in transliteration by means of a placeholder grapheme dedicated to one of the basic functional classes listed in §3.1.2. We choose this solution because individuating peripheral graphemes more precisely can be impractical due to the wide variety of their graphetic shapes, the often vague and/or imprecisely understood nature of their graphemic functions, and their prevailing lack of consistent association between shape and function. This section presumes that you have read §3.1 and understand what *demarcation* and *individuation* mean in this context.

Original spaces are also peripheral graphemes of sorts and are also handled in transliteration by means of a placeholder character. For this reason they are discussed at the end of this section (§5.6), but the encoding of space is different from that of normal peripheral graphemes, and arbitrary placeholders shall not be used for space.

5.1. Representing peripheral graphemes in transliterated text

As an essential minimum of representation, we use a narrow range of default placeholders, with one placeholder dedicated to each functional class of peripheral grapheme (ideogram, punctuation mark, connector and abstract symbol). More granular individuation of peripheral graphemes, when possible and desired, can be optionally achieved as summarised in Figure 5.1.A: through XML markup (§5.1.1) and, especially in contexts where XML markup is not applicable, through arbitrary placeholders (§5.1.2).

Figure 5.1.A. Representation of peripheral graphemes through transliteration and encoding

			individuation	
method	**solution**	**demarcation**	**by form**	**by function**
transliteration	default placeholders	clear	—	basic
	arbitrary placeholders	unclear when multi-character	vague	vague, implied
encoding	<g>	clear	—	—
	@type	—	—	basic
	@ref	—	specific	(implied for ideograms)

5.1.1. Encoding symbols

The encoding of symbols in XML, by means of the `<g>` element and its attributes, is described in the Encoding Guide. Here, we introduce it only to clarify the interaction between symbol transliteration and symbol encoding.

The `<g>` element itself serves as a wrapper for transliterated peripheral graphemes. When the transliteration uses an arbitrary placeholder with more than one target grapheme representing a single source grapheme, this wrapper is put around the entire target sequence, thus clarifying the demarcation of the grapheme in the source.

When this element is complemented with the attribute `@type`, the value of this attribute identifies the basic function of the grapheme (ideogram, punctuation mark, connector or abstract symbol) in the same manner as the selection of a default placeholder. Thus, when applied in tandem with the default placeholders, the encoding of `@type` is redundant, with the difference that it is better suited for computer processing.

When the attribute `@ref` is added, its value is a **token**, which can be looked up in our Glyph Taxonomy, a structured canonical list defining a limited but large number of peripheral graphemes either on the basis

of emic names (primarily for ideograms) or on the basis of etic descriptions. The Taxonomy can be expanded as needed in the future, but the granularity of differentiation on the basis of etic shapes should be kept to a reasonable level.

5.1.2. Using arbitrary placeholders

Arbitrary placeholders may be any Unicode character or string of characters. The subsections on the classes of peripheral graphemes contain some suggestions for arbitrary placeholders, but these are not mandatory prescriptions. Arbitrary placeholders can be single characters chosen to resemble the source grapheme's graphetic appearance or to imitate its perceived graphemic function, as well as strings of characters alluding to an emic name of the source grapheme or describing it in etic terms. The latter kind of arbitrary placeholder can resemble, or even be identical to, the symbol tokens (§5.1.1) used in encoding and serve a similar function, but arbitrary placeholders are interpretable only by human readers, while tokens are for computer processing (and interpretable by human readers conversant with our encoding or viewing a processed output), so the two are not interchangeable.

In choosing your own arbitrary placeholders, care must be taken to eliminate ambiguity, so that the placeholders will not be mistaken for the transliteration of other (especially core) graphemes, and if more than one character is used in as an arbitrary placeholder, it must be made clear that these together represent a single source grapheme. In texts for human consumption alone, this can usually be achieved through spacing before and after the placeholders and the addition of distinctive signs before and/or after the placeholder string. In an XML edition or in a text that will later be converted to an XML edition, it is best to stick to single-character arbitrary placeholders, and to use only characters which do not have any other function in our transliteration system. When using dingbat characters as placeholders for peripheral graphemes, it is wise to avoid characters with poor font support, since they are likely to become gibberish when moved to a different software environment.

5.1.3. Working strategies for representing peripheral graphemes

The methods available for the representation of peripheral graphemes leave us with an *embarras de richesse* of optional combinations. Essentially, we envision two basic working strategies.

5.1.3.1. The formal strategy

- in the **formal strategy**, we rely on XML encoding for representing all relevant information about symbols, in a manner that is fully compatible with computer processing
- the role of transliteration is only to supply placeholders (which, redundantly, also carry some information about symbols)
  - only the default placeholders are needed for this purpose, so arbitrary placeholders are acceptable but discouraged

5.1.3.2. The informal strategy

- in the **informal strategy**, we rely on transliteration (arbitrary placeholders) for representing all relevant information about symbols, which is expected to be reasonably understandable to a human reader, but will not necessarily be processable by computer
- XML symbol encoding is unnecessary, but it can be used
  - to demarcate and disambiguate arbitrary placeholders
  - to encode additional information about symbols

5.1.3.3. Mixing strategies

The two strategies can be **mixed**, so there is a smooth continuum between full encoding and no encoding.

- when mixing strategies, we encourage consistency, for example
  - using arbitrary transliteration for some or all symbols, but always complementing it with `<g>`, without attributes, for demarcation and disambiguation

  - using arbitrary transliteration without encoding (or with encoding only for demarcation) for a limited range of symbols which are common and well-known in a given subcorpus, but using full encoding on default placeholders for all other symbols
- the choice of strategy depends on the kind of work you are doing
  - if you work in an XML environment or are preparing an e-text that will eventually be converted to an XML edition, then `<g>` markup is strongly recommended, even if you use it without attributes, only for demarcation and disambiguation
    - strict adherence to the formal strategy is preferred when creating a sophisticated XML edition on the basis of decent visual material
    - a mixed strategy is preferred when creating a basic XML edition, for example when
      - you are working without visual material
      - you are digitising printed editions *en masse*
      - you are simply disinterested in the granular individuation of symbols
  - a purely informal strategy is only recommended when you are creating an electronic text that will not foreseeably be converted to an XML edition
- when mixing strategies, it is important that **you avoid representing conflicting information** through the channels of transliteration and encoding
  - as shown in Figure 5.1.A, both transliteration and encoding can represent information pertaining to both the shape and the function of a peripheral grapheme
  - the two methods can represent information redundantly, they can complement one another, or they can be contradictory
  - the individuation of function through the default placeholders and through encoding by the `@type` of `<g>` carries the exact same kind of information
    - when default placeholders are used in conjunction with encoding, the function encoded in `@type` should be the same as the function associated with the placeholder
    - note that technically, `<g>` without `@type` is equivalent to an abstract symbol, so when a default placeholder for another class is encoded, the appropriate `@type` should be added explicitly
    - in computer processing, only the encoding will be relevant, so conflicting transliteration will be ignored (but this does not mean that you should not avoid it)
  - the formal and implicit functional individuation provided by arbitrary placeholders can overlap to some extent with the information encoded in `@type` and `@ref`, but the two are different in nature (encoding is more precise and more definite), so conflict does not arise between arbitrary placeholders and encoding

5.2. Ideograms

Ideograms are graphemes that signify words, morphemes or abstract concepts (§2.3.3). In theory, they include numerals, but in our practical approach numerals are treated as core graphemes (§3.1.1.3), and this section is only concerned with non-numeral ideograms.

Ideograms tend to manifest as shapes resembling, and sometimes derived from, core graphs. Consequently, they are not prone to being described in terms of patterns; they do, however, typically have emic designations conventionally associated with these signs in their native writing systems. These designations usually identify both the glyph's graphemic function and its graphetic appearance, and are therefore a practical means of individuating ideograms. Ideograms are not always easily distinguishable from core glyphs; rules of thumb for telling them apart are given in §6.2 below.

The default placeholder for ideograms should only be used when an ideographic function can be asserted with fair confidence. Otherwise, even if the same glyph can be used ideographically in other contexts, identification as an abstract symbol is preferable. These cautions apply likewise to symbols typically found at the beginning and end of texts, which are often called "auspicious" and are sometimes equated to *oṁ* or *siddham*.
- the default placeholder for ideograms is @ (at sign)

  - in any DHARMA-transliterated text, this is declared to represent a generic ideogram
  - more precise individuation of the ideogram, when possible and desired, can be achieved through XML encoding or by substituting an arbitrary placeholder for the default one
- recommended arbitrary placeholders for ideograms include
  - Unicode characters for the actual ideogram or a closely related sign, when available in a modern font, e.g.
    - ॐ (U+0950 Devanagari Om)
    - ௵ (U+0BF3 Tamil Day Sign)
    - ౿ (U+0C7F Telugu Sign Tuumu)
  - short labels indicative of the emic name of the ideogram (with or without transliteration diacritics and abbreviated as needed), prefixed with @ to imply ideographic function, e.g.
    - *@om*
    - *@piḷḷaiyārCuḻi*, *@pillSuli*
    - *@tūmu*

5.2.1. Burmese ideograms

The “abbreviations” used in the Burmese writing system are not merely abbreviations (which would mean that only part of a word is written) but graphic contractions and distortions of originally phonographic graphs. As such, they are ideographic in nature and should be transliterated as such. The recommended arbitrary placeholders for them are those suggested by Lammerts and Griffiths (2016, 3) and shown in Figure 5.2.A, but for better consistency with the rest of the transliteration system, the prefix character may be an @ sign instead of an asterisk.

Figure 5.2.A. Arbitrary placeholders for Burmese ideograms

			
**n*	**r*	**r*	**e*

5.3. Punctuation marks

For the purposes of this Guide, we define punctuation marks in a strict sense as graphic signs employed in the original for segmentation into syntactic or metrical units such as such as sentences, clauses, list items and stanzas or their subdivisions. They are thus similar in function to a modern comma, full stop, question mark, exclamation mark, colon or semicolon. Conversely, signs that typically occur at the beginning of end of entire texts or large semantic units (comparable to paragraphs and chapters) are excluded from the scope of punctuation. Punctuation marks help the reader construct linguistic structure out of the unilinear flow of graphemes.

As regards their graphetic appearance, punctuation marks tend to be simple geometric and abstract (non-iconic) shapes, such as the vertical bars, dots and dashes used widely for punctuation in the Indic writing system. They can, however, be more complex, and sometimes highly ornamental, elaborations of such shapes. Some punctuation marks have emic names, but very few of these designations are strongly associated with a consistent shape across the range of specific writing systems we are concerned with. For this reason, the primary way to individuate punctuation marks more precisely is to describe their shape in etic terms. In many specific texts and in some writing traditions, punctuation marks could also be further individuated with respect to their function (e.g. different marks for punctuation at the level of phrases, sentences, hemistichs and stanzas). However, these more specific functions are often vague and inconsistent, especially across the broad range of texts for which we need a uniform transliteration. For this reason, the DHARMA scheme provides no standardised method for individuating punctuation marks more precisely on

the basis of function. As and when desired, future projects may choose to do so by means of XML markup and/or by dedicating additional placeholders to precisely defined subcategories of punctuation.

The default placeholder for punctuation marks should only be used when the linguistic function of punctuation in the above restricted sense can be asserted with fair confidence. Otherwise, even if the same glyph can be used for punctuation in other contexts, identification as an abstract symbol is preferable.

- the default placeholder for punctuation marks is . (dot, period or full stop)
  - in any DHARMA-transliterated text, this is declared to represent a generic punctuation mark
  - more precise individuation of the punctuation mark, when possible and desired, can be achieved through XML encoding or by substituting an arbitrary placeholder for the default one
- recommended arbitrary placeholders for punctuation marks include
  - modern international punctuation marks primarily for individuation by perceived function, e.g.
    - , (comma)
    - ! (exclamation mark)
    - ? (question mark)
  - easily accessible characters with simple shapes for basic individuation by appearance, e.g.
    - | (U+007C Vertical Line) for signs comprised of a single plain vertical bar
    - || (two iterations of U+007C Vertical Line) for signs comprised of a double plain vertical bar
      - alternatively: ‖ (U+2016 Double Vertical Line), so a <g> wrapper is not required
    - / (slash) for signs comprised of a single vertical bar with a hook, crossbar or ornamental addition
    - // (two slashes) for signs comprised of a double vertical bar with a hook, crossbar or ornamental addition
      - alternatively: \\ (U+244A OCR Double Backslash), so a <g> wrapper is not required
    - , (comma) for short, predominantly vertical and often curved strokes at any height
    - ~ (U+223C Tilde Operator) for signs comprised of a single horizontal dash, plain or with ornamentation

5.3.1. Balinese punctuation marks

The often elaborate punctuation marks of Balinese script, as well as comparable punctuation marks and combinations from related writing systems, can be transliterated with the arbitrary placeholders shown in Figure 5.3.A.

Figure 5.3.A. Arbitrary placeholders for Balinese punctuation marks



carik siki	*carik kalih*	*panten/panti*	*pamada*	*carik agung*	*pasalin*	*carik agung*
|	||	~	//	~0~	||0||	//0//

5.3.2. Supplying punctuation

When transliterating a text, punctuation that is not present in the original must never be supplied silently. Any original punctuation must be preserved in transliteration as it is.

In XML editions, supplied punctuation can be marked up as editorial. Supplying editorial punctuation is recommended for syntactic segmentation, but no editorial punctuation marks should be supplied for metrical segmentation, since metrical segments are identified in other parts of the XML markup. Supplied punctuation should normally take the form of the default placeholder . for punctuation marks.

While preparing a text for conversion into an XML edition, you may want to use private shorthand for supplied punctuation, for instance the combination +. which you will later have to convert to the proper encoding.

5.4. Connectors

Connectors are marks (§2.3.3) which signify that a spatial interruption in the script, such as the end of a writeable zone or a physical defect in the surface, is not to be understood as a break in linguistic structure. In Indic writing systems we are aware of two slightly different usages of connectors. In one, similarly to the modern hyphen, a mark indicates that there is no word break at the end of a line. In the other, a part of the writing surface that could not accommodate the text to be written is filled up with one or more marks in order to preserve the continuity of the writing (regardless of whether a word break is present at the spot or not), while the actual written text continues after the interruption.[97]

Connectors, like other marks, tend to be graphetically simple, although some may resemble phonographic graphemes or manifest as complex abstract shapes. Some have emic names, such as the Balinese *pameneng*. Because of their relative rarity, we have little interest in individuating connectors beyond distinguishing them from other functional classes of peripheral graphemes.

The default placeholder for connectors should only be used when this function can be asserted with fair confidence. Otherwise, even if the same glyph can be used as a connector in other contexts, identification as an abstract symbol is preferable.

- the default placeholder for connectors is § (U+00A7 Section Sign)
  - in any DHARMA-transliterated text, this is declared to represent a generic connector
  - more precise individuation of the connector, when possible and desired, can be achieved through XML encoding or by substituting an arbitrary placeholder for the default one
- arbitrary placeholders for connectors may be selected on the basis of graphic similarity

5.5. Abstract symbols

An abstract symbol (§2.3.3) is a grapheme that we do not classify as primary grapheme, an ideogram, a punctuation mark or a connector. Classifying a grapheme as an abstract symbol is not equivalent to asserting that it has no graphemic function, only that you choose not to assert any function for it. This may happen because you are not confident that a particular instance of a particular glyph serves one of the functions recognised in our classification (and prefer to keep options open); or because the work you are doing does not extend to identifying the function of some or all peripheral glyphs and representing it in transliteration.

Graphetically, abstract symbols may be simple geometric shapes, but they often appear as combinations and/or ornamental derivations of simple shapes, as complex geometric shapes, or as abstract iconic patterns. Some have emic names in some traditions, but very few of these emic names are strongly associated with a particular shape across the entire scope of our project. Therefore, when more precise individuation of symbols is desired, this is best done by describing their shape in etic terms. Since generic symbols are, by definition, those for which we do not assert a graphemic function, they cannot be individuated more precisely with respect to function.

- the default placeholder for abstract symbols is * (asterisk)
  - in any DHARMA-transliterated text, this is declared to represent a generic abstract symbol
  - more precise individuation of the symbol, when possible and desired, can be achieved through XML encoding or by substituting an arbitrary placeholder for the default one
- recommended arbitrary placeholders for abstract symbols include
  - Unicode dingbat characters suggestive of the original symbol's shape, e.g.
    - ◊ (U+25CA Lozenge)
    - ◯ (U+25CB White Circle)
    - ◎ (U+25CE Bullseye)
    - ⁜ (U+205C Dotted Cross)

[97] This latter kind of mark was called a space filler in earlier versions of the DHARMA guides. Since both the space-filling and the word-connecting functions occur in our texts and the two cannot always be clearly distinguished, we now recognise both as belonging to the single, broader category of connectors.

  - ✻ (U+273D Heavy Teardrop-Spoked Asterisk)
  - ✼ (U+2748 Heavy Sparkle)
- short labels indicative of the etic shape of the symbol, prefixed with * to imply an abstract symbol, e.g.
  - **circle*
  - **bullseye*
  - **floret*
  - **squiggle*

5.6. Blank space

As noted in §3.1.3, all spaces in our transliterated texts will be considered editorial. Editorial spaces will be discussed in §7.3.4. Conversely, spaces present in the source and deemed significant must always be represented in transliterated texts. In an XML environment this is done through encoding, but for other working contexts we dedicate the placeholder _ (underscore) to significant space in the original. Whether a space in the source is significant is up to your discretion, but we provide some rules of thumb in the following subsections.

5.6.1. Basic spacing

The typical use of space is for the separation of linguistic units. The placeholder _ will be normally understood to represent a basic space which is broad enough to be perceptible as a space but generally not broader than two average character widths.

- small spaces (less than a typical character width) around numerals, punctuation marks and other symbols are generally not significant
- spaces (including small spaces) used in lieu of punctuation (i.e. for semantic segmentation) are generally significant
- the placeholder _ may, if desired, be iterated in non-XML contexts for a rough representation of the size of a space in the source
  - in an XML edition, the size of spaces must be represented only in the encoding

5.6.2. Spacing for layout

When spacing is a feature of a text's layout rather than its linguistic or semantic structure, it is generally not considered graphemic and shall not be represented in transliteration.

- ignore layout spaces in the transliteration, including the following kinds
  - blank space at the left of a right-aligned line or at the right of a left-aligned line
  - spacing between most or all characters of a line justified to the two margins
  - spacing between metrical units when these result in a column-like arrangement for an entire text or section
- layout may be represented by various encoding methods
- when a layout feature is not encoded, you may choose to represent a layout space in the transliteration in spite of the above

5.6.3. Special spaces

Spaces, usually broader than basic spacing, may appear in the original in other functions, such as blanks meant to be filled later on, or stretches of the support skipped by the writing because of a defect or an artificial feature. Such spaces shall be represented in XML encoding. In contexts not involving XML, you may choose to show them in transliteration with the underscore, but they should also be annotated.

6. Dealing with graphemic ambivalence

Since transliteration is based on graphemes, identification of the graphemes of a source text is a precondition to transliterating it. As the graphic signs of the source can be homographic or otherwise ambiguous, this involves an interpretive choice on the part of the editor. This section considers some cases where this choice is not straightforward.

6.1. Identification of homographs

Most varieties of the Indic writing system involve a few homographs and several near-homographs. As noted in §2.5.3, the cloud of allographs surrounding each theoretically distinct graph results in overlaps. The possibility of alternative interpretations for any given graph or glyph increases when the text was not skilfully written, when it is damaged, or when the available visual documentation is sub-par, and the likelihood of correctly resolving ambiguities decreases when the language is not fully understood, and when the text is short and provides few or no additional instances of the problematic graphs that could be compared. As the editor of the text, you must choose the preferred interpretation, informed by the context and your knowledge of the writing system, the language and the textual genre. Confident interpretive choices shall be made silently. Else, uncertainty may be marked up in XML and/or annotated.

Choosing the interpretation most appropriate to the context can involve extending the benefit of doubt to the scribe or engraver. If a sign seems to be inappropriate to its context but could be plausibly interpreted as an allograph (even an atypical allograph) of an appropriate sign, then feel free to assume that it is the contextually correct sign. However, scribes and engravers do make mistakes and write similar but contextually inappropriate signs, so such cases must be judged individually based on your discretion.

6.2. Core glyph or ideogram?

As noted in §5.2, the graphic shape of some ideograms is derived from core glyphs. In their original function, these glyphs may constitute the morpheme represented by the ideogram (or a part thereof), as is the case with signs for syllable *oṁ* and with the Burmese "abbreviation" signs (§5.2.1). In other cases, there is little or no apparent semantic connection between the glyph's core signification and its ideographic use, such as when a glyph originally signifying the number 1 is employed as an auspicious symbol at the beginning of a text, or one originally signifying the *akṣara* <tha> or <cha> is used similarly at the end of a text.

Occasionally, it may be difficult to tell whether you are dealing with a core glyph or an ideographic one, but a decision must be made before it can be represented in transliteration. Our general rule for such cases is to prioritise the glyph's original signification (§2.7.1) unless a recognisably different graphetic shape is consistently associated with an ideographic function.

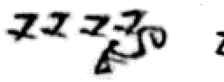
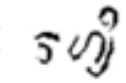
- regular phonographic and numeral graphs of the script of the surrounding text, including ornamental renditions, are to be transliterated normally and not treated any differently from regular text
  - regular signs of another script that could plausibly be used at the time and place associated with the text are also best recognised as phonographic
    - the script difference may be annotated
- signs created by combining regular graphs of the script in a manner that does not follow the writing system's graphotactic rules, shall normally be recognised as ideographic
  - e.g. the sign |ஓ| as opposed to |ஓம்| = |om| in regular Tamil *akṣara*s
- signs which are clearly different from the script of the text, shall normally be recognised as ideographic
  - e.g. the sign |ॐ| as opposed to Devanagari |ओं| = |oṁ|
  - "clearly different" typically involves either or both of the following
    - a palaeographically much older shape, especially one from a script that is not also used for regular text in the relevant writing tradition
    - a stylised and/or graphically simplified shape

6.3. Secondary graphemic functions

Some signs in certain varieties of the Indic writing system have taken on a secondary signification as a context-dependent alternative to their primary signification shared in all varieties of the Indic system. Situations where an identical graphic representation corresponds to several different significations might be interpreted as homography (§2.5.3), where two distinct graphemes incidentally manifest in the same shapes. However, in accordance with our preference to adhere to default interpretations (§2.7.1), in these cases we only recognise a context-dependent secondary function for what we see as a single grapheme. Therefore the transliteration of such signs uses the primary graphemic value that is universal to Indic writing systems, disregarding the innovative value that may be applicable in a given context. Nonetheless, in loose transliteration we prioritise the phonetic value applicable in any given context, i.e. actually transcribe such graphemes rather than transliterating them. In XML editions, moreover, the transliterated primary signification may be normalised through computer markup to the transcribed secondary signification.

Examples of such innovative repurposing in our purview come from maritime Southeast Asia, chiefly from Old Sundanese (Figure 6.3.A). Some of these also affect the way in which an *akṣara* is mapped to a sequence of speech sounds, but this does not affect our treatment. Analogous phenomena in other specific writing systems should be handled in the same way.

Figure 6.3.A. Repurposed graphic signs

	1	2	3	4
				
strict	*sasṭā*	*di jə2niṁ vavaṁṅun:an·*	*ku nu rye*	*rahyiṁ*
loose	*sastra*	*di jəroniṅ vavaṅunan*	*ku nu reya*	*rahiyaṅ*

- when the graphs |  | = |ṭā| are used in Old Sundanese to represent the phonemes /tra/
  - use *ṭā* in strict transliteration and *tra* in loose transliteration
  - e.g. Figure 6.3.A/1 → *sasṭā*; loose transliteration: *sastra*
- when the numeral sign |2| is used in Old Sundanese to represent the phonemes /ro/ (presumably because the word for the number 2 is *ro* or *roro* in Javanese, whose writing tradition heavily influenced Sundanese)
  - use *2* in strict transliteration and *ro* in loose transliteration
    - since the target grapheme <2> does not represent a number in this case, the XML markup for numbers must not be used
  - e.g. Figure 6.3.A/2 → *di jə2niṁ vavaṁṅun:an·* ; loose transliteration: *di jəroniṅ vavaṅunan* ("in the interior of the building")[98]
- when subscript <y> is used in Old Sundanese to make an *akṣara* disyllabic (so that the body consonant and the attached vowel marker are to be pronounced first, followed by /ya/)
  - use *CyV* in strict transliteration and *CVya* in loose transliteration
  - e.g. Figure 6.3.A/3 → *ku nu rye*; loose transliteration: *ku nu reya* ("by many [people]")
  - e.g. Figure 6.3.A/4 → *rahyiṁ*; loose transliteration: *rahiyaṅ*[99]
- when the independent glyph <R̥> is used in Old Javanese for expressing the syllable *rə*
  - use *R̥* in strict transliteration and *rə* in loose transliteration

[98] In the loose transliteration of this example, *ṁ* and *ṁṅ* are both represented by *ṅ* (since both express the phoneme /ŋ/). Simultaneously, *n:* (theoretically denoting /nn/) is simplified to *n*, and *ṁṅ* (theoretically /ṅṅ/) is simplified to *ṅ*, because consonant gemination is not considered to be a phonemic feature of the language, but rather an orthographic peculiarity.

[99] In the loose transliteration of this example, *anusvāra* has been normalised to *ṅ*.

– e.g. *səR̥ḥ* corresponding to *sərəh* in loose transliteration (with the *visarga* also normalised to *h*)
– see also §4.4.4 about *R̥* forming a conjunct like a consonant graph

6.4. The double *kāl* in Tamil

Figure 6.4.A. Double kāl



rā, *r=ā* *ā=r*

Tamil writing sometimes features a fused form of two *kāl* signs, as in Figure 6.4.A, which may signify either of the phonemic sequences /rā/ or /ār/. In our perception this is not a genuine glyph of Tamil script and it does not correspond to a special grapheme; rather, it is a cluster-level graphotactic allograph (§2.5.1.2) of both the grapheme sequences <rā> (the glyph |ரா|) and <ār> (the graphs |ார்|). Accordingly, there is no transliteration equivalent for the double *kāl*. Instead, you must make an interpretive decision as to which grapheme sequence was intended. If ambiguity is present, you may record it in annotation or XML encoding. To indicate the fact that either grapheme sequence was written in the original with a double *kāl*, you may use the character joiner sign = (§3.3.2.1) between the individual graphemes.

– transliterate the preferred sequence as follows
  – for <ār>, add the character joiner sign (and/or the corresponding XML encoding when applicable) to indicate that these graphemes are visually contiguous in original
  – for <rā>, markup is not necessary, since these graphemes would naturally constitute one *akṣara*
    – but you may choose to add the shorthand and/or XML markup nonetheless, to make it explicit that a fused double *kāl* is present
– when the character joiner sign is placed between two graphemes which are separated by an editorial space or hyphen (§7.3), always put the = sign first, and the space or hyphen second

7. Editorial markup in transliteration

DHARMA transliteration is meant to be used in tandem with editorial markup. Depending on your working environment, markup can take place through XML encoding, but even in XML editions we envision including some shorthand (§3.3) in the transliterated text itself. This section provides guidelines for the use of optional (§3.3.2) and essential shorthand (§3.3.1) in DHARMA texts in contexts not already covered in the specific sections above. Keep in mind that human-readable annotation (§3.3.4) is always an option for any feature of the text and its interpretation that is deemed significant but cannot be satisfactorily represented through markup.

7.1. Interpretive alterations in the transliteration

Some of our shorthand notation takes the form of diacritical marks. In this case, the regular target graphemes are replaced by diacritical graphemes which simultaneously represent both a source grapheme and an editorial interpretation associated with it.

7.1.1. Distinction of long *ē* and *ō* from short *e* and *o*

As noted in §4.1.1.2, writing systems of the period we work with very rarely make a graphic distinction between the short and long versions of /e/ and /o/, even when this is a phonemic distinction in the language being written. When the palaeographically primary graph is employed for writing a phoneme known to be long, the rigorous approach would be to transliterate the primary (short) grapheme and optionally add an editorial normalisation in XML. However, since distinct graphs for the long counterparts of these vowels do not, as a rule, occur in the texts in our purview, we have made a projectwide decision to adopt shorthand involving transliteration with a macron. The target graphemes *ē* and *ō* will therefore always be understood in our practice as editorial, equivalent to the XML markup for the normalisation of an originally written short vowel. As a consequence, any graphically distinct long vowels present in the source will have to be marked up in XML explicitly as original. This makes the editing of texts that do make this distinction more cumbersome, but greatly simplifies the editorial markup of the majority of the texts we edit.

- when the phonemes /ē/ and /ō/ are distinctive in a language but are represented by the same grapheme in writing
  - where a long phoneme is expected in the language, use *ē* or *ō* (rather than *e* or *o*) as optional shorthand for these instances of the grapheme
    - in an XML context, you may choose to use encoding for editorial normalisation (of *e* or *o* to *ē* or *ō*) instead of this shorthand
  - where a short phoneme is expected, transliterate the grapheme normally, as *e* or *o*
- if a graphic distinction is present in the source you are transliterating, this should always be explicitly noted in the palaeographic description
  - where the short grapheme is written but the long phoneme is expected, the above optional shorthand may be used
    - alternatively, encode editorial normalisation of the short grapheme to the long one
  - where the short grapheme is written and the short phoneme is expected, transliterate normally, as *e* or *o*
  - but in all other cases, encoding is mandatory in XML contexts:
    - where the long grapheme is written and the long phoneme is expected, encode as original to distinguish it from the shorthand
    - where the long grapheme is written and the short phoneme is expected, encode as editorial normalisation

7.1.2. Vernacular short vowels in Sanskrit loanwords

Where a short vowel is written in place of an otherwise identical long vowel, you may choose to add a breve to the transliterated vowel as shorthand indicating that the vowel in question is the original spelling (and not a typographic mistake in your edition). This practice is recommended for Sanskrit loanwords in Indonesian vernacular documents, following Damais (1955, 15), but need not be applied in such documents if they never make a graphic distinction between short vowels and their long counterparts. In all other writing traditions, we prefer to use only XML markup (editorial normalisation) for similar cases.
– use *ă*, *ĭ* or *ŭ* as optional shorthand when *a*, *i* or *u* is used instead of expected *ā*, *ī* or *ū*

7.2. Sandhi analysis

Since we are concerned with the transliteration of what is written in an original source, sandhi should, as a rule, never be resolved in our texts. Phonographic graphemes must never be added, removed or altered in the process of transliteration. However, when the addition of simple text-based markup can serve to highlight the presence of certain kinds of sandhi, we approve of using hyphens to indicate epenthesis (§7.2.1) and apostrophes to indicate elision (§7.2.2) when such markup is in accordance with the conventions of the subfield. The use of these marks is a form of essential shorthand in our transliteration scheme. Hyphens (§7.3.5) and apostrophes (§4.3.1) are also used for other purposes in our transliteration scheme, so care must be exercised to avoid ambiguity when using them for sandhi analysis.

7.2.1. Epenthetic consonants

When an epenthetic consonant is inserted into certain strings, you may use a hyphen joining that consonant to the preceding grapheme as essential shorthand. The use of this analytical hyphen is optional. The epenthetic consonant is regarded as belonging to the former word, and should be separated from the next word by a space (§7.3.4) or, if in compound, by a hyphen (§7.3.5) or, optionally, nothing.
– examples in Tamil:
  – *I-p-peruṅ-kōyil·* (இப்பெருங்கோயில் *i+perum+kōyil*)
  – *tiru-mēṉi-y āṭa* (திருமேனியாட *tiru+mēṉi + āṭa*)
– example in Kannada: *samavṛtti-y appavōl·* (ಸಮವೃತ್ತಿಯಪ್ಪವೋಲ್ *samavṛtti + appavōl*)
– non-standard Sanskrit sandhi involving an epenthetic *m*, *s* or *r* may be indicated in the same way, e.g.
  – *paṁca-s-triṁśottaratame*
  – *mleccha-rājye-m apūjitaḥ*
– the above hyphenation must not be used in the following cases, for which see also §7.3.2
  – when epenthesis takes place on the phonological level, but its graphic reflection involves the alteration of graphemes rather than the insertion of an additional grapheme (as in Sanskrit *tad dhi* from *tat + hi*), then we have no choice but to treat the altered grapheme as part of the latter word, and an analytical hyphen must not be inserted
  – when in non-standard Sanskrit orthography a single consonant is doubled in conjunction with an *r* (or, occasionally, in other formations), this is not regarded as epenthesis; analytical hyphenation is not applicable, and the doubled consonant is understood to belong with the word to which the single consonant belongs
    – e.g. *tair ggatvā*

7.2.2. Elision of final vowels

The elision of a final vowel, in particular of the "overshort" final *u* of Tamil and short final vowels of other Dravidian languages (*lōpasandhi*), may be indicated by adding an apostrophe in place of the elided vowel as essential shorthand. The use of this analytical apostrophe is optional. The elided vowel is regarded as belonging to the former word, and so the apostrophe should be separated from the next word by a space (§7.3.4) or, if in compound, by a hyphen (§7.3.5). The apostrophe shall not be used if elision takes place inside a compound that you do not segment with hyphens.

The mandatory use of a space or hyphen after this apostrophe also serves as a mechanistic distinction from the apostrophe used for the *avagraha* (§4.3.1). Such a distinction is important because the apostrophe representing elision is a feature of linguistic analysis, whereas the *avagraha* in Sanskrit is an actual grapheme which could conceivably be present in the original, and when absent, it may be supplied by the editor for the sake of normalisation.

- example in Tamil: *arit' en̲ru* (அரிதென்று *aritu + en̲ru*)
- examples in Kannada:
  - *enag' īge* (ಎನಗೀಗೆ *enage + īge*)
  - *aṁt' āgi* (ಅಂತಾಗಿ *aṁtu + āgi*)
- do not use the apostrophe in lexicalised compounds, as well as in any compounds you choose not to segment with hyphens
  - e.g. in Tamil, *koṇṭ-āṭu* (கொண்டாடு *koṇṭu + āṭu*)
- the apostrophe should preferably be ’ (U+2019 Right Single Quotation Mark)
  - if necessary, ' (U+0027 Apostrophe), which is accessible on most keyboards, may be used as an informal alternative

7.3. Editorial segmentation

In addition to transliterating a source text, you will usually want to segment it in one or more ways. Division into larger blocks may be descriptive — on the basis of the physical layout of the original — or interpretive — on the basis of the semantic and/or prosodic structure of the text. Both of these concerns are to be addressed primarily through XML markup, but they are discussed briefly in §7.3.1.

For interpretive analysis on a lower level, namely that of words, we recommend editorial spacing (§7.3.4) of independent words and hyphenation (§7.3.5) of words in compound. The same kind of segmentation could be achieved, in some cases more rigorously, through computer markup, but at present we see no advantage in doing so. In other words, the spaces and hyphens in our texts constitute essential shorthand for interpretive markup. However, because of the ubiquity of such segmentation, using XML for this purpose would greatly increase code clutter and impede working with our files. Basic markup comprised of spaces and hyphens keeps the files more human-readable, and may in the future serve to facilitate the electronic lemmatisation of texts. Furthermore, it not only helps the reader understand the texts you edit, but, in ambiguous cases, expressly communicates your parsing of the text. Finally, it is in many cases advantageous for on-screen display and typesetting.

Since the texts we work with typically use *scriptio continua* — i.e. continuous writing without interword spaces — any and all spaces in our transliterated texts will be considered editorial. The same applies to hyphens — which have no equivalent in the source writing systems[100] — and to new lines. The insertion of editorial segmentation is facilitated by Romanised transliteration, but hampered in some cases by vowel sandhi. We discuss the consequences of both in §7.3.2. We also provide some general advice on when to use editorial spaces and hyphens and when not (§7.3.6). At the end of this section we also address editorial truncation (§7.4) in situations where a string excerpted verbatim from a source text cannot be cut at a word boundary.

7.3.1. Descriptive and interpretive blocks

Editions encoded in XML must accurately represent both the extrinsic structure of the source — i.e. how the source text is broken up into inscribed lines and, if applicable, into pages or other distinguishable inscribed zones — and its intrinsic structure — i.e. how it is constituted of semantic or prosodic units (paragraphs, stanzas and verse lines). When preparing a draft for an XML edition or transliterating a text that will not be encoded, you may choose to use private shorthand for either or both of these purposes, inserting new lines and, if applicable and desired, adding line numbering and/or headers for specific groups of lines. When converting to XML encoding, all such segmentation and labelling must be removed.

[100] See §5.4 about the transliteration of the partially equivalent word joiner signs.

- when a line break, whether representing extrinsic or intrinsic structure, falls inside a word, this must normally be encoded in XML
  - when XML encoding is not applicable, you may choose to add a hyphen at the end of the former line in accordance with modern typographic conventions
    - but if the text is later converted to XML, this hyphen is private shorthand and must be converted to the proper encoding
- see §7.3.5 about handling line breaks that coincide with an editorial hyphen for compound segmentation

7.3.2. Segmentation versus *akṣara*s and *sandhi*

While the boundaries of morphemes are frequently hidden inside an *akṣara* of Indic script and thus not separable, Romanisation allows us to separate any two adjacent graphemes for the purpose of analysis. Thus, wherever interpretive segmentation of any kind is applicable, *akṣara* boundaries can and must be disregarded. Note that while editions of Sanskrit epigraphic texts often employ an = sign (double hyphen) to indicate that the end of one transliterated word and the beginning of the next belong to a single source *akṣara*, we always use a space or hyphen (as applicable) in such a situation. The fact that the letters separated by a space comprise one *akṣara* is indicated by their lowercase spelling (§7.3.2), while the = sign is used only to highlight instances where certain source graphemes comprise a single *akṣara* even though they should not by the standard graphotactic convention of the writing system in question (§3.3.2.1).

However, some of our source languages feature vowel fusion sandhi, whereby the final vowel of one word and the initial vowel of the next word merge into a single vowel phoneme, represented in writing by a single grapheme. In such cases we transliterate the resulting grapheme faithfully and forego interpretive segmentation. We furnish some specific cases and examples below.

- where interpretive segmentation is called for, it can and should be used even if a morpheme boundary is within an *akṣara* in the source text, including
  - independent words, e.g. Sanskrit तदेव → *tad eva*
  - words in compound, e.g. Sanskrit तत्पुरुष → *tat-puruṣa*
- this continues to apply if graphemes on either side of the boundary are altered due to sandhi or orthographic convention, so long as the graphemes can be allocated to one side of the boundary or another, including
  - the reduction of vowels to semivowels, e.g. in Sanskrit
    - गच्छत्येव → *gacchaty eva* (from *gacchati + eva*)
    - मन्वादि → *manv-ādi* (from *manu+ādi*)
    - महर्षि → *maha-ṛṣi* (from *mahā+ṛṣi*)
  - the use of the class nasal where standard orthography would employ an *anusvāra*, e.g. in Sanskrit
    - उक्तञ्च → *Uktañ ca*
    - शरणङ्गतः → *śaraṇaṅ gataḥ*
  - the assimilation of initial /h/ to a preceding stop, altering the /h/ to a corresponding aspirate, e.g.
    - Sanskrit तद्धि → *tad dhi* (for *tat + hi*) and तद्धित *tad-dhita* (for *tat+hita*)
    - Old Javanese *buAt-thajyanya* (derived from *bvat haji*, §7.3.6.1)
    - note that in these cases, the phonetic sequences may be more appropriately segmented into /tadd hi/ and /buatt-hajyanya/, but the written form involves the graphemes <dh> and <th>, which, though transliterated by digraphs, are indivisible
  - the generation of graphemes not originally present in either of the morphemes, to be segmented as in the examples below
    - Sanskrit *putrāṁl lakṣmīḥ* (from *putrān + lakṣmīḥ*)
    - Sanskrit *dīnārair ddaśabhiḥ* (orthographic variation of *dīnārair daśabhiḥ*)
    - Old Javanese *darpaṇa ryy avakta* (orthographic variation of *darpaṇa ry avakta*)
    - Old Javanese *tann inaku* (from *tan + inaku*)
    - Tamil இப்பெருங்கோயில் → *I-p-peruṅ-kōyil* (from *I + perum + kōyil*)
    - Tamil திருமேனியாட → *tiru-mēṉi-y āṭa* (from *tiru + mēṉi + āṭa*)
      - see §7.2.1 for the use of the hyphen in Tamil sandhi analysis

  - the complete elision of one of the phonemes (and the corresponding grapheme)
    - Tamil *arit' eṉṟu* (for *aritu + eṉṟu*)
      - see §7.2.2 about the elision of final *u* in Tamil
    - the elision of initial *a* after a preceding *o* or *e* in Sanskrit does not interfere with morpheme boundaries, so it can be segmented without any difficulty
      - e.g. *tato pi* (for *tataḥ + api*)
      - see also §4.3.1 about the *avagraha*
- conversely, the fusion of a final and an initial vowel into a single vowel must be distinguished from elision, and can never be segmented, including the following cases in Sanskrit
  - *kṛtārtha* (from *kṛta+artha*) and *sāpi* (from *sā + api*)
    - note that editorial *avagraha* (§4.3.1.1) must never be added to indicate the merging of initial *a* or *ā* into a preceding *a* or *ā*
    - should one or two original *avagraha*s be present in such a case, they must of course be transliterated (§4.3.1)
  - *gacchatīva* (from *gacchati + iva*)
  - *seyam* (from *sā + iyam*)
  - *gataujas* (from *gata+ojas*)

7.3.3. Segmentation versus shorthand

When the character joiner sign = (§3.3.2.1) is placed between two graphemes which are separated by an editorial space or hyphen, always put the = sign first, and the space or hyphen second.

7.3.4. Editorial spacing

Since spaces between words or other linguistic units are rare in our source texts, all original space must be explicitly represented by a placeholder (§3.1.3). Conversely, actual spaces in the transliterated text are declared to be editorial, corresponding to no feature of the source text. Editorial spaces can, and by our recommendation should, be inserted in order to segment the text into words.

Editorial spaces shall normally be inserted between words that are not compounded to one another. In addition to obviously independent words — including inflected nouns, finite verbs, verbal derivatives, invariant particles, etc. — fixed expressions should in general be regarded as phrases rather than compounds, and their members should accordingly be separated by spaces. In spacing your texts, make sure you are familiar with §7.3.2 about the interaction of segmentation with *akṣara*s and sandhi. The good practice guidelines in §7.3.6 improve consistency across the project and, in some cases, prevent conflict, redundancy or ambiguity when a text is encoded in XML and rendered for display. The additional subsections contain practical suggestions and rules for using spaces in various contexts and situations. These apply equally to text which is encoded in XML and to text which is not.

7.3.4.1. Space sparingly

Avoid redundant spaces; in particular:
- never start a line with a space
  - when creating a draft for an XML edition, where you simply enter line numbers at the beginning of each line of the source, make sure you remove the intervening space when converting the numbering to XML markup
- never use more than one space at any point

7.3.4.2. Space and numerals

Around numeral signs, use spaces as follows.
- numeral digits 0-9 (§4.2.1) must be separated by an editorial space from any adjacent graphemes
  - except that additional digits belonging to the same multi-digit number written in place-value notation must not be separated by spaces, as in Figure 4.2.A
- all other numeral signs must be separated by an editorial space from any adjacent graphemes

- this includes groups of numeral signs that represent a number together, as in Figure 4.2.B/2-5
- no spaces must be added
  - between the target characters that stand together for a single numeral sign in the source
  - between the target characters and a + sign used as shorthand (§4.2)

7.3.4.3. Space and *avagraha*

The *avagraha*, editorial or original (§4.3.1), is to be spaced as follows.
- *avagraha* must never be separated by a space from the following word, to which the elided vowel belongs
- *avagraha* must always be separated by a space from a preceding independent word, e.g. *so ’bhūt* and not *so’bhūt*
- if the word preceding the *avagraha* is compounded to the next word, then a space must not be inserted, but a hyphen may be used for compound segmentation, e.g. *saro’nte* or *saro-’nte*

7.3.4.4. Space and peripheral graphemes

Around punctuation marks (as defined in §5.3), spaces should be deployed as follows.
- preferably avoid adding a space before a punctuation mark (even though editions of Indic texts (both in transliteration and in Indic scripts) often do so when using vertical bars to represent *daṇḍa*s in the source)
- always add a space after punctuation marks if they are followed by text
  - except when a punctuation mark is (for whatever reason) inside a word
- should several punctuation marks appear in a group, preferably add spaces between them
  - especially if you transliterate some or all of them with arbitrary placeholders

Around all other peripheral graphemes, use spacing as follows.
- symbols must generally be separated by a space from any other graphemes adjacent on either side
  - generally including core as well as peripheral graphemes
    - but groups of symbols appearing together may be transliterated without spacing at your discretion
  - except when a symbol is (for whatever reason) inside a word, in which case there should be no spaces around it

7.3.4.5. Space and original space

Around a significant space in the source, whether it is encoded in XML or represented by shorthand (§5.6), use editorial spaces as follows.
- original spaces must generally be separated by an editorial space from any graphemes adjacent on either side
  - except when an original space is (for whatever reason) inside a word, in which case there should be no editorial spaces around it

7.3.5. Editorial hyphenation

Similarly to editorial spacing (§7.3.4), we declare all hyphens in the transliterated text to be editorial and recommend using hyphens for the segmentation of compounds into their members as well as for sandhi analysis (§7.2). However, modern typographic hyphens — used at line endings that fall inside a word — should be normally avoided (though see §7.3.1). Instead, use XML markup to indicate words broken by line breaks. Actual source graphemes used in a hyphen-like function are covered under connectors (§5.4).

Compound words should generally be segmented with hyphens, particularly in Sanskrit and other compound-heavy languages, while complex morphemes other than compounds — such as new words derived by affixation — should not be segmented analytically. The hyphenation of compounds is optional (though encouraged) and need not be exhaustive. Hyphenation is primarily intended to help the reader reconstruct how you analyse compounds, and should be used at your discretion. In hyphenating your texts, make sure you are familiar with §7.3.2 about the interaction of segmentation with *akṣara*s and sandhi.

The good practice guidelines in §7.3.6 improve consistency across the project and, in some cases, prevent conflict, redundancy or ambiguity when a text is encoded in XML and rendered for display. They apply equally to text which is encoded in XML and to text which is not.

- use hyphens only for the purposes endorsed by this guide, i.e.
  - normally, only for the segmentation of compounds
  - for a particular kind of sandhi analysis (§7.2.1)
  - in specific circumstances, as private shorthand for words cut across inscribed lines (§7.3.1)
- editorial hyphens will normally have phonographic graphemes on both sides
  - in rare cases, an editorial hyphen may have to be placed next to a different feature, which is normally represented by XML encoding, but may in some circumstances be represented by shorthand
    - such features include symbols, original spaces, the character joiner sign = (§3.3.2.1), the ends of epigraphic lines, and the ends of verse lines
    - in all of these cases, the editorial hyphen must be placed after the feature

7.3.6. Good practice in editorial segmentation

It is in some cases difficult to decide whether certain morphemes of a source language are to be considered dependent (bound) or independent (free), and whether a dependent morpheme is a compound member or an affix. The prevailing conventions vary considerably, so — depending on your background — you may be tempted to segment differently than what we suggest. We cannot provide exhaustive rules, much less a consistent and comprehensive rationale for such rules, but in order to increase homogeneity of practice across the project, we propose some rules of thumb and considerations for editorial analysis and recommend that you override these only with good reason. Most of our discussion primarily concerns Sanskrit, while some additional (and partly different) rules are given for Tamil and Old Javanese. When working with other languages, make an informed choice after considering any parallelisms with the guidance given here.

7.3.6.1. Phrases

On the whole, words constituting a phrase must always be separated with spaces. However, phrases may become parts of compounds, or may be subject to morphological derivation with affixes. In both of these cases, they are to be treated as compounds. Thus,

- in Sanskrit:
  - *iti kartavyam* → *iti-kartavyatā*
- in Old Javanese:
  - *tahi tikus* → *manahi-tikusa*
  - *bvat haji* → *buAt-thajyanya*
    - for the positioning of the hyphen in this second example, see also §7.3.1

7.3.6.2. Grammaticalised structures

Combinations of verbs and substantives with other words should be understood as phrases wherever reasonably possible and, accordingly, segmented with spaces. This includes grammaticalised structures (performing a function like that of conjugation or declension). For example, in Sanskrit,

- periphrastic perfects, e.g. *varayāṁ cakāra*
  - especially since other words may intrude inside such a construction, e.g. *saṁraṁjayāṁ ca prakṛtīr babhūva*
- past tense formed with imperfect and *sma*, e.g. *samādiśati sma*
- verbal prefixes used as prepositions with substantives, e.g.
  - *ā samudrāt*
  - *anu gaṅgām*

7.3.6.3. Multiple function words

Pairs or groups of function words (mainly conjunctions) are to be considered separate, even when frequently used together in a meaning that is not evident from the individual meanings of these words. For example, in Sanskrit,
- *atha vā*
- *kiṁ ca* and *kiṁ tu* (even if spelt *kiñ ca* and *kin tu*)
- *api ca* and *api vā*
- *anyac ca*
- *tad yathā*
- *na hi*
- etc.

However, for such **Sanskrit structures borrowed into other languages**, forego segmentation if the structure is listed as a single word in the relevant dictionaries, where applicable. For example,
- Old Javanese *kimuta*
- Old Cam *kintu*

7.3.6.4. Repetitive structures

When words (chiefly pronouns or substantives) are iterated with the same inflectional ending in order to express a generalised or distributive meaning, each iteration is to be treated as an independent word. Repetitions of this kind should therefore not be hyphenated, even though they are sometimes classified as a special type of compound (*āmreḍita*).
- words iterated with the same inflectional ending shall be spaced, for example
  - *yasya yasya*
  - *dine dine*
- when the first iteration does not have an inflectional ending, the formation is of course to be treated as a compound and accordingly hyphenated (if sandhi allows), for example
  - *ekaikam* (from *eka+eka*, not segmentable)
  - *pūrva-pūrvāḥ*
- in more complicated cases, proceed according to your discretion
  - hyphenation may be preferred in the following cases
    - if the word forms used may be either inflected or uninflected, as in *ahar-ahar*
    - if the iterations have different inflectional endings, as in *ekam-ekāḥ*
  - hyphenation (or no segmentation) shall be preferred when such a doubled word is followed by other compound members or a suffix, e.g. *dinaṁ-dinaṁ-vivardhamāna*
  - no segmentation is to be preferred when an iterative formation is well established in a particular meaning, as in *paramparā*
- however, **repetitions in Old Javanese** are to be considered compounds (§7.3.6.10)

7.3.6.5. Quasi-compounds

In non-standard Sanskrit, strings of nouns may appear without (or with occasional) inflectional endings. Such strings constitute neither a proper compound nor a syntactically correct phrase without emendation. They are much like compounds, but in order to avoid an interpretation that their author probably did not mean, we generally prefer to separate them with spaces.
- for example,
  - *lamvoṣṭha dedamita mahādeva guṇṭhaka ity evam-ādibhyo*
    - compare *lamvoṣṭha-dedamita-mahādeva-guṇṭhakety-ādibhyo*, *lamvoṣṭho dedamito mahādevo guṇṭhaka ity evam-ādibhyo*, etc.
  - *samrāṬ vākāṭakānāṁ mahārāja śrī-pravarasenasya*
    - compare *samrājo vākāṭakānāṁ mahārājasya śrī-pravarasenasya*, *samrāḍ-vākāṭaka-mahārāja-śrī-pravarasenasya*, etc.
- do feel free to hyphenate such structures if you feel that this is helpful for the reader

– such structures may be normalised or emended in XML markup, but this is only recommended if minor editorial intervention can render them into standard language, or if their interpretation would be ambiguous without intervention

7.3.6.6. Verbal formations

Sanskrit structures whose posterior member is a verb (typically from the roots *kṛ*, *as*, *bhū* and *gam*) or its derivative may often be perceived either as verbs with special prefixes, as compounds involving verbs, or as adverbial phrases. The segmentation of such structures is left to your discretion, based on whether you deem the structure to be separable. We offer the following specific considerations.

– structures whose anterior member is an adverbial morpheme that only occurs when prefixed to verbs — such as *tiras*, *tiras*, *puras* and *āvis* — may be hyphenated or treated like regular verbal prefixes and left unsegmented, e.g.
  – *āvir-bhavati* or *āvirbhavati*
  – *tiro-bhūta* or *tirobhūta*
– structures whose anterior member is a noun without any affix — such as *alam*, *namas* and *śrad* — should normally be hyphenated
  – e.g. *namas-kṛtya*, *śrad-dadhāmi*
  – but may at your discretion be treated as phrases (and thus spaced) when the nominal part could occur without the verbal part
    – e.g. *astaṁ gacchati*, *astaṁ gataḥ* (compare *astaṁ ca savitā yātaḥ*)
– formations expressing becoming or making into something, formed with a nominal anterior member derived with the ending *-ī* or *-ū* (the *taddhita* suffix *cvi*), should be preferably hyphenated, e.g.
  – *svī-karoti*, *svī-kṛtya*
  – *vaśī-bhavati*, *vaśī-bhūta*
  – *tanū-karoti*
– structures with adverbial anterior members formed from nouns with suffixes (*taddhita*) other than those listed above are best treated as phrases and thus segmented with spaces, e.g.
  – *brāhmaṇasād gatāḥ*
  – *khaṇḍaśaḥ karoti*

7.3.6.7. Nominal compounds

Nominal compounds should, as a rule, be segmented with hyphens unless they are proper names (§7.3.6.9). Compound hyphenation is optional and intended mainly to help the reader understand the text and follow your interpretation. General considerations for doing so are addressed here, while §7.3.6.8 discusses particular cases where hyphenation is not always desirable. Note that words derived from nominal compounds (§7.3.6.10) should be left unsegmented.

– regular nominal compounds are to be segmented with hyphens where sandhi permits (cf. §7.3.2)
  – Sanskrit *mahā-devī*, *nānā-śāstrābhyāsopabṛṁhita-niśita-vimala-buddhiḥ*
  – Tamil *tiru-makaḷ*
– the scope of Sanskrit nominal compounds includes those where the anterior member retains a declensional ending (*aluk samāsa*)
  – e.g. *ante-vāsin*, *bhayaṁ-kara*
  – however, in classical Sanskrit, most such compounds tend to be basic compounds (§7.3.6.8) and names (§7.3.6.9), so it is often preferable not to segment them
    – e.g. *dhanaṁjaya*, *puraṁdara*
– in **long compounds**, it is generally preferable to leave smaller, close-knit subunits — especially basic compounds (§7.3.6.8) — without analysis, e.g.
  – *aneka-tulābhāra-dāna-dhārā-prakṣālita-kalmaṣaḥ* (not *an-eka* or *tulā-bhāra*)
  – *paramabhaṭṭāraka-mahārājādhirāja-rājaparameśvaraḥ* (not *parama-bhaṭṭāraka-mahā-rājādhirāja-rāja-parameśvaraḥ*)
  – *snāna-pavitrīkṛta-śirasāṁ* (not *-pavitrī-kṛta-*)

– in **complex compounds** involving subordinate relationships between members which are themselves compounds, hyphenate in the way you deem best suited for interpreting the compound, including
  – hyphenating where you would otherwise prefer not to hyphenate, e.g.
    – *a-cāṭa-bhaṭa-prāveśya* (not *acāṭa-*)
    – *aśva-gaja-śāstra-jña* (not *-śāstrajña*)
    – *asi-carman-vat* (not *-carmanvat*, cf. §7.3.6.10)
  – not hyphenating where you would otherwise prefer to hyphenate
    – e.g. *brahmadeyī-kṛtya* (not *brahma-deyī-*)
– in a poetic text where a **bitextual reading** (*śleṣa*) depends on two alternative segmentations of the same sequence of graphemes, choose one of the following options at your discretion
  – forego all or part of the segmentation so as not to impose either segmentation on the text
  – prioritise the meaning you translate as primary, segment according to that, and optionally mention the alternative segmentation in a note to your translation

7.3.6.8. Basic compounds

By 'basic' compounds we refer to short compound nouns whose meaning is either straightforward or widely known. Such compounds do not necessarily have to be analysed into their members, and are in fact better left unsegmented when they are part of a longer compound. They include in particular the following categories.
– compounds with a conventional meaning that cannot be derived straightforwardly from the meaning of the members, e.g.
  – *mahārāja* (a particular kind of ruler) vs. *mahā-rāja* ('great king')
  – *nṛpati* (a king) vs. *nṛ-pati* ('man-lord')
  – *dinamaṇi* (the sun) vs. *dina-maṇi* ('day-jewel')
  – *turaṁ-gama* ('quickly-goer') or *turaṁgama* (a horse)
– compounds whose anterior member cannot occur on its own and is thus rather a prefix, i.e. *a-/an-*, *sa-*, *su-* and *ku-*
  – e.g. *an-artha*, *sa-putra*, *ku-rājan*
– compounds whose posterior member is a verbal root (with or without an explicit *kṛt* suffix such as *-t*)
  – e.g. *śāstra-jña*, *śatru-jit*
  – especially if such a compound has a non-evident conventional meaning, e.g.
    – *dvija* (a bird; a member of the upper classes) vs. *dvi-ja* ('twice-born')
    – *madhukara* (a bee) vs. *madhu-kara* ('honey-maker')
– compound cardinal numerals, e.g.
  – *dvā-daśa*
  – *pañca-viṁśati*
  – see also §7.3.6.10 about ordinals
– nominal derivatives of verbal formations for which hyphenation is recommended in §7.3.6.6, e.g.
  – *namas-kṛta*
  – *āvir-bhūta*
  – *svī-kṛta*

7.3.6.9. Proper names and styles

Proper names should not normally be segmented into compound members.
– e.g. *jayasiṁha*, *puṇḍravardhana*, not *jaya-siṁha*, *puṇḍra-vardhana*
– compound proper names may be segmented on a case by case basis when this is deemed helpful for interpretation, for example when the literal meaning of a name or part of a name is foregrounded
– do use hyphens to separate honorifics and titles from names
  – e.g. *śrī-jayasiṁha-vallabha*, *viṣṇuvardhana-mahārāja*
  – long sequences of such items may be treated as quasi-compounds (§7.3.6.5)

- when such words seem to be used as part of a name rather than styles attached to it, segment on the basis of what you interpret to be the name, e.g.
  - *bhīmarājākhyaḥ* (a person named Bhīmarāja rather than a Bhīma named rāja)
  - *śrīdharaśarman* (a Brahmin named after Śrīdhara rather than 'the honourable Dharaśarman')

7.3.6.10. Derivatives of compounds

Secondary derivatives of Sanskrit compound nouns are not themselves compounds and should therefore not be hyphenated.
- for example, *cāturvarṇya* is not a compound
  - it is derived from the compound *catur-varṇa*, but is itself not composed of the members **cātur* and **varṇya* and should not be segmented
- when the derivation does not involve a *vṛddhi* alteration of a vowel, it is still better to forego hyphenation in order to help with the interpretation
  - e.g. *asicarmanvat*, 'having a sword and a shield,' consists of the suffix *-vat* added to the compound *asi-carman* as a whole, rather than being a compound of *asi* and *carmanvat*
    - at your discretion, if you feel this is helpful for the analysis of a complex compound, instead make an exception and separate the suffix, as well as all members, with a hyphen (cf. §7.3.6.7)
- compound ordinal numbers are also derivatives of compounds and should not be hyphenated, e.g.
  - *caturviṁśa*
  - *caturviṁśatitama*

7.3.6.11. Affixes and clitics

Morphemes which are not considered independent in a given language should normally not be separated by any segmentation from the word or stem with which they are associated. In particular:
- the following Tamil formations **must not be separated** from the words to which they are attached
  - enclitic particles (e.g. *ē*, *ō*)
  - forms of the verb *āku-tal* (e.g. *āṉa*, *āy*, *āka*) when used adverbially
- the following Old Javanese formations **must not be separated** from the words to which they are attached
  - enclitic pronominal suffixes (*-(ṅ)ku* etc.)
  - possessive constructions built with the linker *-ni* (*-nikaṅ*, etc.)
  - the definite article *-ṅ*
  - the conjunction *-n*

7.4. Truncation

We use ° (the degree sign) to indicate editorial truncation at the beginning or end of a verbatim excerpted word or other string that is not cleanly separable in its original context. This use of the degree sign follows widespread convention ultimately based on analogy with the Devanagari abbreviation mark. We transliterate abbreviation marks with the same target grapheme, so in the rare situations where both might occur, they need to be disambiguated as described in §4.3.2.
- use ° (U+00B0 Degree Sign) as essential shorthand to indicate editorial truncation at the beginning or end of a string excerpted verbatim from a source text
  - truncation is not applicable in actual editions of source texts
- truncation should only be indicated when a string of characters must be cut at a point where segmentation with a space or hyphen is not applicable, i.e.
  - when only part of a word is lifted from a text, for instance at the beginning or end of a line
    - e.g. *pari°* (cited as a lemma from the word *parihāra* broken across epigraphic lines)
  - when a word lifted from a text is fused in vowel sandhi to an adjacent word
    - e.g. *°oddyotita* (for the word *uddyotita* fused in vowel sandhi to a preceding *a*)

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